Robotics Research Lab
Department of Computer Science
University of Kaiserslautern-Landau

Master Thesis

Learning-Based Online Adaptation for Control of a Hydraulic Manipulator

Kevin Kudakwashe Murera

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Hereby I declare that I have self-dependently composed the Master Thesis at hand. The sources and additives used have been marked in the text and are exhaustively given in the bibliography.

Monday 7th September, 2026 – Kaiserslautern

(Kevin Kudakwashe Murera)

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Abstract

Hydraulic manipulators are hard to control precisely. The actuator response is nonlinear, depends on load and supply pressure, and varies between machines. Simulating those effects faithfully is possible but far too slow for reinforcement learning, while commanding joint positions directly discards the behaviour that makes the machine difficult.

This thesis asks how hydraulic actuator dynamics learned from a high-fidelity Simscape model of a verge mowing arm can be reduced, transferred into an efficient robotic simulation and used for reinforcement-learning control. Two actuator representations are identified from the same data: a modular inverse and forward chain, which transfers frozen, and a structurally constrained reduced-order model, which receives a bounded calibration against the deployment plant. A model can score well on supervised prediction while ignoring the valve command a controller must use, so excitation design and command ablation are established as admission tests for a learned actuator. Both representations carry a distilled policy that replaces the classical teacher entirely, and a matched comparison over held-out meshes, trajectory families and reserved plant configurations shows that the choice of representation changes tracking, command behaviour and robustness.

Online adaptation is examined last. The deployed policy carries a payload estimate computed at runtime from the task observation it already receives, never from a privileged read. Only the context columns of an otherwise frozen policy are trained, so zeroing the estimate recovers the frozen controller exactly. All three training seeds place their entire context weight on the payload channel unprompted. Where a four to fivefold payload costs the frozen controller ground contact, the adaptive policy improves cut height, lateral tracking and ground contact together, paired over a held-out mesh with every pooled confidence interval excluding zero, though two seeds of three carry the effect. At a payload the frozen controller still handles, the same conditioning costs lateral accuracy, so the mechanism belongs in the loaded regime rather than everywhere. Permuting the payload across environments, which preserves its distribution and destroys only its pairing, is beaten by the correct estimate but recovers most of the gain alone: the benefit is part conditioning on a plausible load and part online identification. The true payload is not an upper bound: at the largest load it can drive the tool off the ground, because the estimate's gradual arrival is itself part of the mechanism. Effective control of this machine therefore depends not only on model accuracy and policy robustness, but on conditioning the controller on a load it infers for itself.

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1. Introduction

1.1 The Role of Mechanised Verge Maintenance

Road verges are cut because they are infrastructure. Vegetation left to grow closes sight lines at junctions and bends, obscures signs and barriers, blocks the drainage channels that keep water off the carriageway, and encroaches on the running surface itself. A national road network carries tens of thousands of kilometres of verge, and each kilometre is cut several times in a growing season, so the work is continuous, repetitive and performed almost entirely by machine.

It is also performed from a moving vehicle. Stopping to cut each section would require closing the lane, so a carrier drives the road at a few kilometres per hour while a boom-mounted mowing head works the verge alongside it. The head must hold a cut height of a few centimetres over ground that rises and falls continuously, follow a grass line that curves with the road, and do so without striking the surface hard enough to damage the tool or lift the machine.

The demands on the manipulator follow from that. The machine studied here, shown in Figure 1.1, carries a heavy rotating tool at the end of a long boom, works in intermittent contact with the ground, and must place that tool accurately while its own base is in motion. Hydraulic actuation is the standard answer at this scale: it supplies the force density and the tolerance to shock loading that electric drives do not, and it does so with components rugged enough for continuous outdoor operation. Precision, however, is not what hydraulics are good at, and that tension is the subject of this thesis.

1.2 Advances in Learning-Based Control for Hydraulic Manipulators

Two developments have changed what is achievable on machines of this kind.

The first is that policy optimisation in simulation has become practical for contact-rich manipulation. GPU-parallel physics engines run thousands of environments simultaneously, which makes reinforcement learning feasible on tasks that would take years of real-machine

experience, and randomising the simulator across a family of plants produces policies that transfer without being tuned to one machine. The second is that the actuator itself can be learned. Rather than deriving valve and cylinder behaviour from first principles and fitting its parameters, a network trained on recorded motion can represent the command-to-motion relationship directly, including the nonlinearities that analytical models approximate away. Chapter 2 sets out both lines and the work they rest on.

Combining the two is attractive and not straightforward. A learned actuator model must be cheap enough to evaluate inside the training loop, which rules out the high-fidelity models used for offline analysis, and it must remain valid across the family of machines the policy will meet. This has motivated a line of work in which a compact code describing the current plant is inferred from recent interaction history and supplied to the controller, so that a single policy can serve a family rather than a single machine. Such methods report substantial gains in legged locomotion and in manipulation, and they are the direct antecedent of the architecture examined here.

Figure 1.1: MULAG FME 500 front mower on a Unimog during roadside verge maintenance. Image source: MULAG Fahrzeugwerk, FME 500 product documentation.

1.3 Challenges of Precise Tracking with Hydraulic Mowing Arms

A valve command does not produce a proportional cylinder motion. The relationship is nonlinear, depends on load and supply pressure, and is subject to dead time between the command and any measurable response. It also differs between machines: seal condition, fluid temperature, hose compliance and the attachment fitted all change it, so two machines of the same model respond differently and one machine responds differently across a season. A controller that inverts only the geometry therefore leaves the largest source of tracking error unaddressed.

That difficulty admits two responses, and they are routinely conflated. The first is robustness: train across a randomised family so that a single fixed model is adequate everywhere. It costs nothing at run time and gives up the difference between the family and the individual. The second is adaptation: infer which member of the family is present from the interaction history and condition the model or the policy on that estimate. It promises to recover that difference, at the cost of an estimator that must work from observable signals alone, since a deployed machine has no access to its own plant parameters.

Whether adaptation repays that cost depends on three things. Plant identity must be present in the signals produced under the available excitation. The robust controller must leave a control-relevant deficit to recover. The correction path must retain enough authority to change the realised command. This thesis measures these conditions on a hydraulic mowing arm and then evaluates whether live history adds value beyond matched residual capacity.

1.4 Contribution of This Thesis

The difficulty described in Section 1.3 is a modelling problem before it is a control problem. A hydraulic actuator can be simulated accurately, but the models that do so are too slow to sit inside a reinforcement-learning loop requiring millions of environment steps, while the usual alternative of commanding joint positions directly removes the load and pressure dependence that makes the machine difficult to control. What is needed is a representation cheap enough to train against and faithful enough that the resulting policy controls something close to the real actuator.

Learned actuator models and payload-conditioned adaptation are both established techniques. What remains unclear is how such models should be identified so that they retain the command dependence a controller needs, whether they transfer between simulators without retraining, and what a controller must know about its load once the models are used in control. Representation and transfer are established first because the adaptation acts through an identified plant rather than replacing one. Online adaptation is evaluated last so that any remaining gain can be interpreted against the transferred model and the robustness of the fixed controller. The order follows the dependencies among the claims. The research statement is as follows:

Comparison notation.

Three policy–actuator stacks are compared, each carrying a short label used throughout. *R* (PPO-ROM) uses the command-conditioned reduced-order model. *IF* (PPO-ID-FD) uses the frozen inverse and forward pair. *AF* (Adaptive Forward stack) drops the inverse model and commands the valve directly, which is where command authority remains. The labels name complete stacks rather than actuator models alone, and Section 6.7 gives their construction. One point is worth fixing here because the name invites the opposite reading: the offline AAM of Chapter 5 is a diagnostic estimator and is not loaded in the AF command path.

To determine how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, and under what conditions online adaptation improves on the robustness those learned models already provide.

The statement decomposes into six research questions, which are answered in the order given. The first three concern how the actuator is represented, the fourth concerns control through that representation, and the last two concern what remains for adaptation once the first four are settled. The six questions are:

1. How can data from a high-fidelity Simscape model be used to learn computationally tractable representations of hydraulic actuation while preserving the command dependence a controller requires?
2. To what extent do actuator models identified from Simscape transfer to an IsaacLab manipulator without deployment-specific retraining?
3. Can a structurally constrained reduced-order hydraulic model supply dynamics that are accurate and stable enough for GPU-parallel policy training?
4. Can a policy trained from a privileged classical teacher become the sole deployed controller while acting through learned actuator dynamics?
5. How does the choice of actuator representation affect tracking, command behaviour, latency and robustness under plant variation?
6. Where plant variation survives robust training, can a controller estimate its own payload online and condition on it, and in which regime does that conditioning improve closed-loop tracking rather than cost it?

RQ6 concerns one deployed mechanism. The policy receives a payload estimate computed at runtime from the task observation it already carries, and only the context columns of an otherwise frozen policy are trained, so zeroing the estimate recovers the frozen controller exactly (Section 6.7.3). What the policy is given is a single physical quantity rather than an unconstrained code, and it is tested as such. The AAM of Section 5.5 is not part of this path. It is an offline diagnostic used to judge whether the inverse model can be trusted, and the frozen stacks are robust controllers rather than adaptive ones.

The first question is answered in Chapters 4 and 5, the second and third in Chapter 5, the fourth in Chapter 6, and the last two in Chapter 7.

RQ6 is the question named in the title and is deliberately placed last, since it is only worth asking once the preceding questions have established that the learned models transfer, that a policy can control through them, and how much plant variation the closed loop already absorbs unaided. Its answer separates two levels. Online plant context is usable at the model level, and whether that benefit reaches the closed loop depends on the interface the correction acts through: plant variation must remain control-relevant after

robust training, be observable from the available history, and reach a correction path with sufficient authority. Any apparent gain must also be separated from the effect of adding residual capacity. The work rests on the following contributions:

Transferable Hydraulic Actuator Learning: A pipeline that identifies hydraulic actuation from high-fidelity Simscape data and transfers it into IsaacLab through two representations, a modular inverse and forward chain and a structurally constrained reduced-order model. The inverse and forward pair transfers frozen, without deployment-specific retraining. The reduced-order model instead receives a small bounded calibration against the deployment plant, so it serves as a reference condition rather than an independent competitor (Chapter 5).

Control-Relevant Identification and Validation: A demonstration that predictive accuracy can conceal the loss of exactly the property a controller needs, together with the procedure that detects it. A forward model reached a near-perfect supervised score while learning to ignore the valve command. The validation loss did not show this; ablating the command did. Excitation design paired with input ablation is established here as a required test before a learned actuator model is admitted to control (Chapters 4 and 5).

Reduced-Order Actuator Model: A command-conditioned recurrent actuator surrogate whose transition is written as an explicit hydraulic chain, so the predicted step stays differentiable in named quantities and bounded in form, with learned heads covering valve realisation and unmodelled loading. It is accurate enough to serve as the deployment plant, stable in recursive rollout, and cheap enough to evaluate inside GPU-parallel policy training. It is not an identified plant: the fitted physical constants settle on their clamps and the realised motion comes from the learned heads, which is measured and reported rather than assumed (Section 5.4).

Policy-Only Control Through Learned Hydraulics: A damped least-squares controller supplies privileged demonstrations during training and is then removed, leaving the distilled policy as the entire deployed controller, acting through learned actuator dynamics. Both representations support closed-loop terrain following in that configuration (Chapters 6 and 7).

A Decision Procedure for Online Adaptation: A matched protocol that asks, in order, whether plant identity is present in the identification data, what a privileged plant encoding does in the closed loop, and only then fits a deployable estimator of it. Either of the first two measurements can show that a correction would have nothing to act on (Chapters 5 and 7).

Where Online Adaptation Is Effective: An online payload-conditioned controller measured on one machine under one protocol, with the regime in which it helps identified rather than assumed. It improves cut height, lateral tracking and ground contact together where a heavy payload costs the frozen controller ground contact, and costs lateral accuracy where the frozen controller is not failing. Permuting the payload across environments separates what correct identification contributes from what conditioning on any plausible load contributes (Chapter 7).

Chapter 8 restates these six with the measurements attached. They are stated here as claims and there as results, so the evidence is given once.

The causal plant estimator, the bounded adaptation path and the eight-gate evaluation harness are described where they are used, as technical enablers rather than separate claims.

Figure 1.2 condenses the logic that connects these contributions. The sequence is a cycle rather than a one-way pipeline: evidence from robustness, identifiability and adaptation feeds back into the next actuator representation and the transfer assumptions made for it.

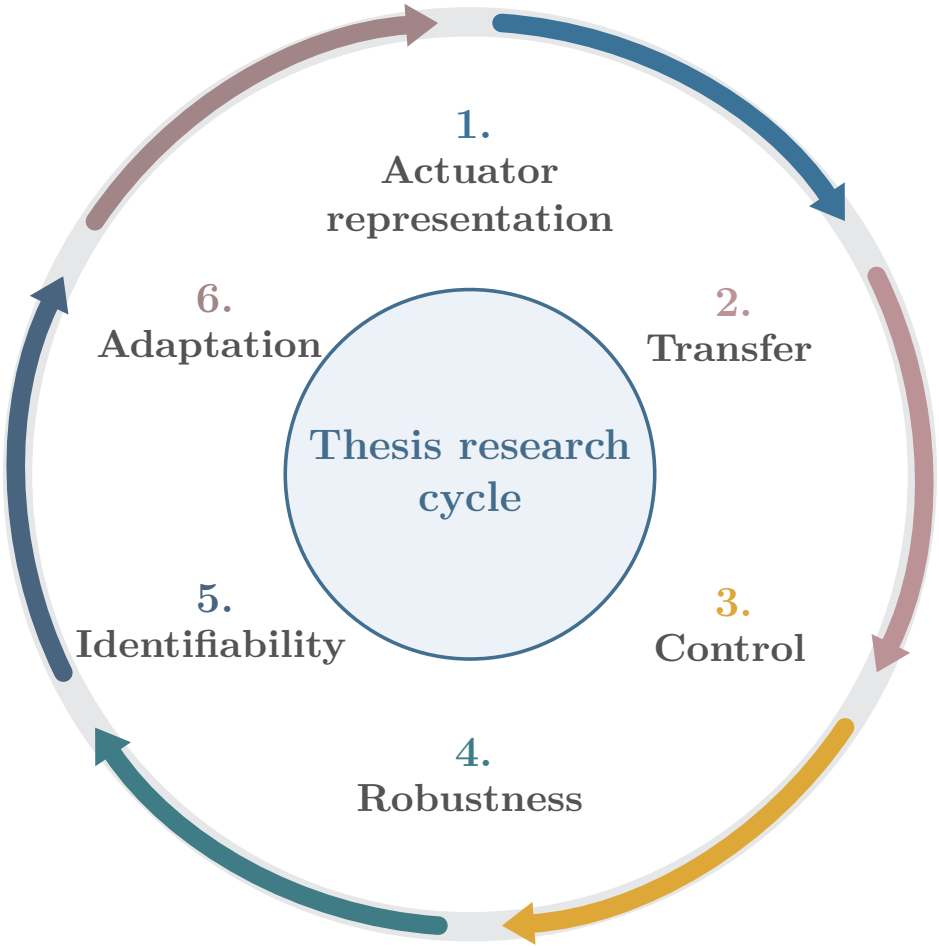

Figure 1.2: Research cycle connecting actuator representation, transfer, closed-loop control, robustness, identifiability, and adaptation. Evidence from the final stages informs the next actuator representation and its transfer assumptions.

1.5 Thesis Structure

This thesis investigates how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, grounding that investigation in the identifiability of the actuator models and in the robustness of the controllers built on them. The document is organised as follows:

Chapter 2 establishes the background against which the remainder is read. It covers the actuation chain, the kinematic tools underlying the classical controller, conventional model-based control of hydraulic uncertainty, learned dynamics models and the pitfalls

of identifying them from closed-loop data, domain randomisation and latent adaptation. The chapter closes by stating the gap that remains and where the three control stacks sit against it.

Chapter 3 analyses the platform, taking the MULAG FME 500 on a Unimog carrier as the case study. It describes the rigid bodies and the serial chain they form, the mappings between task space and cylinder space, and the high-fidelity Simscape model the actuator data derives from. It also fixes what the controller may observe and the evaluation gates every later result is scored against.

Chapter 4 presents the data generation strategy and answers the first research question. It describes the excitation design, the randomised factor set that defines the family of plants, and the procedures that keep training, validation and reserved test configurations separate. It then reports the identifiability audit, including the measured finding that excitation gathered under a task controller does not separate plants and the open-loop collection that repairs it.

Chapter 5 develops the learned actuator models and answers the next two research questions. It presents the inverse model, the forward model, the evidence that supervised accuracy alone does not establish command authority, and the reduced-order model that serves as the deployment plant. It closes on the AAM as a diagnostic for whether the inverse model can be trusted, and on the online payload estimator the deployed controller conditions on.

Chapter 6 describes the control framework and its integration with the simulator, and answers the fourth research question. It presents the damped least-squares teacher, the observation and action interfaces the stacks hold in common, the distillation of the teacher into a student policy, and the reference-anchored fine-tuning that keeps further optimisation admissible. It then assembles the three deployed stacks and states which comparison each one can carry.

Chapter 7 reports the experimental evaluation and answers the last two research questions. It compares the matched pair on tracking, command behaviour and dynamic response, and measures how much of the randomised factor draw reaches each plant. It closes on online adaptation, measured inside the adaptive stack against its own frozen base and a capacity-matched control.

Chapter 8 concludes the thesis. It summarises what the work establishes and what it does not, states the design implications that follow, and records the limitations of the evidence. It ends with the future work, which begins with the simulation experiments that would most change what the thesis can claim and ends with transfer to the physical machine.

2. Fundamentals and Related Work

Chapter 1 argued that precise task-space tracking with a hydraulic mowing arm is limited less by kinematics than by the actuator. The path from a valve command to actual cylinder motion is nonlinear and load dependent, and it differs from one machine, season, or attachment to the next. This chapter collects the background needed to judge that argument and to understand the design decisions made later in the thesis. Section 2.1 introduces the machine class and its electro-hydraulic actuation. Section 2.2 recalls the kinematic tools behind the classical controller, and Section 2.3 reviews how conventional model-based control deals with hydraulic uncertainty. The remaining sections cover the learning methods this work builds on. Section 2.4 treats data-driven dynamics models and the pitfalls of identifying them from closed-loop data. Section 2.5 covers reduced-order and grey-box modelling, Section 2.6 domain randomisation and latent adaptation, and Section 2.7 policy optimisation with structured priors. Section 2.8 then states the gap that remains and how the three control stacks of this thesis (R, IF, and AF) address it. As defined in Chapter 1, the first two abbreviate PPO–ROM and PPO–ID–FD; AF denotes the deployed Adaptive Forward stack rather than the offline AAM. Only material that is needed for the later chapters is included. Background and results are kept apart: where a later measurement is the reason a line of work was left behind, it is quoted forward and marked as such, and no argument in this chapter rests on it. The concrete system definition and problem formulation follow in Chapter 3.

2.1 Manipulator Actuation and Control Fundamentals

2.1.1 Application Context and Implement Carriers

The MULAG FME 500 mounts a multi-link hydraulic boom and a 1.2 m flail head on a Unimog carrier [MULAG Fahrzeugwerk 26]. It reaches about 6.5 m laterally, below road level and over obstacles, while an operator adjusts cutter height, tilt and lateral position as the vehicle moves. Automating this continuous tracking task is the practical motivation of the thesis.

Hydraulic cylinders suit the required force, shock tolerance and existing vehicle power supply, but their command gain changes strongly with load and configuration. This trade-off between force density and precision is central to hydraulic manipulator control [Mattila 17].

2.1.2 The Electro-Hydraulic Actuation Chain

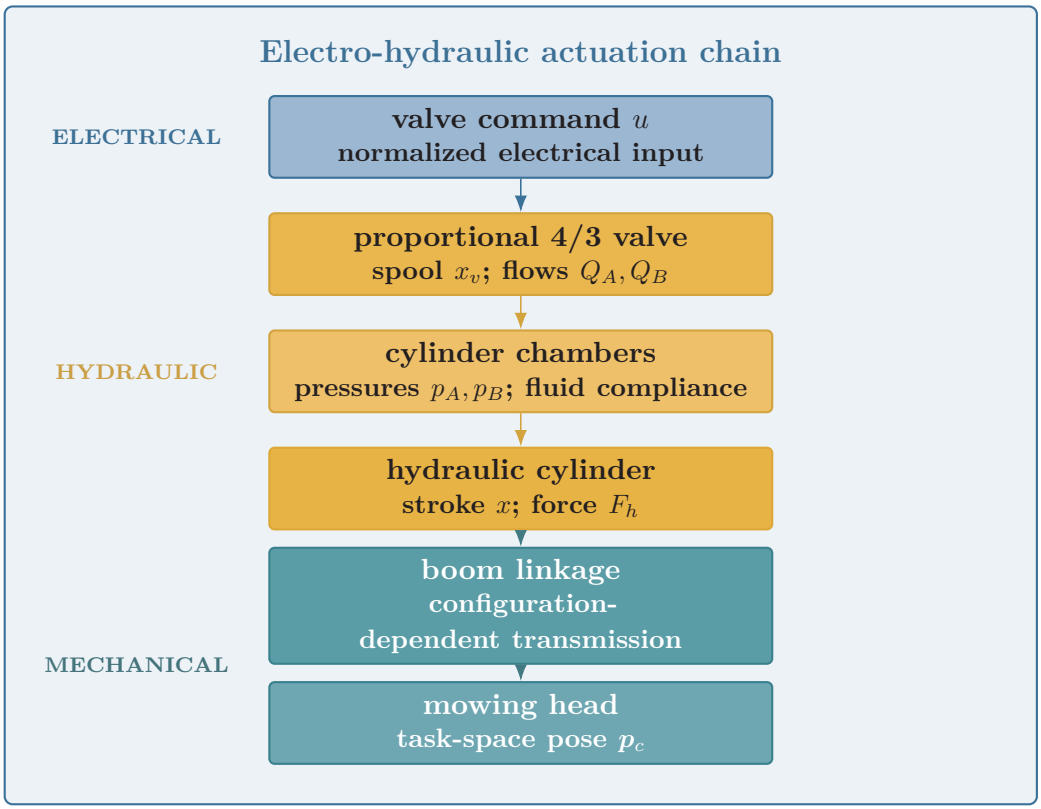

Figure 2.1: The electro-hydraulic actuation chain from valve command to mowing-head motion. The block colors distinguish the electrical, hydraulic and mechanical stages; each subtitle identifies the principal quantity passed to the next stage.

In Figure 2.1, an electrical command moves a proportional valve spool and meters flow to the cylinder chambers. The orifice law [Merritt 67] makes flow depend on spool position and the square root of the pressure margin. Valve authority is therefore load-dependent and directional; overlap creates a dead zone near null and the spool saturates at large commands [Jelali 03].

Compressibility, changing chamber volume, relief limits, leakage and seal friction further separate flow from motion. Several of these effects are non-smooth, while bulk modulus, leakage and friction drift with temperature and wear. The mechanical linkage then converts cylinder translation through configuration-dependent lever arms. Chapter 5 consequently models the hydraulic realisation in cylinder space, before kinematic effects are mixed into cutter error.

2.1.3 High-Fidelity Simulation as a Data Source

The component-level Simscape plant [The MathWorks, Inc. 24a] resolves the stiff hydraulic dynamics but runs one machine at small integration steps. Policy optimisation instead

benefits from thousands of GPU-parallel plants [Makoviychuk 21]. The detailed model is therefore used offline to generate identification data; compact actuator models carry its behaviour into parallel training, as summarised in Figure 2.2.

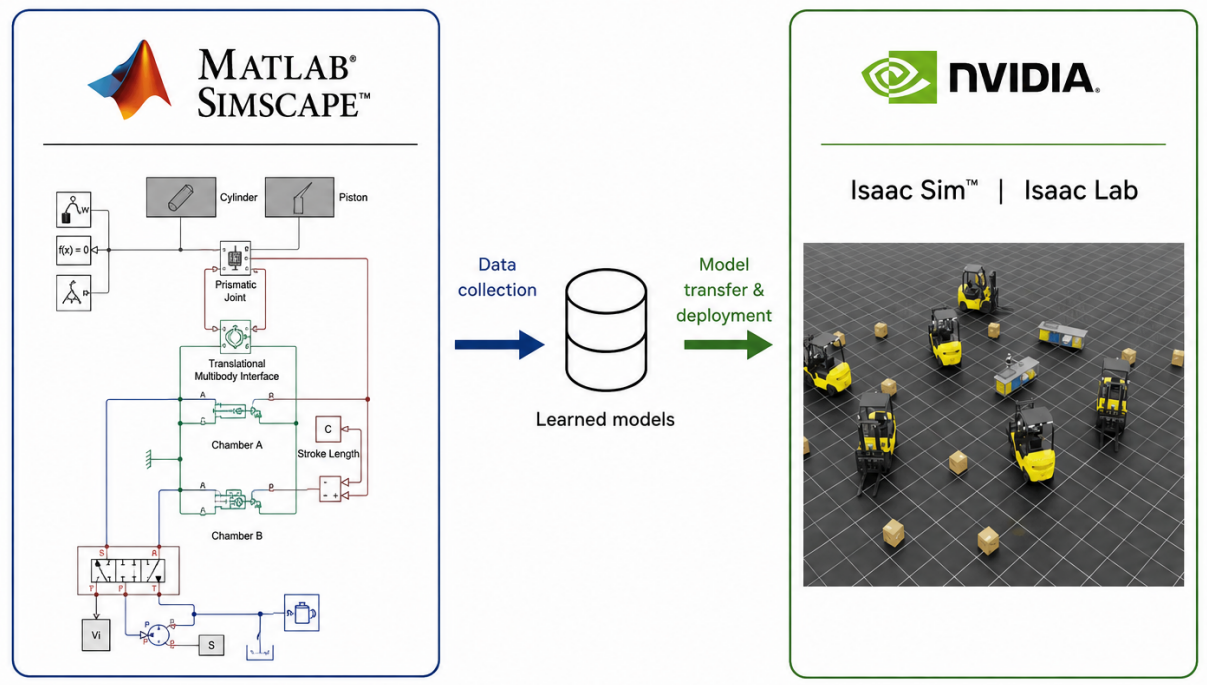

Figure 2.2: Simscape supplies high-fidelity hydraulic data; compact actuator models transfer that behaviour to GPU-parallel training in IsaacLab. Product images: MathWorks and NVIDIA.

2.2 Kinematics and Task-Space Tracking

2.2.1 Differential Kinematics

The cutter path is specified in vehicle coordinates, so the manipulator Jacobian maps joint rates to the required tool velocity [Siciliano 09]. Near singular configurations the ordinary pseudo-inverse produces excessive commands. The damped least-squares inverse of Wampler [Wampler 86] and Nakamura and Hanafusa [Nakamura 86] trades some local accuracy for bounded commands and is retained as the classical controller.

2.2.2 From Joint Space to Cylinder Space

The actuated coordinates are cylinder lengths, and their nonlinear cylinder-to-joint map has a configuration-dependent lever arm. A passive head joint further limits independently controllable directions. The calibrated cutter Jacobian in Chapter 3 composes these mappings. It still assumes that commanded cylinder motion is realised; the learned actuator models address that remaining gap.

2.3 Model-Based and Conventional Control Approaches

Three families of conventional control are relevant here: fixed-gain cascades, nonlinear cancellation, and online parameter estimation. Each of them works on hydraulic manipulators. Each also carries a maintenance or excitation cost, and those costs are the reason this thesis learns the actuator instead.

2.3.1 Fixed-Gain Cascades

Industrial implements commonly map joystick input directly to valve current. The standard improvement is a cascade in which an outer position loop sets a velocity or pressure reference and a fast inner loop tracks it [Jelali 03]. The construction is cheap, needs no plant model beyond a few tuned gains, and is what a commercial machine realistically ships with.

The cost is that the gains are tuned around one operating point. The orifice law makes command-to-velocity gain scale with the square root of the pressure margin, so it changes with load and with direction of travel, and the linkage lever arm changes with configuration. A gain set tuned at the stiff end of that range responds weakly at the compliant end; tuned at the compliant end it oscillates at the stiff end. Scheduling the gains over configuration and load recovers some of the range, but the schedule is machine and attachment specific and has to be remeasured whenever either changes.

2.3.2 Nonlinear Cancellation

Nonlinear model-based control removes the dependence on an operating point by cancelling the known dynamics rather than tuning around them [Slotine 91]. For a hydraulic actuator the cancellation has to span the orifice nonlinearity, chamber compressibility and the rigid-body load, which places the valve command several integrations away from the position error and makes backstepping the usual construction [Jelali 03]. The survey of Mattila et al. [Mattila 17] reports this line, together with virtual decomposition control, as the state of the art for heavy-duty hydraulic arms, and it does deliver accurate tracking on instrumented machines.

Cancellation is only as good as the model it inverts, and the terms it depends on are the terms that drift. Bulk modulus, leakage and seal friction change with temperature and wear over a working day. Some of the structure is non-smooth and cannot be cancelled at all: valve overlap produces a dead zone that has to be inverted explicitly, as Yang et al. [Yang 26] do alongside state constraints. The learned actuator models of Chapter 5 carry an explicit dead-zone term for the same reason. Maintaining this identification per machine, season and attachment is what makes the family expensive on a commercial implement [Mattila 17], not any weakness in the control law.

2.3.3 Online Parameter Estimation

Adaptive control estimates the uncertain parameters during operation instead of fixing them in advance [Åström 95]. Yao's adaptive robust control (ARC) pairs a projection-bounded parameter estimator with a robust feedback term, so that a guaranteed transient

tracking bound does not wait for the estimate to converge, and it is the established result for single-rod hydraulic actuators [Yao 00]. On the face of it this is the closest classical answer to the problem of this thesis, because it targets exactly the drifting parameters just described, and because it is the one classical family that ships a proof rather than an empirical trend, in direct contrast to the adaptation studied here (Section 6.7.7). ARC buys that guarantee by assuming a known parametric structure for the plant: the orifice, chamber and friction terms must be derived correctly in advance, with only their coefficients left uncertain. That is exactly the assumption Chapter 5 finds hard to hold on this machine, where the reduced-order model's own orifice and friction parameters settle at their optimisation clamps rather than at physically plausible values (Section 5.4).

Two properties of the task limit it. Parameter convergence requires persistent excitation, and mowing supplies smooth, low-bandwidth commands that do not provide it; the redundancy between command and state that Section 2.4.2 describes for learned models constrains a recursive estimator in the same way. The valve also saturates for much of the scored window on this machine (Section 5.6), where the regressor stops reflecting the applied input and adaptation has to be suspended or explicitly compensated [Sofrony 24]. Both limits belong to the task rather than to the estimator, so they bear on any correction applied to the command on this machine. They are measured (Section 5.6.2) rather than assumed away, and they are part of why the deployed mechanism conditions the policy instead of correcting its command.

The scope of that argument should be stated plainly. No classical adaptive baseline is run in this thesis. The experiments of Chapter 7 compare actuator representations inside one reinforcement-learning stack, so nothing here shows that adaptive robust control tracks worse than the learned route on this machine. What is claimed is narrower: the two conditions the classical estimator needs, informative excitation and an unsaturated regressor, are not supplied by the mowing task, and the maintenance burden the family carries per machine, season and attachment is the burden the thesis is trying to remove.

The experiment that would settle the comparison is concrete rather than hypothetical. It needs four things. The first is a per-cylinder differential-pressure loop, because ARC's estimator is driven by the same pressure signal this simulation only produces as an uncalibrated model state (Appendix B.5). The second is an open-loop identification campaign on the physical implement, to fit the orifice and friction structure ARC assumes rather than the clamped values the reduced-order model settles on (Section 5.4). The third is a projection bound and adaptation gain tuned against that identified structure. The fourth is a second control implementation, run on the same excitation schedule and evaluation gates as Chapter 7, so that the two families are compared under one protocol rather than one against the literature's reported numbers for the other. None of that instrumentation exists on the target machine yet (Chapter 8), which is why the comparison is left open rather than run.

2.3.4 What Is Retained

The classical methods are not replaced wholesale. Kinematics is reliable and is kept: the DLS controller of Section 2.2 acts as teacher and as the data-collection controller. What is learned is the part the model-based families struggle to maintain: the mapping from cylinder command to realised motion, and the policy that issues that command (Section 2.7).

2.4 Data-Driven System Identification and Learned Dynamics

2.4.1 Learned Forward and Inverse Dynamics

Robot model learning distinguishes forward models, which predict the next state from state and command, from inverse models, which predict the command for a desired transition [Nguyen-Tuong 11]. Section 5.1 fixes both interfaces. One-step fit is insufficient when a forward model is recursively rolled out, because small errors compound [Nagabandi 18]. The models here are therefore trained with multi-step objectives. Probabilistic ensembles represent growing rollout uncertainty [Chua 18]; recurrent architectures instead retain unobserved pressure and valve history [Hochreiter 97, Cho 14], as used by the reduced-order model.

2.4.2 Closed-Loop Identifiability

Informative identification requires inputs that excite the plant independently of feedback [Ljung 99, Isermann 11]. In closed-loop data, command and state history are correlated; a flexible predictor can score well by ignoring the command, and validation from the same distribution will not reveal the shortcut [Schoukens 19].

That failure occurred here: a closed-loop forward model reached one-step $R^2 \approx 0.999$ yet showed almost no command dependence under ablation. Chapter 4 therefore uses open-loop valve excitation, and every learned model is checked with zeroed and shuffled commands. Prediction score alone is not treated as evidence of control authority.

2.5 Reduced-Order Modelling

Reduced-order models retain dominant behaviour at lower cost [Benner 15]. Here the full Simscape plant is reduced to per-cylinder extension, velocity and pressure difference. The construction is grey-box: known structure is embedded so that data fits what remains unknown [Karniadakis 21]. As in the structured dynamics network of Lutter et al. [Lutter 19], explicit physical layers improve interpretability. The hydraulic ROM retains the orifice, pressure and integration structure; learned terms represent valve realisation, configuration-dependent load and bounded residuals.

Hwangbo et al. [Hwangbo 19] combined analytic rigid-body mechanics with a learned actuator response, a pattern later scaled in GPU-parallel simulation [Rudin 22]. Yamamori et al. [Yamamori 26] instead shaped the physical actuator to match the simulation. A commercial valve and pump cannot be redesigned in that way, so this thesis learns the hydraulic realisation while PhysX retains the rigid-body mechanics (Figure 2.3).

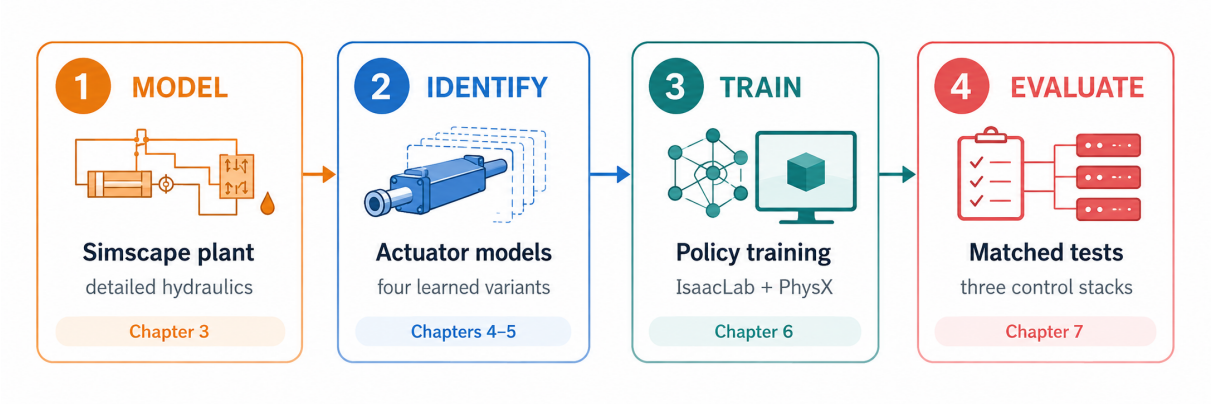

Figure 2.3: Workflow from the hydraulic plant model to matched evaluation. The separation between rigid-body physics and learned actuation follows the actuator-network approach in [Hwangbo 19]. This figure was generated using AI.

2.6 Domain Randomisation and Latent Adaptation

2.6.1 Robustness Through Randomised Training

A model trained on one simulated plant inherits its parameters as assumptions. Domain randomisation instead trains across a family, first for visual appearance [Tobin 17] and then for dynamics such as mass, friction, gain and latency [Peng 18]. The hydraulic family in Chapter 4 varies bulk modulus, valve gain, leakage, friction and payload, producing robustness to physically plausible plant variation.

2.6.2 From Robustness to Adaptation

Figure 2.4 contrasts a policy trained for one robust response across the plant family with an adaptive stack that conditions its response on the current plant.

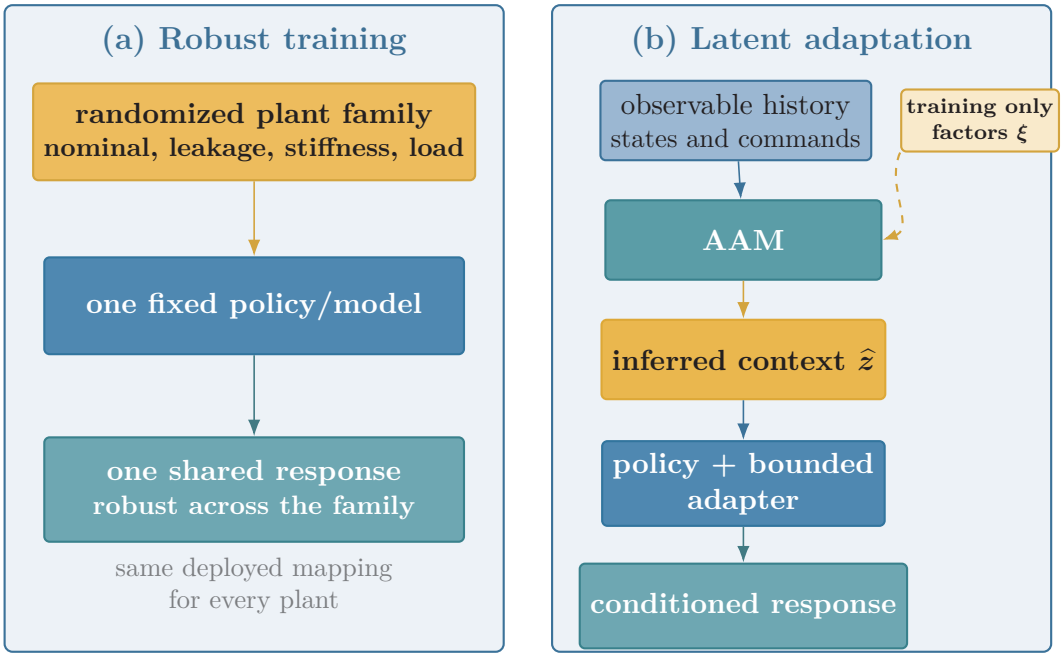

Figure 2.4: Canonical latent-adaptation architecture contrasted with robust training. The architecture motivates the offline identifiability study; the final AF implementation instead learns its context representation end to end from the control objective.

Teacher–student methods encode known simulation factors into a privileged latent and train an estimator to recover it from observable history. The construction generalises the wider privileged-learning idea, in which a teacher with access to unavailable state supervises a student restricted to deployable observations [Chen 19]. Lee et al. [Lee 20] used it for hidden terrain properties; Kumar et al. [Kumar 21] developed rapid motor adaptation; Miki et al. [Miki 22] extended the same teacher–student route to fusing proprioception with exteroception under unreliable perception. Distillation from a privileged teacher is the mechanism this thesis uses in Chapter 6, with a classical controller in the teacher's place rather than a privileged policy [Ross 11]. All of them assume that the factor directions the estimator retains are the directions the controller needs. Liu et al. [Liu 25] report the failure when it does not hold, where a low-dimensional adapter drops factors that act through separate channels, and add training phases to recover them. Chapter 5 finds the same mismatch here in a sharper form, and the finding is quoted forward because it is why the deployed stack departs from this line: what an unconstrained estimate carries here is mainly seal condition, which acts through pressure, and pressure is the one channel the controller does not integrate (Section 5.5.1). The deployed controller therefore conditions on a single physical quantity it can act on, the payload, rather than on a learned code.

Unlike classical online parameter estimation [Åström 95], the learned latent need not have a direct physical interpretation. It can be given one. Hu et al. [Hu 26] estimate named physical quantities on an excavator arm with Bayesian reinforcement learning, which buys

interpretation at the price of the explicit plant model that Section 2.3 identified as the maintenance burden. The latent used here is anonymous and is therefore audited for content rather than read as parameters. A latent can also condition a dynamics model instead of a policy, as in the contextual neural ODE of Yu et al. [Yu 26]. That is the nearest relative of the route taken here, with one difference that decides the outcome: in AF the conditioning enters the command path rather than the predictor, because a better prediction helps only if something downstream acts on it.

The offline AAM developed in Chapter 5 tests whether one second of observable history contains recoverable plant information. It is not the encoder deployed in AF, which trains its own history encoder end to end with the residual under the task objective rather than against an offline target (Section 6.7.5 details the training scheme). The two routes therefore answer different questions: the former tests identifiability; the latter tests control-relevant adaptation. The experiments also keep robust randomised training distinct from both routes.

Randomised training produces a single controller for a family of plants. Identification of that family is what makes a member-specific correction possible. In canonical latent adaptation, adaptation is the online use of that identification: the controller infers which plant is present from observable history and conditions the command on that estimate. The two responses are therefore sequential, not alternatives.

2.7 Reinforcement Learning and Structured Priors

2.7.1 Policy Optimisation

Reinforcement learning optimises a policy from observations to actions for expected cumulative reward [Sutton 18]. This thesis uses PPO [Schulman 17] with generalised advantage estimation [Schulman 16]; Section 6.2.5 gives the exact objective. GPU-parallel simulation supplies the required throughput [Makoviychuk 21, Rudin 22], which is why compact actuator surrogates replace the full hydraulic plant during training.

2.7.2 Structured Priors for Policy Learning

Learning from scratch would discard reliable kinematics. Residual RL instead adds a bounded learned correction to a controller at runtime [Johannink 19, Silver 18]. Policy distillation moves the prior to training: a privileged DLS teacher supplies demonstrations, then the student acts alone [Ross 11]. This thesis takes the second route. The teacher is distilled and PPO updates are anchored to the frozen student (Section 6.4), and the deployed actor remains the sole controller: no analytic controller runs underneath it at deployment, so the runtime correction of residual RL is not the construction used here.

One clarification belongs here, because the vocabulary collides. The adaptive stack of Chapter 6 does modify its own frozen actor, but not by adding a residual in the sense of this subsection. There is no classical controller beneath it to correct, and its purpose is not to supply a prior that accelerates learning. It is an online adaptation mechanism internal to the reinforcement-learning policy, whose only function is to carry the dependence on inferred plant context that a frozen actor cannot express. It therefore belongs to the

adaptation line of Section 2.6.2 and is compared against that literature, not against residual policy learning.

The idea parallels real-world fine-tuning that retains a simulation-trained value function [Yin 25], but here the retained object is the reference policy. Anchoring was introduced because unconstrained PPO improved reward while degrading the physical tracking criterion (Section 6.3).

2.7.3 Reinforcement Learning on Hydraulic Machines

Deep RL has reached full-size hydraulic machines. What distinguishes those stacks from the one built here is not scale but the level at which the policy commands the machine, because that level decides how much of the hydraulic chain of Section 2.1.2 the policy has to represent at all.

Egli and Hutter [Egli 20] trained an excavator-arm policy in simulation and transferred it to a full-size walking excavator, then generalised the construction across excavation tasks [Egli 22]. The policy acts on joint-level velocity references, and a low-level tracking layer on the machine converts them into valve currents. Spinelli et al. [Spinelli 24] keep that division on a commercial material handler and add an underactuated tool, so the policy has to exploit passive tool dynamics rather than command them, while the actuated joints are still driven through the same kind of tracking layer. In both cases the orifice nonlinearity, the dead zone, the directional asymmetry and the load dependence live below the learned policy, in a layer that was identified and calibrated for that machine before the policy existed.

That division is sound where the tracking layer already exists. It is not available here for two reasons. The FME 500 is a series road-maintenance implement with no calibrated joint-velocity layer to inherit, and building one is precisely the per-machine identification that Section 2.3 identified as the maintenance cost of the model-based families. More importantly, a velocity interface hides the quantity this thesis sets out to measure. If the layer beneath the policy already owns the command-to-motion mapping, the question of whether a learned actuator retains command authority under saturation and dead zone cannot be posed at all; it has been answered by construction, once, by hand. The stacks here therefore command the actuator directly, either as a cylinder increment realised by a learned model or, in AF, as the valve command itself.

Maitan et al. [Maitan 26] take the opposite position and train on the physical cylinder, bounding exploration with a contraction-metric filter. That work does command the actuator at the low level, and training on hardware removes the transfer question outright. The result is force control on a single actuator on a test bench, and it does not obviously scale to a multi-link boom on a moving carrier, where the exploration the policy needs is the exploration the filter has to prevent.

These studies settle scale and feasibility. In each of them the actuator representation is a fixed design choice inside the stack rather than a variable, so none separates the transfer result from the representation that produced it. That separation is criterion V below.

None of these stacks varies the plant within a deployment. They are trained for robustness across one machine and its payload or soil conditions, and a plant that changes during a run is outside the construction. What remains open is the combination: a randomised

plant family, online actuator adaptation, a privileged kinematic teacher, and a matched comparison of actuator representations under one protocol.

A policy can also compensate for a poor actuator model. Chapter 7 therefore reports rollout error, command magnitude and smoothness, and cylinder-level residuals alongside task tracking. Equal task error alone is not evidence of equal model quality.

2.8 Research Gap and Thesis Positioning

The literature leaves a specific combination unresolved. Hydraulic control provides structure but requires maintained plant models; learned dynamics replace derivation with data but depend on identifiable excitation; actuator surrogates and latent adaptation are established mainly for electric legged robots; and hydraulic RL studies do not isolate actuator representation under a matched protocol. Table 2.1 summarises this position.

The five criteria are: (I) learned hydraulic actuation, (II) data audited for causal input identifiability, (III) online adaptation to plant variation at deployment, (IV) a classical-control prior transferred through policy distillation, and (V) a matched actuator comparison.

What the table does and does not assert is worth fixing before it is read. The rows are lines of work, not individual papers, and the criteria are the ones this thesis needs rather than a neutral taxonomy of the field. Every criterion is individually established somewhere, several of them for decades, and single works cover more of a row than a coarse mark can show. The latent adaptation row in particular samples a large and active literature; it is read here for the assumption those constructions share, that the factor directions an estimator retains are the directions the controller needs, and not for coverage. The claim is the combination on one machine under one protocol, and Column III shows how narrow even that claim becomes once it is measured rather than designed for.

Table 2.1: Positioning against the reviewed lines of work. Parenthesized marks denote partial or implicit coverage. Column III is parenthesized for this thesis because the deployed route conditions on a single estimated physical quantity rather than on the learned latent of [Lee 20, Kumar 21], and because its benefit is confined to the loaded regime (Section 7.5).

Line of work	I	II	III	IV	V
Model-based hydraulic control [Mattila 17, Jelali 03]	—	(✓)	(✓)	✓	—
Actuator networks for legged robots [Hwangbo 19, Rudin 22, Yamamori 26]	—	✓	—	—	—
RL for hydraulic machines [Egli 20, Egli 22, Spinelli 24, Maitan 26]	✓	(✓)	—	—	—
Latent online adaptation [Lee 20, Kumar 21, Yu 26, Liu 25, Hu 26]	—	—	✓	—	—
Residual RL and policy distillation [Johannink 19, Silver 18, Ross 11, Yin 25]	—	—	—	✓	—
This thesis (R / IF / AF)	✓	✓	(✓)	✓	✓

Column III carries a parenthesis, and the reason is specific. What AF deploys is a policy conditioned on one estimated physical quantity, the payload, recovered online from the

task observation it already carries, rather than the anonymous latent of the canonical architecture. Section 7.5 measures what that conditioning recovers, in which payload regime it helps, and how much of the gain survives when the payload is permuted across environments.

The contribution remains the complete experimental chain: identifiable cylinder data, structurally constrained models fast enough for parallel training, online adaptation and a common evaluation protocol. Adaptation is reported as a conditional mechanism, with positive and negative training seeds kept in the same analysis. The three tested stacks are:

R. A command-conditioned ROM realises the policy's desired cylinder motion. It is the reference and the only actuator calibrated to the deployment plant (Section 6.9).

IF. The same policy interface drives the frozen inverse model followed by the strict forward model, forming the robust learned baseline.

AF. The policy emits the valve command directly and the inverse model leaves the path. A causal history encoder conditions a bounded residual on that command, which the same forward model realises ahead of PhysX.

All stacks share the task, terrain, trajectories, deployment factors and evaluation gates. R and IF additionally share the action contract and form the matched pair; AF does not, because its policy chooses a valve command rather than a cylinder increment, and it is therefore compared with its own frozen base rather than ranked against the others. The ROM is calibrated to the deployment plant, so causal architectural claims rely on the matched pair and on within-stack paired ablations rather than on a three-way contrast (Section 6.9.2). Chapter 3 now grounds this comparison in the concrete machine.

The remaining gap concerns the interaction between these established techniques: whether hydraulic actuators identified for control retain command authority after transfer, and under which plant conditions a history-conditioned correction on the identified valve path adds value beyond randomisation and matched residual capacity.

3. System Fundamentals and Problem Formulation

Chapter 2 established the control and modelling concepts used in this thesis. This chapter converts that background into a fixed system definition for the remaining work. It specifies the physical mechanism, coordinate mappings, available measurements, actuator models, task geometry, and evaluation interfaces before the controllers are compared. Keeping these definitions together prevents later chapters from changing the plant, observation set, or success criteria when interpreting results.

The physical machine and its two simulation representations are distinguished explicitly. Simscape is the high-fidelity hydraulic source used for actuator identification. Here, high fidelity is relative to the reduced actuator representations used for policy training. The Simscape model is a component-level hydraulic source model; this thesis does not claim that its complete hydraulic state, particularly absolute pressure, has been validated against measurements from the physical machine. IsaacLab/PhysX is the parallel environment used for policy training, adaptation-data collection, and closed-loop evaluation. Both use the cylinder ordering, kinematics, action contract, and cutter-tip task variables defined here, but they are not assumed to be identical plants.

Sections 3.1–3.3 define the mechanism, coordinate spaces, and signals. Sections 3.4 and 3.5 describe the two simulation models. The remaining sections define the verge-following task, the R, IF, and AF control stacks, the control objective, and the evaluation gates used in the experiments.

3.1 Manipulator Platform

The MULAG FME 500 is modelled as a serial-link mechanism with five revolute joints. Four hydraulic cylinders actuate the slew, two boom, and head-pivot joints. The *Messerkopf* rotation has no independent actuator and is locked at the mowing configuration; it is therefore not a controlled degree of freedom. The cylinder order used throughout the thesis is slew (*Drehzapfen*), boom I (*Ausleger I*), boom II (*Ausleger II*), and head pivot (*Messerkopf-Schwenk*).

Table 3.1 gives the cylinder stroke ranges used in the training data and the fixed limit of the *Messerkopf* joint. Task-space errors are measured at the cutter tip, which lies 0.79 m below the *Messerkopf* body origin.

Table 3.1: Cylinder stroke ranges in the training data (Chapter 4) and the fixed *Messerkopf* joint limit used for the mowing task.

i	Joint	Function	Limit
1	*Drehzapfen*	slew	0.018–0.493 m
2	*Ausleger I*	boom I	0.275–0.791 m
3	*Ausleger II*	boom II	0.309–0.857 m
4	*Messerkopf-Schwenk*	head pivot	0.269–0.689 m
5	*Messerkopf*	head rotation (locked)	$q_{5,\min} = q_{5,\max} = q_{5,\text{lock}}$

Figure 3.1 identifies the links, joints, and four actuator channels used throughout the model.

Figure 3.2 separates the rigid bodies that form the assembled mechanism. The catalog makes the L0–L5 body assignment explicit and shows the representative barrel–rod geometry used for cylinders C1–C4.

Manufacturer CAD supplies the link geometry, joint locations, masses, centres of mass, and inertia tensors through Phobos [von Szadkowski 20]. The same CAD-derived bodies and joint frames form the mechanical basis of the Simscape and IsaacLab models, keeping dimensions, axes, masses, and cylinder lever arms consistent while the hydraulic implementations remain model-specific.

3.2 Kinematic Structure and Coordinate Mappings

3.2.1 Joint Space and Cylinder Space

The configuration is $\boldsymbol{q} \in \mathbb{R}^5$; the actuated coordinates are the four cylinder extensions $\boldsymbol{x} \in \mathbb{R}^4$. Cylinder i joins CAD attachment points $\boldsymbol{p}_A$ and $\boldsymbol{p}_B$ on adjacent links. Its length and extension are

$$\ell_i(\boldsymbol{q}) = \Big\|\boldsymbol{p}_B(\boldsymbol{q}) - \boldsymbol{p}_A(\boldsymbol{q})\Big\|, \qquad x_i = \ell_i(\boldsymbol{q}) - \ell_{0,i}, \tag{3.1}$$

with $\ell_{0,i}$ the retracted reference length. Differentiating (3.1) gives the scalar mapping between joint motion and extension rate,

$$\dot{x}_i = J_{\text{cyl},i}(\boldsymbol{q})\,\dot{\boldsymbol{q}}, \qquad J_{\text{cyl},i} = \boldsymbol{e}_i^{\top}\Big(\boldsymbol{J}_B - \boldsymbol{J}_A\Big), \tag{3.2}$$

where $\boldsymbol{e}_i$ is the cylinder-axis unit vector and $\boldsymbol{J}_A$ and $\boldsymbol{J}_B$ are attachment-point Jacobians. Virtual work gives $\boldsymbol{\tau}_{\text{act}} = \boldsymbol{J}_{\text{cyl}}^{\top}\boldsymbol{F}_h$. The mapping varies with configuration and stroke.

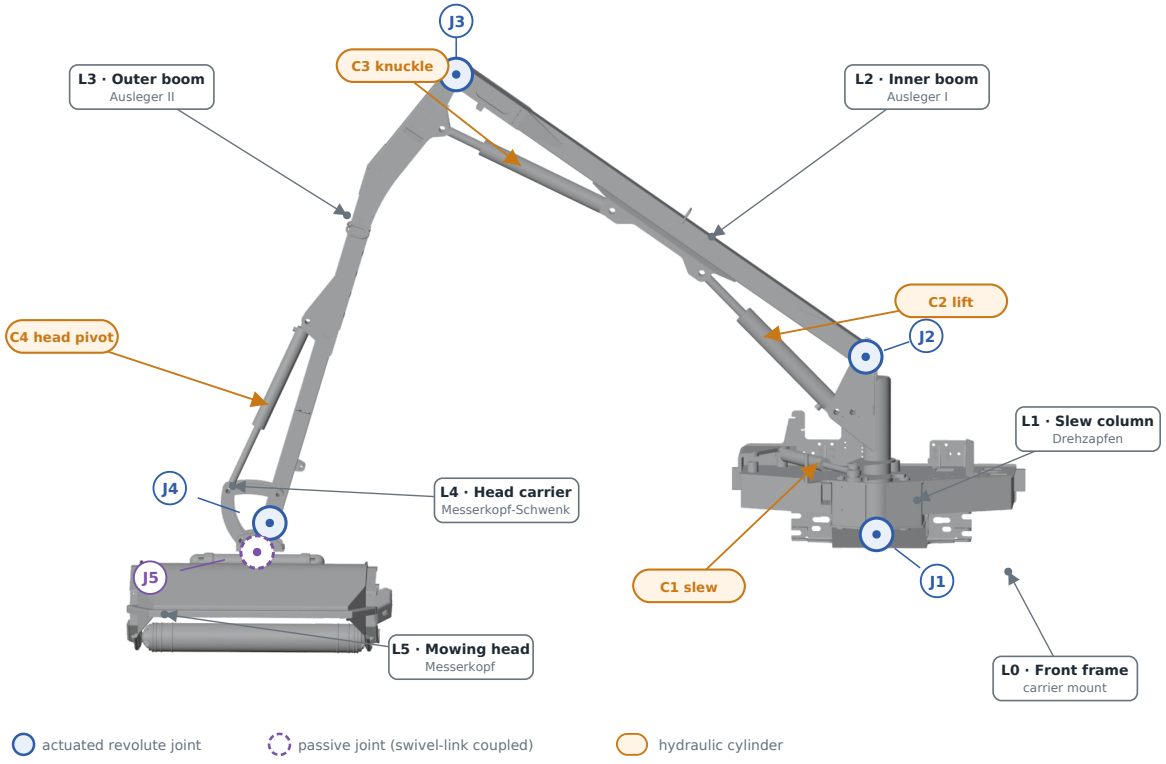

Figure 3.1: Annotated CAD view of the MULAG FME 500. Links L0–L5, joints J1–J5, and the four hydraulic cylinders C1–C4 are identified; J5 denotes the *Messerkopf* rotation, which is locked in the mowing configuration used in this thesis.

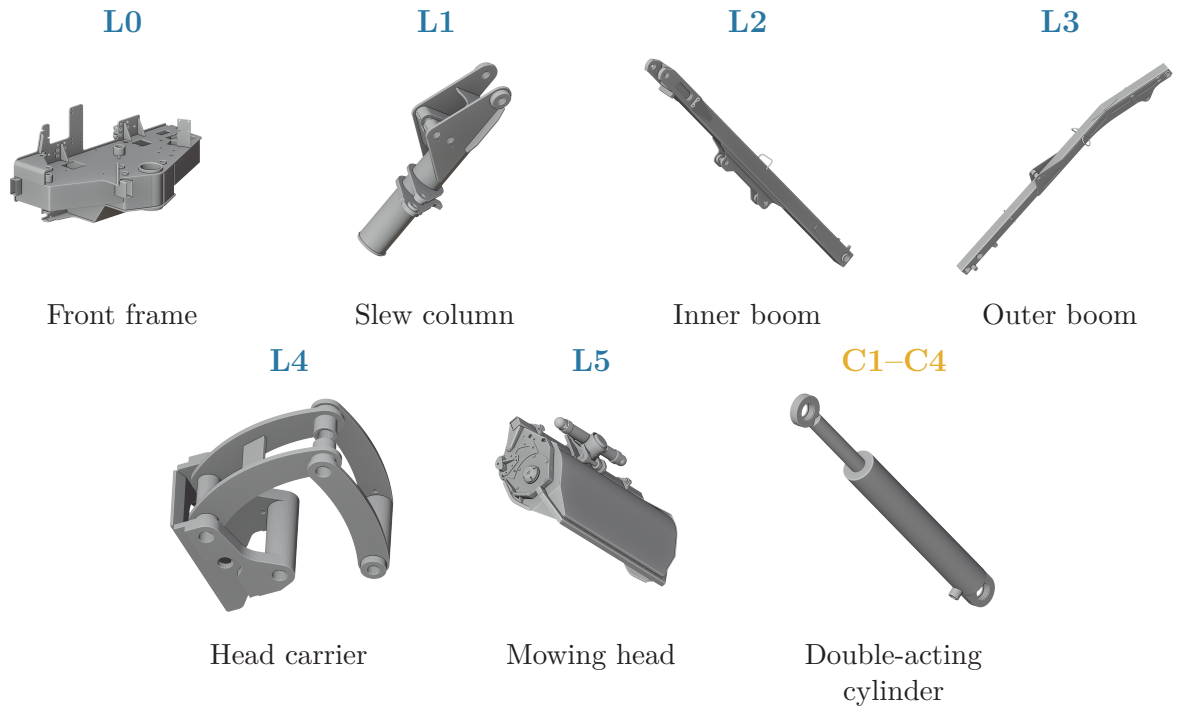

Figure 3.2: Individual CAD-derived bodies of the MULAG FME 500 shown at a common orthographic scale. L0–L5 are the front frame, slew column, inner boom, outer boom, head carrier, and mowing head. The C1–C4 panel shows the representative double-acting cylinder barrel–rod pair.

3.2.2 Task Space and the Calibrated Cutter Jacobian

The cutter tip is expressed in the truck chassis frame (`base_link`); using the moving slew-base frame would corrupt the tracking error. The controlled point includes the $-0.79\,\mathrm{m}$ tool-tip offset. Vehicle motion supplies the along-track direction.

The calibrated Jacobian $\boldsymbol{J}_c \in \mathbb{R}^{3\times4}$ maps cylinder extension rates to cutter-tip velocity. It was measured at the head-down mowing pose by stepping each cylinder target and recording the settled cutter displacement. Its singular values are 8.24, 5.69, and 2.79, giving rank 3. The damped least-squares controller in Chapter 6 uses this same Jacobian.

3.3 Actuation and Measured State

Table 3.2 lists the signals and their availability.

In IsaacLab, valve commands and differential pressures are learned-actuator states, not direct measurements. R produces them internally; IF obtains them from the inverse/forward pair, and AF from the frozen forward model alone. The physical factor vector $\mathbf{e}$ and privileged latent $\mathbf{z}_{\mathrm{SD}} = \mu(\mathbf{e})$ are unavailable to every deployed controller. The estimated latent $\mathbf{z}_{\mathrm{AAM}}$ is used only in the offline identifiability and delivery studies of Chapter 5; it is not an input to final AF. Final AF instead receives an online payload estimate regressed from the same 66 observable channels it already reads.

Table 3.2: Signal inventory of the modelled system. "S" marks signals present in the Simscape data exports, "I" signals available live in the IsaacLab environment, "M" quantities that exist there only through the learned models of Chapter 5, and "P" *privileged* quantities that exist in simulation but must never enter a deployable policy or estimator input.

Signal	Dimension	Availability	Meaning
$\boldsymbol{u}$	$[-1,1]^4$	S, M	normalised proportional-valve commands
$\boldsymbol{x}$, $\dot{\boldsymbol{x}}$	$\mathbb{R}^4$	S, I	cylinder extensions and velocities
$\Delta\boldsymbol{p}$	$\mathbb{R}^4$	S, M	chamber pressure differences
$\boldsymbol{F}$	$\mathbb{R}^4$	S	logged rod forces
$\boldsymbol{q}$, $\dot{\boldsymbol{q}}$	$\mathbb{R}^5$	S, I	joint positions and velocities
$\boldsymbol{p}_c$	$\mathbb{R}^3$	S, I	cutter-tip position, truck frame
$\boldsymbol{h}$	$\mathbb{R}^9$	I	forward terrain-height profile ahead of the cutter
$\boldsymbol{c} = [\Delta\boldsymbol{x}^*, \Delta\dot{\boldsymbol{x}}^*]$	$\mathbb{R}^8$	I	desired cylinder command
$\mathbf{e}$	$\mathbb{R}^6$	P	physical randomisation factors of the active plant
$\mathbf{z}_{\mathrm{SD}} = \mu(\mathbf{e})$	$\mathbb{R}^{d_z}$	P	privileged dynamics latent, $d_z = 2$
$\mathbf{z}_{\mathrm{AAM}}$	$\mathbb{R}^{d_z}$	M	offline diagnostic estimate of $\mathbf{z}_{\mathrm{SD}}$; not deployed in AF

3.3.1 Sensing and Observability

The deployable observation contains articulation state, terrain rays, contact signals, the desired cylinder command, and actuator-model pressure state. Plant parameters are not observed; AF infers their effect from causal history.

Figure 3.3 separates deployable measurements from privileged plant information. Articulation state is read from the physics engine without encoder noise or quantisation.

3.4 High-Fidelity Simscape Model

The actuator models use a multi-domain Simscape model of the FME 500 developed in the preceding master project. Table 3.3 distinguishes the data source of each frozen component.

Simscape supplies the frozen ID and strict FD checkpoints. The ROM receives an additional bounded IsaacLab calibration (Section 5.4.1). The adaptation models are trained from IsaacLab data collected through the ROM and PhysX.

3.4.1 The Command ROM as Deployment Actuator

The command ROM plus PhysX is the deployment plant. Simscape is too slow for policy training and bare PhysX has no hydraulics. The ROM is therefore both the plant model and the actuator of R:

- R uses the deployment plant itself and is the reference condition.
- IF replaces it with an independently identified inverse/forward pair.
- AF drops the inverse model, has the policy emit the valve command directly, and adds online adaptation on the forward model.

The R–IF gap measures the transfer cost of the independently identified pair; it is not a general ranking of the two model classes.

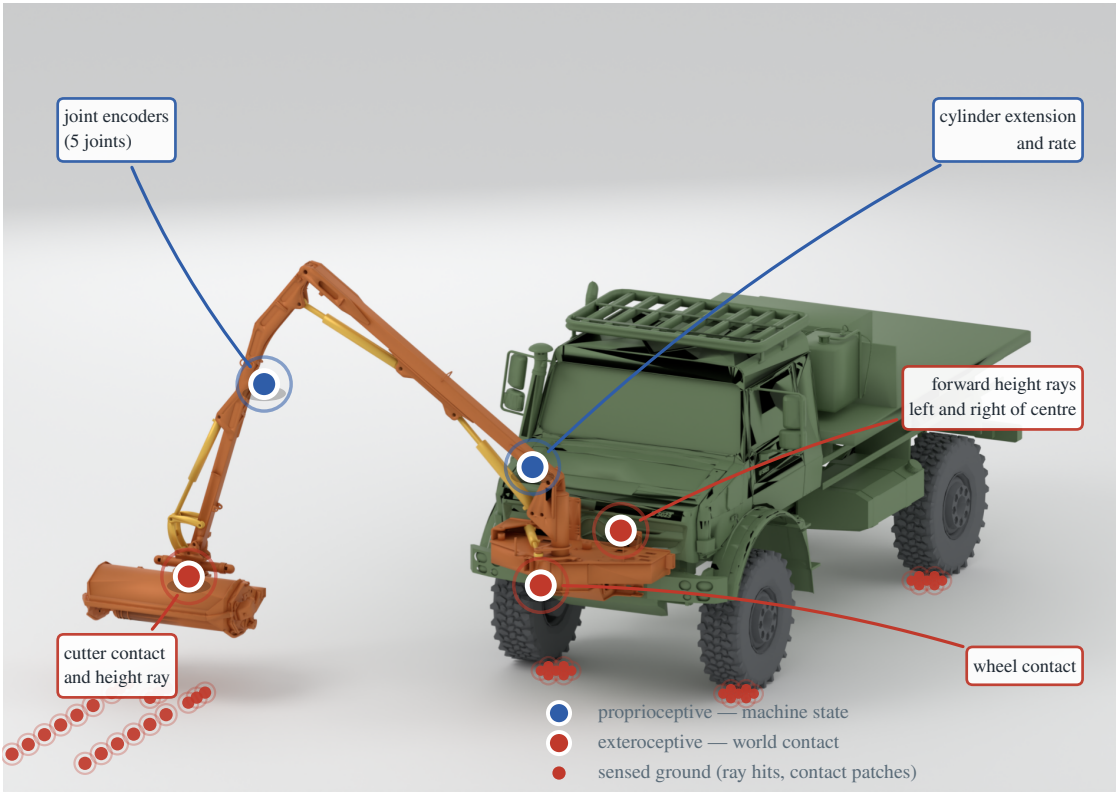

Figure 3.3: Sensor and observability map for the deployed controller. Blue callouts identify proprioceptive inputs: joint encoders and cylinder extension and rate. Red callouts identify exteroceptive inputs: forward terrain-height rays and cutter and wheel contact. The small red ground markers show where those exteroceptive inputs are taken: the tyre contact patches, and the strip and forward ray rows the cutter senses ahead of itself. They are placed on a ground frame fitted to the three visible tyre contact points, so their spacing carries the perspective of the render.

Table 3.3: Provenance of each frozen actuator component. The command ROM is the only one that has been calibrated against the deployment plant; the inverse/forward pair is Simscape-identified throughout. This asymmetry is a property of the comparison and is carried into the discussion of Chapter 7, not glossed over.

Component	Identified on	Calibrated on
Command ROM	Simscape closed-loop export (grey-box, full model)	IsaacLab/PhysX closed-loop rollouts, via a bounded command-gated correction head only
Control ID	Simscape closed-loop export	—
Strict FD	Simscape closed-loop export, retrained on the strict open-loop excitation export	—
SD_0, $\mu(\mathbf{e})$, SD_z, AAM	IsaacLab/PhysX adaptation collection (from scratch)	—

3.4.2 Mechanical Subsystem

The mechanism uses the CAD-derived links, joints, and cylinder attachment frames of Section 3.1. It is simulated in Simscape Multibody [The MathWorks, Inc. 24b]. The joint-space dynamics take the standard manipulator form [Siciliano 09]

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) + \boldsymbol{\tau}_{\text{fric}}(\dot{\boldsymbol{q}}) = \boldsymbol{\tau}_{\text{act}} - \boldsymbol{J}^{\top}(\boldsymbol{q})\,\boldsymbol{F}_{\text{ext}}, \tag{3.3}$$

with inertia $\boldsymbol{M}$ from the imported CAD. Friction combines viscous, Coulomb, and Stribeck terms. The cutting load scales with feed velocity and vegetation density. Each cylinder uses a spherical–prismatic–cylindrical loop and hard stops enforce the limits in Table 3.1.

3.4.3 Hydraulic Subsystem

Figure 3.4 shows four 4/3 valve-driven asymmetric cylinders fed by a variable-speed, fixed-displacement pump ($Q_p = D_p\omega$), accumulator, and pressure-relief valve.

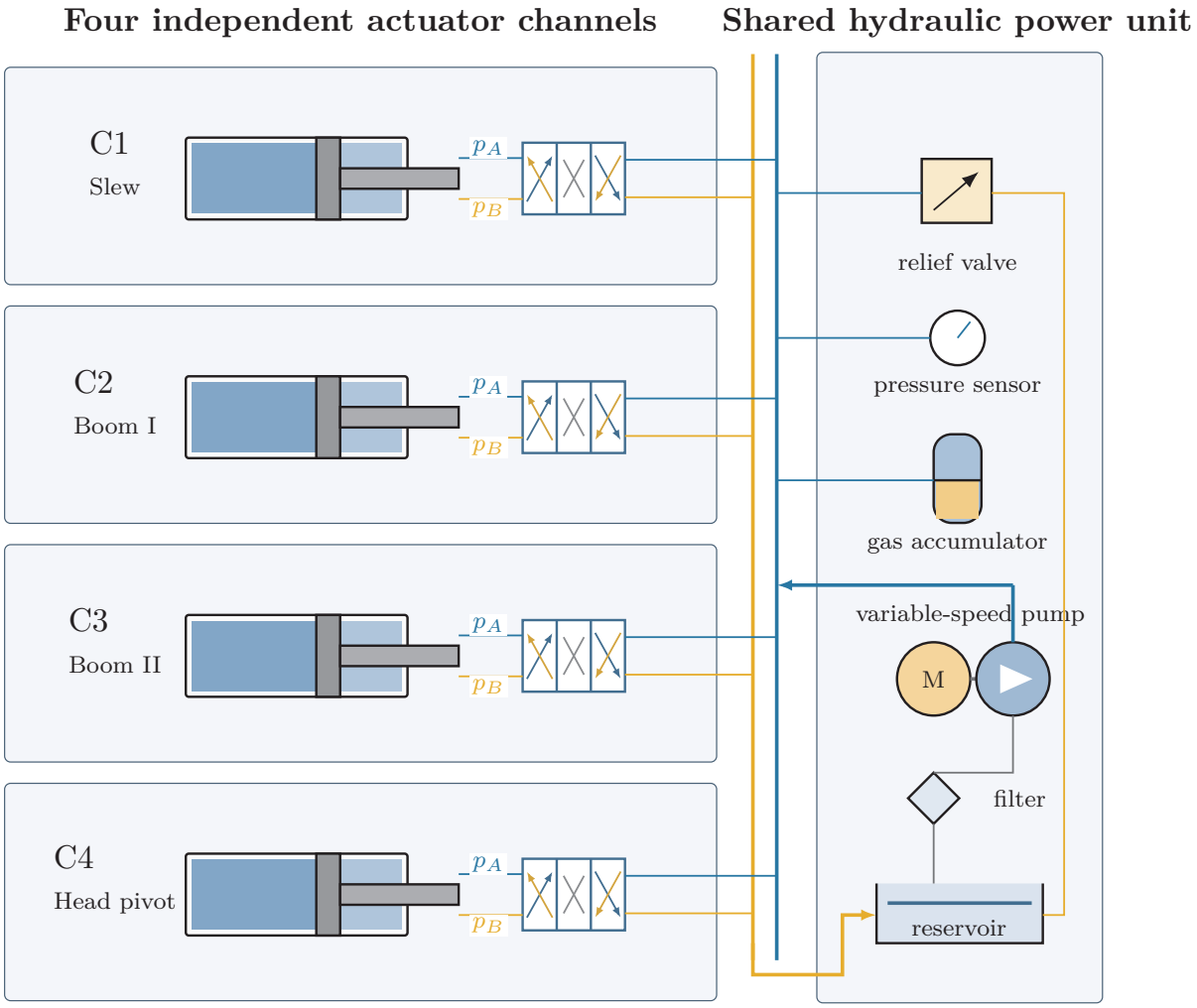

Figure 3.4: Hydraulic system schematic. Four proportional-valve cylinder channels share a variable-speed pump, reservoir, accumulator, sensor, and pressure-relief valve. Blue and orange rails denote supply and return.

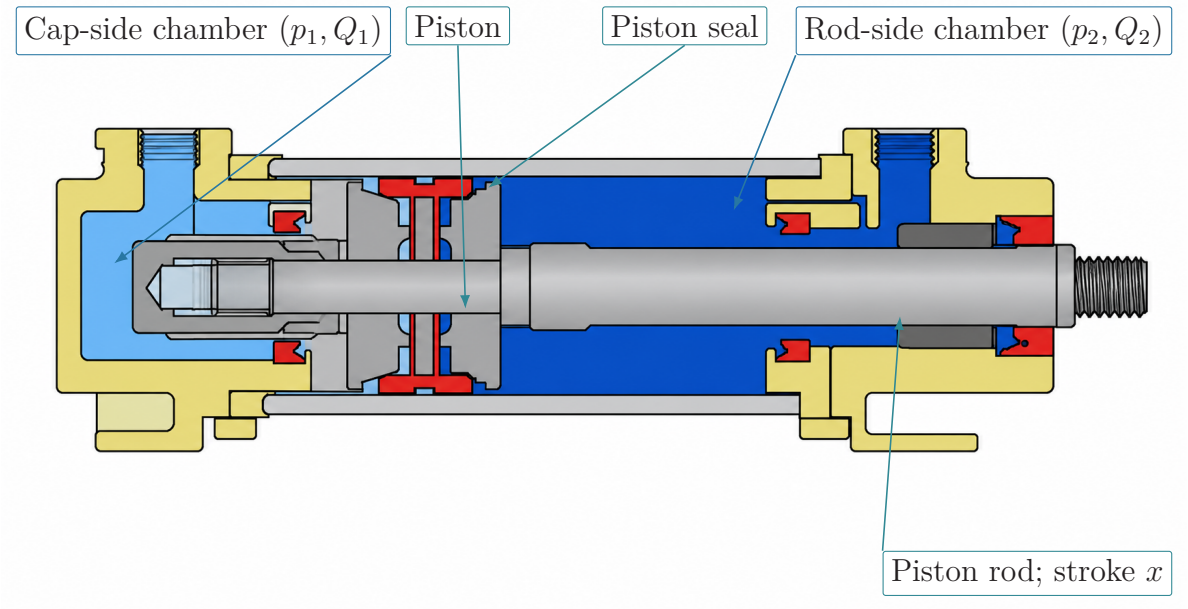

Figure 3.5: Annotated cutaway of an asymmetric double-acting hydraulic cylinder. The cap-side and rod-side chamber pressures generate the piston force and rod stroke. The labels are the states of (3.4), and the four cylinder channels of Figure 3.4 are of this type.

Figure 3.5 identifies the cylinder chambers, piston, seal, and rod used in the actuator model.

The 4/3 valve in Figure 3.4 meters flow to the two chambers of each asymmetric double-acting cylinder, cap side (p_1, Q_1) and rod side (p_2, Q_2). The chamber pressures follow the standard continuity relations [Merritt 67, Jelali 03]

$$\dot{p}_1 = \frac{E_{\text{eff}}}{V_1(x)}\big(Q_1 - A_1\dot{x} - C_{\text{leak}}(p_1 - p_2)\big), \qquad \dot{p}_2 = \frac{E_{\text{eff}}}{V_2(x)}\big(-Q_2 + A_2\dot{x} + C_{\text{leak}}(p_1 - p_2)\big), \tag{3.4}$$

where $V_{1,2}(x)$ include dead volumes, $A_1 \neq A_2$, and C_{leak} is internal leakage. Entrained air is represented by the usual effective bulk modulus [Jelali 03]

$$E_{\text{eff}}(p) = \left(\frac{1}{E_f} + \frac{\alpha}{p}\right)^{-1}, \tag{3.5}$$

Valve flow follows the orifice law of Section 2.1.2; spool dynamics are $\dot{x}_v = (u - x_v)/\tau_v$. The piston force is $F_h = p_1 A_1 - p_2 A_2 - F_{\text{fric}}(\dot{x})$. The coupled model is stiff and uses a variable-step implicit solver, which prevents RL-time simulation.

3.4.4 Validated Scope of the Model

Two checks fix what this model may be used for. The first is a parameter sensitivity study carried out in the preceding master project. Under a fixed integral-backstepping controller at 0.01 s, sweeping $\beta \in \{0.5, 1.0, 1.5, 2.0\}$ GPa moves slew-axis delay from about 0.85 s to 0.65 s, the system delay index from 0.45 s to 0.35 s, and task-space tracking error by about 11 %, with the improvement saturating beyond 1.5 GPa. The model responds to

bulk modulus in the direction and with the saturation that the compressibility term of (3.5) predicts.

The second check asks whether the exported behaviour is what the configured plant should produce. It runs on the strict open-loop excitation export that identifies the forward model (Chapter 4): four runs, 25 600 rows on a 100 Hz grid, 93.7 % of rows valid. The reference is the collector's own declared parameter set, so the check tests the assembled model rather than the parameter file. Table 3.4 reports it.

Table 3.4: Behavioural checks on the strict open-loop excitation export against the parameters the collector configures. Values are pooled over the four runs; ranges span the four cylinders. The last two rows are the limits of the validated scope, not passes.

Property	Measured	Expected
Motion onset after command onset	within two log samples, $\leq$ 20 ms	the nominal 8 ms spool constant plus transport
Rise to 63 % of peak velocity	30 ms–80 ms	spool constant plus chamber fill
Velocity sign follows command	100 % above 0.05 m/s, $\geq$ 95.8 % above 0.01 m/s	no inversion once the cylinder moves
Speed against rising load at $\lvert u\rvert > 0.5$	falls by two orders of magnitude across $\lvert\Delta p\rvert$ terciles	orifice law: flow follows the pressure margin
Stroke visited	13 %–34 % of the configured stroke	bounded by the assembled linkage, not by the valve
$\lvert\Delta p\rvert$ ceiling	116 MPa–122 MPa, repeatable per channel	relief valve set at 27.6 MPa

Timing, direction and load dependence hold. Motion starts within two log samples of the command, velocity reaches most of its plateau inside 80 ms, the velocity sign never inverts against the command once the cylinder is actually moving, and speed collapses as the pressure the cylinder works against rises. That last property is the orifice law acting through the pressure margin, and it is the reason a fixed command gain does not describe this plant. These are the behaviours the learned actuator models have to reproduce, and the export has them.

Two properties are outside the validated scope. Stroke coverage is narrow: the excitation visits 13 % to 34 % of the configured cylinder stroke because the assembled linkage cannot reach the rest, and the head pivot is the narrowest channel. Chapter 4 carries that forward, and it is the channel the strict forward model predicts least well (Section 5.3).

Absolute pressure is not validated at all. Differential pressure does clip at a ceiling that repeats per channel, which is the signature of a relief valve, but the ceiling is 116 MPa to 122 MPa against a configured setting of 27.6 MPa, and about half of the valid rows sit above twice that setting. The pressure signal has the right shape and the wrong level, and the level is inherited from this source rather than introduced by the learned models. No result in this thesis is stated in absolute pressure units: every learned model consumes Δp through a fitted normaliser, and Appendix B.5 reads the deployed pressure state for shape only, for this reason.

A normaliser keeps the wrong level away from most of the results, but not all of them, and the exception matters. Each model standardises the pressure channel as $(\Delta p - \mu)/\sigma$ with μ and σ fitted on this same export; the inverse model's four pressure channels carry σ between 50 MPa and 70 MPa, and the forward model, the reduced-order model and the offline AAM each store their own. Wherever pressure enters only through that standardisation, scaling every pressure by a constant and scaling μ and σ with it leaves the input unchanged, so the trained network and the numbers it produces are unchanged. That covers the inverse model, the strict forward model and the AAM.

It does not cover the reduced-order model's explicit chain. There pressure also enters unstandardised, under a square root and against a fitted supply pressure, in Equation (5.7), and that expression is neither linear nor homogeneous in the pressure scale. With Δp reaching 116 MPa to 122 MPa against supply pressures the fit drives onto clamps at 1 MPa on two cylinders and 50 MPa on two, the radicand is degenerate over most of the operating range. That is a candidate cause of the finding in Section 5.4 that the explicit chain produces no net travel and its parameters settle on their clamps, and the two should be read together rather than separately. Correcting the level in the export would therefore change that model's fitted constants, though it would not by itself repair the channel: pressure-increment R^2 is 0.06 to 0.09, so the state follows the level poorly in any units. Elsewhere the wrong level costs interpretation rather than correctness, and that cost falls where the pressure channel is read as physical, which is the seal-condition decoding of Section 5.5.1.

Beyond these checks the model represents one machine and omits thermal dynamics. The logged rod-force channel is not carried into the learned actuator models: force is generated internally by the PhysX plant at deployment, and the logged channel also fails the piston-area consistency audit (Section 5.4). Chapter 4 introduces variation through parameter randomisation.

3.5 IsaacLab Environment

Policy training, adaptation-data collection, and evaluation run in IsaacLab on PhysX. IsaacLab is the successor to Orbit, and the Orbit paper [Mittal 23] remains its citable description; the underlying GPU-parallel simulator is Isaac Gym [Makoviychuk 21]. The USD articulation contains the Unimog carrier, five arm links, and four prismatic cylinder joints. Learned actuator models supply the hydraulic behaviour between the controller and PhysX cylinder targets.

3.5.1 Simulation and Control Rates

Four clocks run concurrently. PhysX integrates at 120 Hz and the policy acts at 60 Hz under a decimation of two. The learned actuator models and the AAM update every three policy steps, on the 20 Hz grid at which the identification data were sampled, and their prediction is held across the intervening PhysX updates. Data collection uses that same 50 ms grid. "The control frequency" is therefore not a well-defined quantity for this system and no scalar rate is reported for it. Table B.1 summarises these rates and distinguishes the offline AAM timing from the deployed AF history path.

3.5.2 Environment Configuration Details

The arm root is spawned at $z = 1.26\,\mathrm{m}$; ground-height spawning buries the vehicle. Cylinder targets are integrated in reference space so gravity does not erase commanded increments. Vehicle motion is scripted because the authored tyres and suspension do not provide reliable PhysX traction; a planar fixture holds lateral position and heading.

3.6 The Verge-Following Task

3.6.1 Task Definition

The task is moving verge following. The truck drives a 26 m strip at a speed sampled from 2 to 7 km/h; development uses 4 km/h. Episodes allow 2808 steps (46.8 s), while scoring stops when the strip is completed. The arm must keep the cutter at the target height, on the verge centre line, in contact with the ground, and within its travel limits. Terrain height is randomised per environment (Figure 3.6).

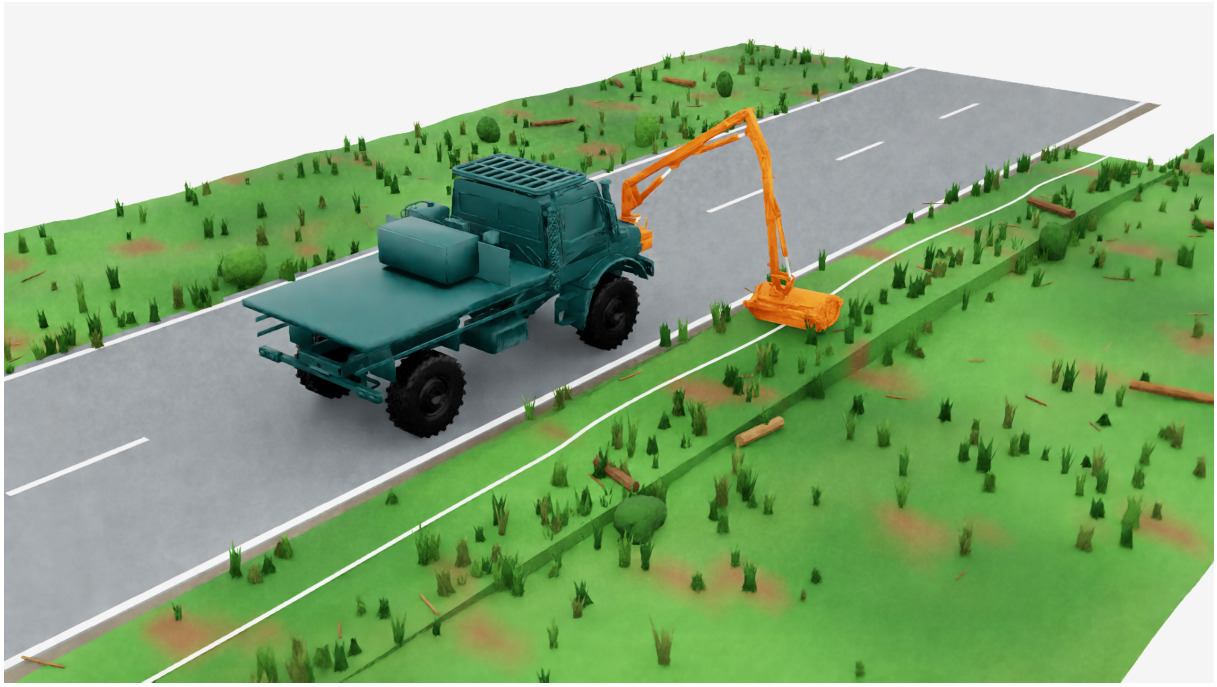

Figure 3.6: IsaacLab realisation of the moving verge-following task. The scripted truck travels on the level road while the four-cylinder arm places the cutter above the procedurally generated verge and follows the white reference path. Randomised grass, bushes, soil patches, twigs, and logs form a non-colliding visual layer; the underlying verge mesh supplies the terrain disturbance and height-sensing geometry used by the controller.

Eleven reference-path families span straight, periodic, spline, sweep, and lane-change motion. Results use the worst path family and terrain mesh rather than a pooled score. A regression test verifies that all eleven families are distinct.

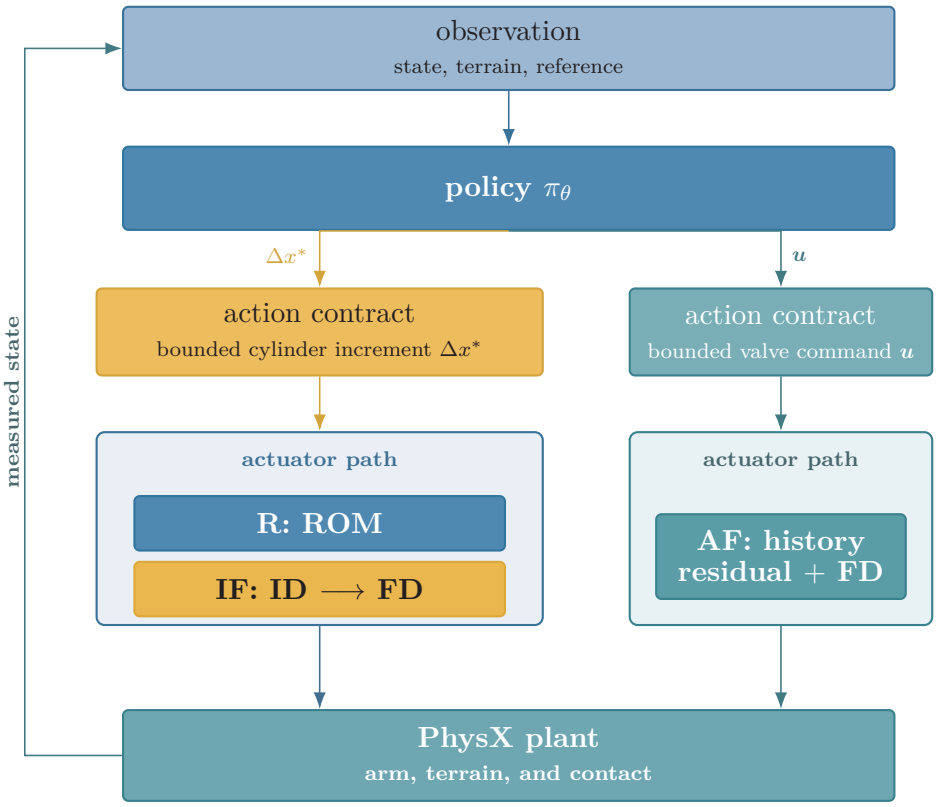

Figure 3.7: Deployed control stacks. R and IF share one action contract, a bounded cylinder increment, and differ only in how they realise it. AF does not share it: the policy emits the valve command itself and a bounded residual conditioned on causal history acts on that command before the frozen forward model. All three set the same PhysX cylinder target, and measured state closes the loop. The differing contract is why AF is compared within itself rather than against the other two (Section 7.1.2).

3.6.2 Task Feasibility and Plant Capability

The action contract limits cylinders 1–3 to 4 mm per policy step, which bounds lateral cutter speed at

$$v_{\text{lat}}^{\max} = \frac{4\,\text{mm}}{1/60\,\text{s}} = 0.240\,\text{m/s}. \tag{3.6}$$

This uses the 1/60 s policy interval. Early path settings exceeded this limit; the worst case (`boom_sweep`) required 3.7× the available rate. Path and terrain parameters are now derived from (3.6), and a preflight check rejects infeasible tasks.

Equation (3.6) is a property of the cylinder-increment contract, so it describes the action chains of R and IF directly. AF bounds its command in valve space instead, and the same arithmetic does not apply to it. The paths and terrain derived from the ceiling are nonetheless shared by all three stacks, so what the task *demands* is identical in every case; what differs is only the space in which the request is bounded.

3.7 Control Architecture

The vehicle advances at the scripted speed and the reference generator supplies the verge centre line and cut height. The policy maps normalised observations to four bounded

actions and has no runtime baseline controller. The DLS controller is used only for teaching and identification.

3.8 Problem Formulation

The reference is the verge centre line $y_{\mathrm{ref}}(t)$ and cut height h_{ref}, expressed in the truck frame. With cutter-tip position $\boldsymbol{p}_c(t)$ and terrain height $g(\cdot)$, the scored errors are

$$e_{\mathrm{lat}}(t) = p_{c,y}(t) - y_{\mathrm{ref}}(t), \qquad e_{\mathrm{cut}}(t) = \Big(p_{c,z}(t) - g(\boldsymbol{p}_c(t))\Big) - h_{\mathrm{ref}}. \tag{3.7}$$

The objective is to minimize the mean and tail of both errors across path families and terrain while maintaining contact, head alignment, and smooth motion. Learned models are evaluated on one-step and recursive cylinder predictions. The stacks are tested across the randomised plant family. R and IF are compared with each other on held-out factors, and AF is compared with its own frozen base under a plant change, for the reasons given in Section 6.9.2.

Constraints are cylinder limits and headroom (Table 3.1), the action contract, checkpoint output limits, and the head-down condition.

3.9 System-Level Evaluation Criteria

Controllers and models are evaluated separately. Controller scores can hide actuator-model errors through policy compensation.

The system-level criteria cover three aspects of operation. Cut-height and lateral errors measure task accuracy. Contact, head alignment, impact rate, and cutter acceleration describe tool–terrain interaction and motion quality. Cylinder headroom and safety terminations check whether the task is completed without driving the mechanism into its limits. The metrics are evaluated over the scored part of each episode and are kept in physical units where possible. This makes a failed gate traceable to a specific limitation rather than to a single combined reward value.

3.9.1 The Eight System-Level Evaluation Gates

Table 3.5 defines the eight physical gates used throughout this thesis. Their role is to supply a criterion that is *not* the reward: a policy is optimised against the scalar return of Section 6.2.5 but selected and promoted against this vector, and Section 6.3.2 shows the two point in measurably different directions. They rank and promote candidates; they do not certify a stack as accepted, and no such claim is made anywhere here.

All stacks are evaluated under the same path, terrain and randomisation conditions. A candidate is preferred when it satisfies more gates at a wider margin, and strong mean tracking does not compensate for loss of contact, inadequate headroom, excessive impacts or safety terminations. Worst-case aggregation stops easy terrain or path cases from masking a local failure, the percentiles catch short alignment, clearance and acceleration events a mean hides, and contact and termination measures are fractions so episodes of different completed length stay comparable.

Table 3.5: Eight system-level evaluation gates. Each gate uses at least three terrain meshes and the worst mesh; tracking-critical gates also use the worst path family. The last column states where each number comes from, because a threshold read off an earlier baseline does not carry the same weight as one fixed by the task or by the machine.

Gate	Metric	Threshold	Aggregation	Origin
Cut height	mean $\lvert e_{\text{cut}} \rvert$	$\leq 0.040\,\text{m}$	worst mesh	mowing spec
Lateral tracking	mean $\lvert e_{\text{lat}} \rvert$	$\leq 0.300\,\text{m}$	worst mesh, worst family	earlier baseline
Ground contact	contact fraction	≥ 0.700	worst mesh, worst family	mowing spec
Head alignment	p01 head-down alignment	≥ 0.900	worst mesh	tool geometry
Hard impacts	hard-impact step fraction	≤ 0.350	worst mesh	wear tolerance
Tool acceleration	p95 cutter acceleration	$\leq 120\,\text{m/s}^2$	worst mesh	measured floor $\times 1.2$
Cylinder headroom	p01 travel margin	$\geq 0.005\,\text{m}$	worst mesh	machine limit
Safety terminations	termination rate	≤ 0.080	worst mesh	engineering tolerance

The origins are not equivalent, and the difference decides how much each gate can be asked to do. Cut height and ground contact come from what verge mowing requires of the tool. Head alignment follows from the cutting head's geometry, which must stay down to cut at all. Cylinder headroom is a machine limit: the arm must not drive into its own travel stops. Tool acceleration and safety terminations are engineering tolerances, the first set at roughly 1.2 times the acceleration floor measured on a working stack, the second at the rate of raw envelope violations judged acceptable per control step.

The lateral gate is the exception and is treated as one throughout. Its 0.300 m was anchored on an earlier and narrower trajectory library, and once that library was widened the number stopped describing the task and started describing whichever family the widening made hardest. It still ranks candidates usefully, because every candidate meets the same number, but it is not a task requirement and no claim in this thesis rests on passing it. Section 7.2 shows both stacks failing it on a single family each, and a different family in each case, which is what a stale threshold looks like.

3.9.2 Cylinder and Model Checks

Cylinder checks use fresh ± 10 mm commands, coordinated 80 mm motions, return error, and saturation. Model checks use one-step and rollout accuracy, input ablation, and Jacobian conditioning (Chapter 5). Chapter 7 reports these with the evaluation contract. The small command checks local response and deadband; the coordinated motion checks coupling between actuator channels. Return error tests reversibility, and saturation checks whether the learned model respects the command and stroke limits. These diagnostics are interpreted alongside the system gates, not as substitutes for closed-loop task performance.

The system is now specified. The next chapter turns to the data the learned models are built from.

4. Data Generation and Preparation

Chapter 3 defined two plant representations: detailed hydraulics in Simscape and fast rigid-body dynamics in IsaacLab. This chapter documents the datasets that connect them, including collection, auditing, feature construction and splitting. Table 3.3 records their provenance.

Simscape campaigns. Closed-loop runs provide realistic operating data for the inverse and reduced-order models. Separate open-loop valve excitation provides the causal command information needed by the strict forward model (Section 4.4).

ROM calibration. After Simscape training, the command ROM was fine-tuned on five closed-loop IsaacLab reference shapes. Only its bounded correction head remained trainable, with Simscape replay retained (Section 4.6). The ID–FD pair was not calibrated in IsaacLab.

IsaacLab deployment data. The system-dynamics surrogate, the privileged encoder and the adaptation module use two 118-run collections recorded in the deployed command ROM–PhysX plant, including dedicated PRBS windows (Section 4.7). No earlier adaptation or system-dynamics model was reused. Figure 4.1 summarises the common preparation workflow.

4.1 Dataset Objectives and Design Requirements

Four requirements governed collection and preparation.

R1: realistic operating distribution. The learned models are deployed inside a tracking controller, so they must be accurate in the states that controller visits, namely coordinated multi-axis motion, gravity-loaded poses and moderate speeds.

R2: causal identifiability of the valve. A forward model is only useful to a policy if it responds to the command. This requires that the valve signal in the training data carry information that the state history does not already carry, which is a property of the *experiment*, not of the network [Ljung 99, Isermann 11].

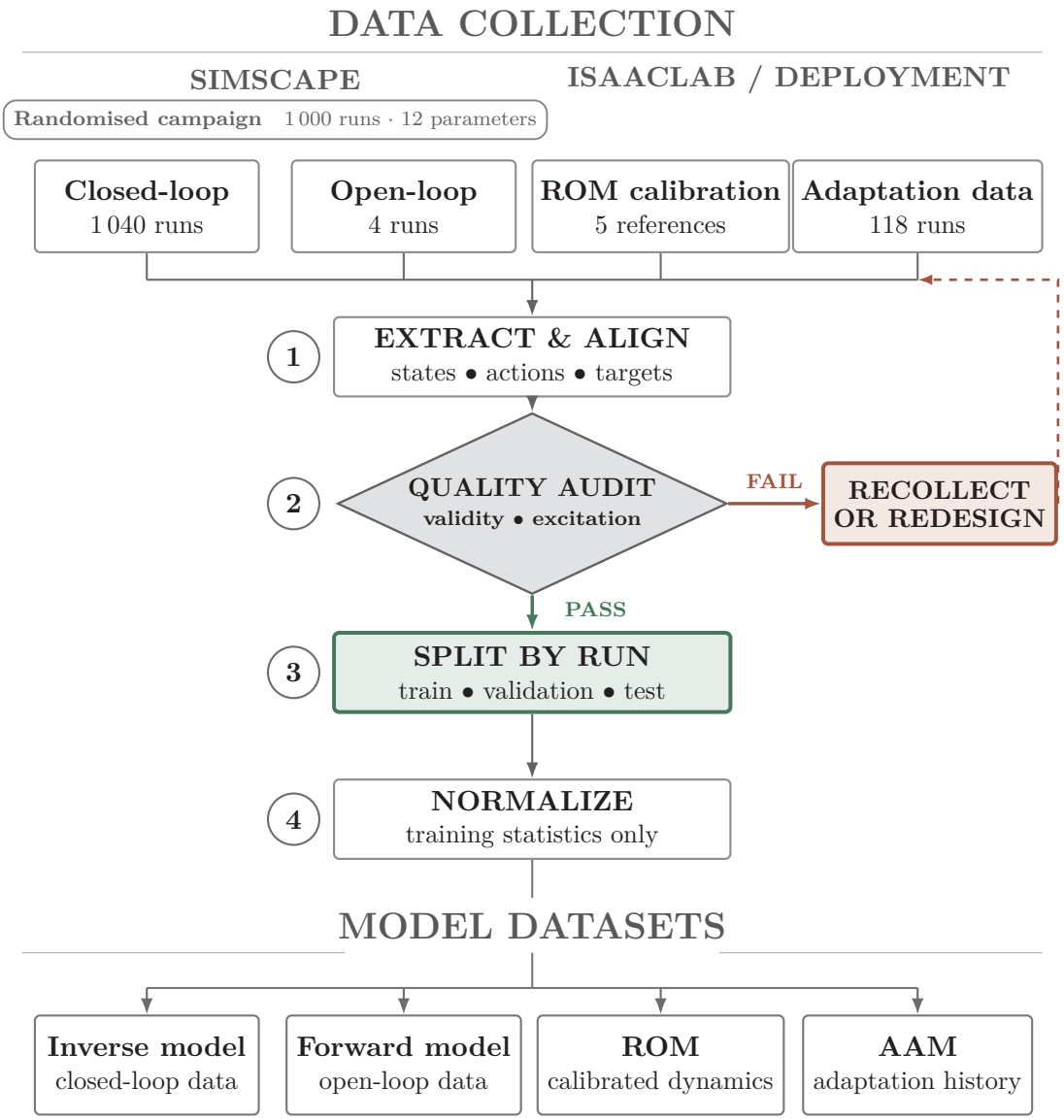

Figure 4.1: Data-preparation workflow. Two Simscape collections and two IsaacLab collections pass through common extraction, audit, run-level splitting and training-statistics normalisation stages. The randomised Simscape campaign is provenance metadata rather than a pipeline stage. A failed audit returns the experiment to collection; accepted material forms the inverse, forward, ROM and AAM datasets.

R3: variation across a plant family. Robust training and online adaptation both need many plants rather than one. The variation must be physically meaningful, must demonstrably change the measured response, and must be recorded per run, because the privileged plant context of Section 4.11 is computed from those factors.

R4: clean separation. Train, validation and test material must be separable at the level of complete runs, so that no transition window straddles a split boundary and no run contributes to two splits.

R1 and R2 require different experiments. Closed-loop control gives realistic motion but correlates command with state; open-loop excitation breaks that correlation but is less representative of deployment. The two campaigns are therefore retained and combined as described in Section 4.5.

4.2 Randomised Parameter Space of the Simscape Campaign

This section records the Simscape randomisation design as collection provenance. No reported model is trained on this curriculum. The export the actuator models are actually fitted on is a separate closed-loop collection on the nominal plant (Section 4.5). The curriculum is kept because its compliance analysis informed the IsaacLab factor design of Section 4.7.2.

Plants are drawn from a twelve-dimensional space by Latin hypercube sampling [McKay 79], fixed once before any simulation runs, so the parameter table can be audited independently of the data it produced. The batch is a curriculum rather than a homogeneous pool: run 1 is the nominal plant, runs 2–251 randomise only the hydraulics, runs 252–501 only the mechanics, and runs 502–1000 both (Figure 4.2(a)), so a modelling failure can be attributed to one axis by retraining on one block. Appendix A lists the axes and Appendix A.2 the generation procedure.

Two design choices carry into the deployment factors. Seal wear is one sampled number driving three parameters together, so a worn cylinder becomes leakier and slower at once: it scales piston clearance linearly, valve leakage area with the cube of the same factor, and the spool time constant from 8 ms to about 13 ms at the top of the range. Sampling the three independently would have produced machines that are very leaky and yet perfectly fast. The entrained-air and dead-volume coordinates randomise chamber dead volume over 505 to 579 mL, which reaches the pressure dynamics through the chamber volume in Equation (3.4), so compressibility enters the batch as capacitance variation. The mechanical axes perturb all fourteen rigid bodies: mass by up to $\pm 20\,\%$, centre of mass by up to $\pm 5\,\%$ of its nominal offset, and the inertia tensor consistently with the mass change, so each body remains physically realisable. Joint stiffness and damping model compliance in the bearings and linkage rather than in the fluid, and set how much of a pressure transient appears as boom oscillation instead of clean motion.

4.3 Closed-Loop Data Collection

Algorithm A.1 in Appendix A.2 gives the procedure that produced the curriculum above. The parameter table is fixed before simulation and each run receives a new reference path.

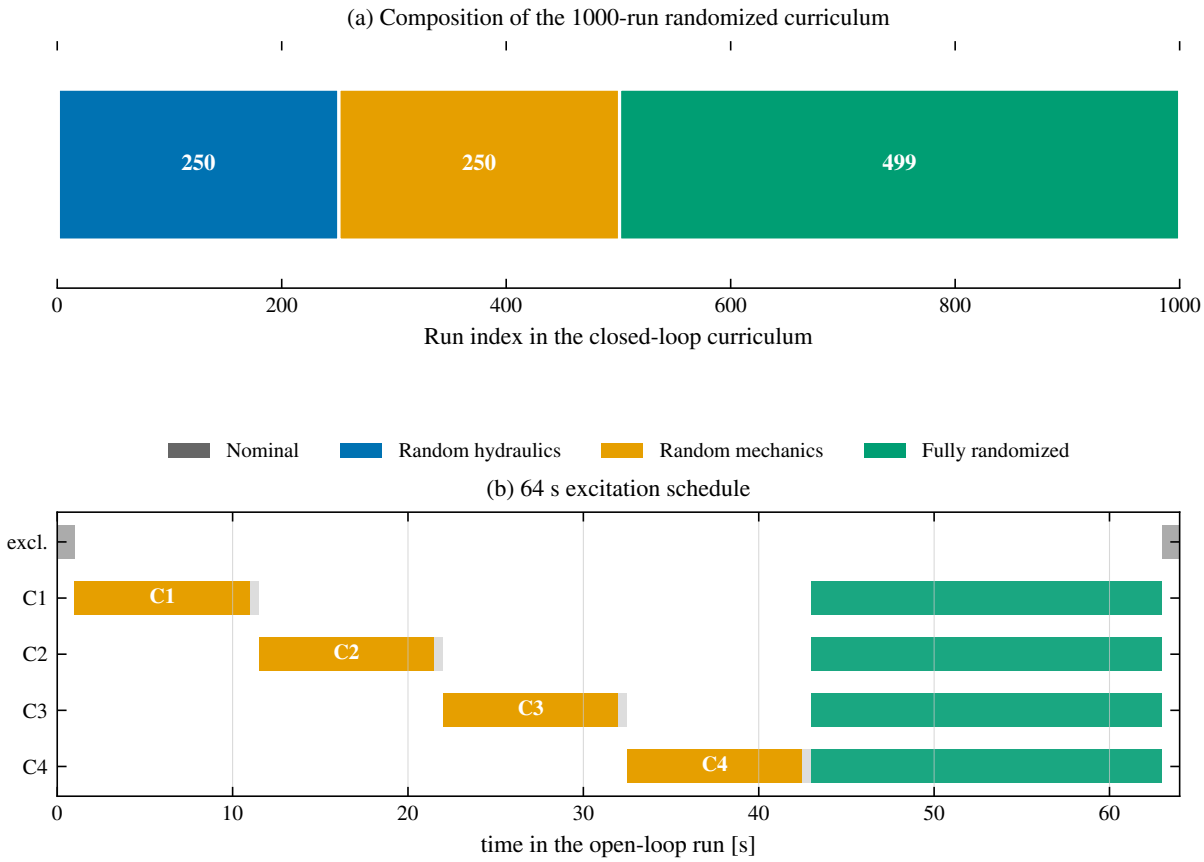

Figure 4.2: Design of the two collection campaigns. (a) Composition of the closed-loop curriculum: one nominal run, then 250 hydraulics-only, 250 mechanics-only and 499 fully randomised draws. (b) Timeline of one open-loop excitation run.

4.3.1 The Controller as a Data-Collection Instrument

Every closed-loop run uses the same integral-backstepping controller at 0.005 s. Its gains remain fixed, so response differences are caused by the plant draw rather than controller retuning.

Joint 2 uses PI control with velocity damping and gravity bias because it carries the boom load. A barrier term enforces the URDF limits. Per-axis command limits are [0.95, 0.95, 0.80, 0.80, 0.90]; the first four are also stored in the inverse-dynamics checkpoint (Section 5.2).

4.4 Open-Loop Excitation Experiments

4.4.1 Limitations of the Closed-Loop Batch

The first forward model achieved one-step $R^2 \approx 0.999$ on closed-loop data but was almost insensitive to valve input: zeroing the input changed long-horizon error by only a few percent (Section 5.3.1). In closed loop, command is largely determined by the state history already given to the model, so the data do not separate command causality from state correlation [Ljung 99].

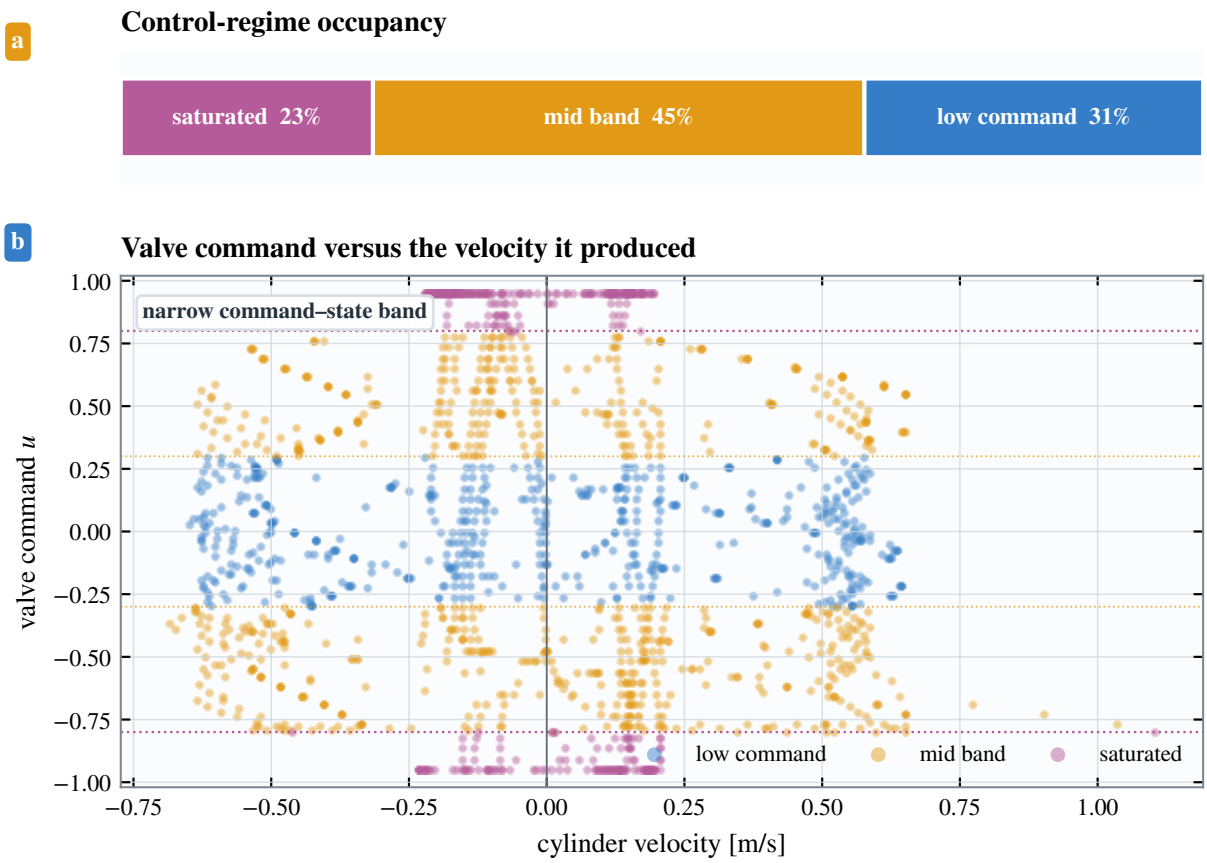

Figure 4.3: The closed-loop batch is diverse in control effort but not identifiable. (a) Fraction of time the valve spends saturated, in the mid band and in low-command creep. (b) The same run in the command–state plane, where the correlation that blocks identification is visible.

Figure 4.3 confirms that coverage alone is not the issue. The valve is saturated for about one quarter of a pooled run, operates at mid-authority for roughly half, and otherwise remains in low-command creep. Despite this spread, command and cylinder velocity remain concentrated along a narrow band. The data are diverse in control effort but not identifiable, motivating a separate open-loop experiment.

4.4.2 Design of the Excitation Signal

The excitation run lasts 64 s and follows the fixed schedule in Figure 4.2(b). One second of startup is discarded, because those samples describe the initialisation rather than the dynamics. Four sequential slots follow, each giving one cylinder ten seconds of excitation followed by a half-second zero-command gap that makes the segment boundaries unambiguous. A twenty-second phase then drives all four channels at once, and a final second of zero command closes the run. Commands are limited to ± 0.90 of full authority.

An initial rapidly switching signal met the formal persistence condition but kept each piston near one stroke position. The final design instead uses long full-authority plateaus in the sequential phase and slow, high-authority motion in the simultaneous phase. A slow near-square cycle is combined with distinct medium-frequency channel components, with raised-cosine entry and exit ramps. This provides both input excitation and useful stroke coverage.

4.4.3 Plant Draws and Acceptance

The same excitation is applied to four plant draws from the parameter space of the closed-loop campaign: nominal, randomised hydraulics, randomised mechanics and fully randomised. Holding the input fixed and varying only the plant makes the four runs directly comparable.

Each run is checked for stroke coverage before acceptance, the collector requiring at least 10 % of available stroke on cylinders 1–3 and 8 % on cylinder 4. The retained runs cover 33 % of the stroke on cylinder 1 and 12 % on cylinder 4, the narrowest of the four. Every channel clears the floor, and Section 4.10 confirms the material is informative at these coverages.

The acceptance checks in Section 4.10 verify that every export contains independent state trajectories.

4.5 Combining Closed-Loop and Open-Loop Datasets

The inverse model uses only closed-loop data because its target is the controller command for a measured state. The forward model is retrained on the strict open-loop export while retaining closed-loop replay at weight 0.25. Without replay, four excitation runs do not provide enough pose coverage.

Figure 4.4 shows the difference: closed-loop data covers the reachable range, while open-loop data concentrates at identifiable valve plateaus. The forward model therefore uses both.

For ROM training, the inverse and forward exports are joined by run identifier, time index and sample time. The join is accepted only when their sixteen-dimensional cylinder states match sample for sample. It contains 452 400 transitions at 0.05 s from 1 040 runs, split into 936/52/52 complete runs. Valve command is used only to supervise the ROM's internal valve head; it is not a runtime input (Section 5.4).

This aligned nominal-plant export trains the inverse, forward and ROM models and ships with the project, so every model-level result can be recomputed from the file that produced it. Exact alignment, increment consistency and the identifiability failure were checked directly on that export.

4.6 Closed-Loop Calibration Data for the Command ROM

The ROM drives a PhysX articulation whose compliance and contact response are not present in Simscape. Further Simscape training cannot remove the resulting deployment offset.

The calibration set therefore comes from closed-loop ROM–PhysX operation on the native 0.05 s grid. Its 4 084 transitions store the six-step runtime history and measured PhysX transition.

Only the command-gated correction head is trainable; all other parameters are frozen and the correction is capped at 3 mm. Original-data replay at weight 0.25 preserves wider pose coverage. The gate makes the correction zero at zero command, retaining stationary hold.

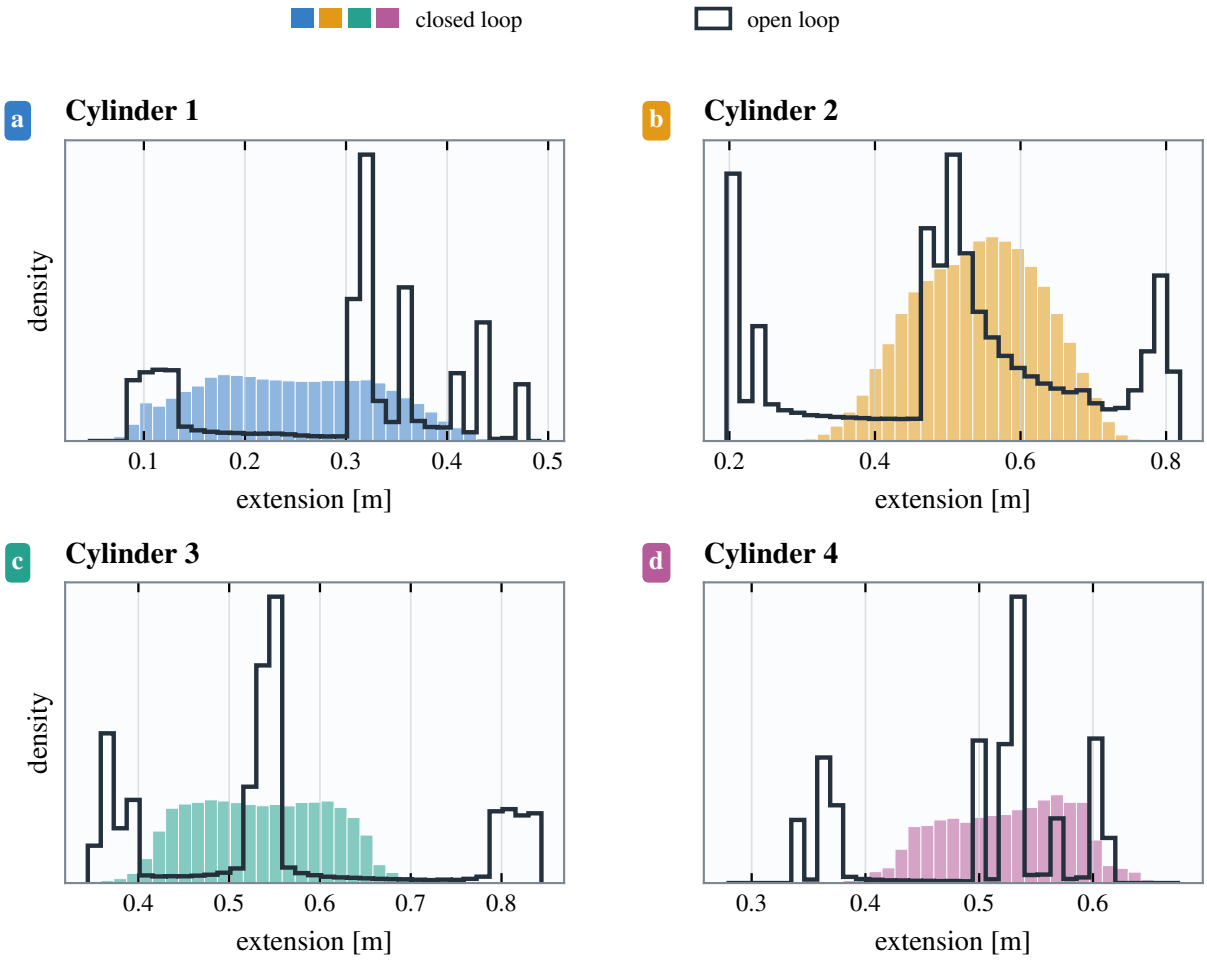

Figure 4.4: Extension coverage of the two campaigns, per cylinder and pooled over runs (closed-loop filled, open-loop outline). The closed-loop data fills the reachable range smoothly; the open-loop data concentrates into the plateaus where the valve is identifiable.

The trajectories predate the moving-verge benchmark and exclude cutter–ground contact. The bounded head limits, but does not remove, this coverage gap. Section 3.4.1 explains the consequence for the three-way comparison.

4.7 The IsaacLab Deployment Collections

Two collections are recorded in the deployment plant. They share their factor design, their domains, their splits and their freezing procedure, and differ in one respect only: how the arm is driven. The first is driven by the task controller and is the collection the inverse and forward models are scored against in Section 5.2.2; the second is driven by open-loop excitation and is the one the adaptation module is trained on. Why both exist is itself a result, and it is given in Section 4.7.5.

4.7.1 Provenance of the Deployment Data

The models that carry the identifiability argument, namely the system-dynamics surrogate, the privileged factor encoder and the offline history encoder, describe the plant the adaptive controller is closed around. That plant is the command ROM plus PhysX, not the Simscape model. A surrogate identified on Simscape data would be a model of a model, and the latent it defines would describe variation the deployment plant does not necessarily exhibit. Both collections are therefore made *in* the deployment plant, and the frozen ROM, ID and FD checkpoints enter this chain only as fixed components.

4.7.2 The Physical Factor Vector

The randomised quantity is a six-dimensional physical factor vector

$$\mathbf{e} = \Big[e_{\text{flow}},\ e_{\text{compliance}},\ e_{\text{seal}},\ e_{\text{pressure}},\ e_{\text{mass}},\ e_{\text{COM}_z}\Big], \tag{4.1}$$

with the ranges of Table A.3. Four coordinates are hydraulic and two mechanical, which is what makes the hydraulic-only, mechanical-only and combined domains of Section 4.7.3 separable. Each candidate was admitted only after a one-factor sensitivity screen confirmed that moving it alone produces a measurable change in cylinder extension, velocity, pressure or end-effector motion.

One omission is deliberate. Actuation delay is not a randomised coordinate, because delay here is an *emergent* property of flow, compliance and load rather than a free parameter. It is measured as a response property of each run (Chapter 7) rather than supplied to any model as a known factor: handing an estimator a factor the real machine would never publish is the same mistake as handing it the true payload.

4.7.3 Domains and Factor Design

Runs are organised into five domains (nominal, hydraulic-only, mechanical-only, combined-training, and an *unseen* combined-test domain), and one factor vector is fixed for a complete run, so that no run mixes plants. Factor configurations are drawn with deterministic scrambled Sobol sequences [Sobol' 67] rather than independent uniform sampling, with explicit boundary anchors so that the corners of the space are represented and not merely approached, and with fixed per-domain seeds so the design is reproducible independently of the data it produced. Combined draws that compound into an excessive response gain are rejected at draw time rather than being collected and discarded later.

The resulting design is auditable: 98.3 % of the factor keys are unique, the active-factor matrix is full rank, every factor quartile is populated, and the factor keys of the training, validation, and test splits are disjoint. That last property is the one that matters most for the claims of Chapter 7: a test run does not merely come from a different episode, it comes from a plant the models have never seen.

4.7.4 Collection Procedure: The Task-Controller Runs

Every run is collected through the same interface the policy uses. The DLS controller closes the loop on measured cylinder state and issues its command through the four-dimensional

action port into the command ROM and PhysX. No joint-target bypass is used and no PPO policy collects identification data, because a policy trained on the same plant would reintroduce the closed-loop correlation that Section 4.4 spent a campaign removing.

Independent excitation is supplied by dedicated four-channel PRBS windows through that same action port, at small amplitude and a hold of ten control steps, so the excitation is causally independent of the state without leaving the safe envelope. Probe actions are logged separately from controller output. Each run follows a fixed eleven-segment structure (settle, three postures, three PRBS windows, DLS task motion, reach-and-return and stow). As recorded, each run is 11 520 control steps and 192.0 s of simulated time.

Figure 4.5 summarises the IsaacLab collection configuration and its frozen run-level outputs.

Figure 4.5: IsaacLab adaptation-data collection. Each run uses one fixed six-factor plant draw and combines DLS task motion with independent PRBS probes. State, action and factor keys are logged, audited and frozen into disjoint train, validation and held-out test splits.

4.7.5 The Open-Loop Context-Identification Collection

A second collection of 118 runs was recorded in the same deployment plant, with the same six-factor design and the same domain and split structure, and it is the offline context-identification collection used to train the AAM. It differs from the collection above in a single respect, and that respect is the point of it: there is no task controller. Each run is driven by open-loop excitation alone, the vehicle is stationary, and no policy is in the loop. The excitation is the same four-channel PRBS used as a probe above, at amplitude 0.02 and held for ten control steps, but it now drives the whole run rather than three windows inside one. Each run is 11 040 control steps, giving 1 302 720 steps across the collection.

The reason for collecting twice is structural. Under a task controller the command is computed from the state, so command and response are close to redundant and a plant descriptor has little left to explain; a model asked to identify the plant from that data is being given a question with no answer in it, and what it reports is a property of the data rather than of the architecture. Removing the controller makes the question well posed, which is why every model-level result in this thesis is fitted on this collection. How rich the excitation must be, and the amplitude beyond which plant identity stops being recoverable, remain outstanding measurements.

What each consumer takes from the recorded feature contract is set out in Section 4.8.2 for the adaptation module and Section 4.8.3 for the prediction models.

4.7.6 The Frozen Dataset

Both collections contain 118 complete runs under the identical domain design: 2 nominal, 40 hydraulic-only, 12 mechanical-only, 48 combined-training and 16 unseen combined-test. Collection ran in four vectorized IsaacLab environments with per-environment factors and logging, and slots that tripped a safety or validity check were discarded and retried individually. All 118 runs passed the final collection gate, which checks finite physics, exact model-grid alignment, command rank, factor-response authority, state coverage and saturation limits.

Each dataset is then *frozen*: the raw runs are checksummed and the run-level split is fixed at 82 training, 20 validation and 16 reserved combined-test runs, with zero factor-key leakage between them. The reserved sixteen do not influence history length, checkpoint, hyperparameter or architecture selection; they are scored only after each model is fixed, and test results are reported with validation results rather than in place of them.

4.7.7 Learning Timebase and Hydraulic State

The learning timebase is the actuator model grid, not the control clock. Samples are exact 50 ms ROM transitions taken at control indices $0, 3, 6, \ldots$: state row t and command row t predict state row $t + 3$. Nothing is interpolated. This is the same alignment rule as Section 3.5.1, and it is enforced rather than assumed, because a model trained across a mismatched interval is wrong by a fixed factor in the quantity it exists to supply.

The hydraulic state used by the system-dynamics surrogates is

$$\boldsymbol{s}_t = \left[\boldsymbol{x}_t^{1:4},\ \dot{\boldsymbol{x}}_t^{1:4},\ \Delta\boldsymbol{p}_t^{1:4}\right] \in \mathbb{R}^{12}, \tag{4.2}$$

with joint position and velocity carried as *context* only, and the prediction target the next 50 ms state delta. Differential pressure is not a measured signal. At runtime it is the actuator model's recursive state, generated by the command ROM or frozen forward model (Section 3.3). The system-dynamics surrogate and AAM may use it because it exists inside the deployed loop, but it inherits actuator-model error. Causal history is therefore retained to represent pressure transients that one sample cannot summarise.

4.8 Feature and Target Construction

4.8.1 The Actuator Models

The three learned actuator models take cylinder-space features, in the ordering and notation fixed in Section 5.1. Their input contracts differ because their questions differ. The inverse model receives the measured cylinder state, the logged rod forces, the desired increments, and the previously applied command, twenty-eight numbers in all. The forward model receives the applied valve command and the measured state, twenty numbers. The reduced-order model receives a six-sample history of state, previously predicted valve, and desired command, twenty-four numbers per step.

Three conventions apply throughout, and they are the same ones the models enforce at inference time.

Increment Targets

Every dynamic target is a within-run one-step increment, $\Delta \boldsymbol{x}_k = \boldsymbol{x}_{k+1} - \boldsymbol{x}_k$, never an absolute state. This removes the large static component of the signals and turns the trivial persistence solution into an explicit baseline rather than a hiding place.

Causal Alignment

Features at step k contain only quantities available at step k. The extraction stage verifies this by recomputing every logged target from the within-run difference of the states and requiring exact agreement, a check reported in Section 4.10.

Time-Base Conversion

The open-loop campaign is logged at 0.01 s but the models run at 0.05 s, so five consecutive raw samples are combined into one transition. The valve input attributed to that transition is the command held at the start of the window, which is the causally correct choice; the window mean is recorded for analysis only. Cylinder extension velocity is not logged directly in the closed-loop archive and is obtained by differentiating extension on the output grid, consistent with the trapezoidal identity that the strict data satisfies to $R^2 = 0.994$–0.9997 (Equation 5.6).

4.8.2 Features and Targets for the Adaptation Module

The adaptation module is trained on the open-loop collection of Section 4.7.5. Its history window carries only quantities the deployed stack produces at runtime: the five joint positions, the five joint velocities, the four cylinder pressure differences and the four

previously applied valve commands, giving the eighteen-value row of Section 5.5. The pressure entry is the learned actuator's own recursive state rather than a measurement (Section 3.3, Table 3.2); it is admitted because the model that generates it runs inside the deployed loop, and the consequence of that choice is examined where the estimator is evaluated. Four classes of signal are excluded by construction, and the exclusion is enforced by an explicit input mask rather than by convention:

- the physical factors $\mathbf{e}$ and the true latent $\mathbf{z}_{\mathrm{SD}}$, because recovering them is the estimator's job;
- any future observation, because the estimator is causal;
- the hidden internal state of the command ROM, because the deployment interface does not expose it;
- the desired task reference, so that the latent describes the plant rather than the trajectory being asked of it.

Windows are contained within a single run, and each carries the stopped-gradient target $\mathbf{z}_{\mathrm{SD}} = \mu(\mathbf{e})$ of its own run together with the run identifier, domain, split and valid history bounds. Applied to the 118 frozen runs this yields 434 240 rows, of which 431 998 carry a complete twenty-step window (3 661 per run; the first nineteen ticks of a run cannot fill one). What is built from them belongs to Chapter 5.

The residual encoder deployed in AF uses no collection at all, for the reason given in Section 4.8.4, and it does not read this row. It reads the whole 66-value policy observation together with the four previously applied valve commands, 70 values per step. The row defined here is the contract of the explicit estimator only. What the wider input costs is examined in Section 7.5.

4.8.3 Features and Targets for the Prediction Models

The prediction models read the same runs through a third contract: a transition at the 50 ms tick rather than a history window, and the privileged factor vector $\mathbf{e}$, which the adaptation module never sees and which is what allows them to define the target the module is then asked to recover. That contract is set out with the dataset it produces in Section 4.11, because what it is for is only visible alongside the audit it makes possible.

4.8.4 On-Policy AF Adaptation Rollouts

The context columns that carry the adaptation result in AF are not trained from the frozen collection above. They have no dataset and no run-level splits, because they are optimised under the reinforcement objective, so their training data is the on-policy experience the stack generates as it learns.

What replaces the dataset is a plant design, since on-policy training makes the training distribution a property of the environment rather than of a file. The cutter mass is resampled per episode across a range spanning one to five times nominal. One fixed compromise is then unlikely to suit the whole family, which is what gives a controller

something to gain from knowing which member it is driving, without assuming in advance that it will.

The estimator that supplies that knowledge at deployment does have a dataset, and it is collected after policy training rather than before: a mixed-payload rollout of the trained policy, with the 66 task channels regressed onto the payload that produced them. It is refitted for each training seed, because the regression is on the observation distribution that policy induces and a different policy induces a different one. Whether the correct payload contributes beyond conditioning on any plausible one is tested against a permuted-payload control in Section 7.5.

One property of the frozen collection is inherited unchanged: the plant is still the deployment plant, so the change of training regime does not substitute a different system. The history row is not inherited. The residual encoder reads the full policy observation and the previous valve command rather than the eighteen-value row, so the exclusion contract above does not apply to it. That is stated where the residual is defined (Section 6.7.3) and carried into Section 7.5, where it becomes the leading candidate explanation for what the adaptation experiment finds.

4.9 Preprocessing and Data Splits

Splitting is done at the level of complete runs, never of individual samples, because a random sample-level split would place transitions from the same trajectory on both sides of the boundary and report a score that mostly measures interpolation within a run. For the command-conditioned dataset the 1 040 runs are divided into 936 training, 52 validation and 52 test runs, giving 407 160, 22 620 and 22 620 transitions, with no run identifier in more than one split. The IsaacLab deployment collection is split by run *and* by factor configuration (82/20/16), the stronger condition: separating runs alone would let two runs of the same plant fall on opposite sides of the boundary and turn a claim about unseen plants into a claim about unseen episodes.

The strict open-loop export needs different treatment because it contains only four runs, and holding one out would remove an entire plant draw from training. Each run is instead divided into twenty temporal blocks, every fifth block is assigned to validation, and a five-sample guard interval is discarded at each boundary so that no 0.05 s transition window spans the split. This preserves comparable target distributions in training and validation.

Normalisation statistics are computed on the training split only and stored inside each checkpoint, so a model always normalises with the statistics it was fitted with; the loader enforces the stored contract rather than recomputing it. Samples are carried with an explicit validity flag rather than silently dropped, 93.7 % of the strict export being marked valid and the remainder being the excluded startup and shutdown seconds and the zero-command gaps. Keeping them with a flag means the time base stays contiguous and the within-run difference checks remain meaningful.

4.10 Dataset Diagnostics

No dataset in this thesis is used before it passes an explicit audit, for the reason given at the end of Section 4.4. The audit of the strict open-loop export is reproduced in Table A.4.

The two rank figures speak directly to requirement R2. The instantaneous command matrix has rank four, so no channel is a linear combination of the others, and the lagged regressor matrix built from the command history has rank eighty, so the excitation is persistently exciting to the order the models need. The condition number of that lagged matrix is about 4.6×10^3, finite and workable rather than comfortable, and it is the price paid for the stroke coverage the long plateaus buy.

The three increment checks return a maximum error of exactly zero, which is expected rather than suspicious: the targets are computed from the logged states by the same within-run difference the check recomputes, so the check verifies that the extraction stage did what it claims rather than that the simulation is accurate. It catches misalignment, off-by-one indexing and rows reordered or concatenated across run boundaries. Physical fidelity is addressed separately by the validation study of Section 3.4.4.

Two properties of the Simscape material are recorded here because later chapters refer to them. Stroke coverage is 12 % on the head-pivot cylinder, the narrowest of the four channels and also the one the strict forward model predicts least well. The rod-force channel carries zero supervised weight in the reduced-order model, a design choice rather than a data defect: force is generated internally by the PhysX plant at deployment, and the logged channel additionally fails a piston-area consistency check on this export. No result rests on it.

The IsaacLab collection is audited against different failure modes: design adequacy before use (unique factor keys, full active-factor rank, disjoint keys across splits), physical validity per run (finite physics, exact model-grid alignment, full command rank, bounded saturation), and leakage structurally, with privileged channels removed by an explicit input mask rather than by trusting the feature list. Its limitation is scale, 118 runs over six factors being a sparse design, which is why the paired test of Section 4.11.1 is reported with bootstrap intervals and per-run detail rather than as a pooled point estimate. That limitation does not bound the closed-loop adaptation result, which is estimated from paired environments generated on-policy under the plant design of Section 4.8.4.

4.11 The System-Dynamics Dataset

The prediction models read the deployment collection through a third contract, built once and frozen with the rest. It is a derived view rather than a separate collection: the same 118 runs, resampled from the 60 Hz control log to the 20 Hz model tick by taking every third step, giving 3 840 transitions per run and 453 120 in total. The run-level split is inherited unchanged, 82 training, 20 validation and 16 test runs, so a run contributes to exactly one split here and in every other view of this data.

Each sample is one transition. The input is the 12-dimensional hydraulic state, the 10-dimensional joint context and the valve command pair, previous and current, for 30 channels; the target is the hydraulic-state increment to the next tick in the same 12 dimensions. Training draws five consecutive transitions at a time, so the objective is a short rollout rather than a one-step fit. Four per-tick flags record cylinder limit, command saturation, valve saturation and non-finite physics, so a model can be scored on valid ticks alone. Each run carries its own six-factor vector $\mathbf{e}$, which is privileged: it is what allows a target to be defined for the adaptation module of Section 4.8.2, and it is masked out of every deployable input.

4.11.1 Dataset Separation Across Plants

One property of this dataset cannot be checked by an audit of its structure. The checks of Section 4.10 establish that the runs are valid, that the factor design is full rank and that the splits do not leak, but not that the runs *differ by plant* in a way any model could use. If they do not, nothing built on this dataset can recover the difference.

The test is a paired one on the dataset itself. Two prediction models are trained on identical samples, splits and objective; one receives a two-dimensional plant context from an encoder $\mathbf{z}_{\mathrm{SD}} = \mu(\mathbf{e})$ trained jointly with it, and the matched control receives no such input. Neither is an actuator and neither enters a control stack.

The context helps, and narrowly. Pooled rollout error falls by 7.20×10^5 on the 20 validation runs, with a 95 % interval of $[-1.03 \times 10^6,\ -4.30 \times 10^5]$, and by 4.41×10^5 on the 16 test runs, interval $[-7.61 \times 10^5,\ -1.74 \times 10^5]$. Both exclude zero, so the dataset does separate plants. Where it separates them is the more useful result: the per-channel decomposition puts the whole gain in pressure, and the extension and velocity intervals cross zero on both pools.

Table 4.1: Per-channel decomposition of the paired advantage at a 20-tick horizon, pooled over 36 held-out runs. Values are the mean paired difference between the context-conditioned model and the matched control, with a 95 % bootstrap interval. The advantage is confined to pressure.

Channel	mean paired difference	95 % interval	excludes zero
extension	5.2×10^{-5}	$[-3.6 \times 10^{-5},\ 1.4 \times 10^{-4}]$	no
velocity	-2.4×10^{-4}	$[-7.1 \times 10^{-4},\ 2.4 \times 10^{-4}]$	no
pressure	1.75×10^5	$[2.1 \times 10^4,\ 3.3 \times 10^5]$	yes

Two limits travel with this result and both matter later. It compares separately trained models, so it establishes that the context is informative, not that an estimator could recover it or that a controller could use it; Section 5.6.3 substitutes contexts within one model, which is the tighter test. And the informative channel is pressure, which the controller does not integrate. That is the first appearance of the delivery problem the rest of this thesis runs into: this dataset carries plant identity in the one channel that reaches the command path least directly.

With the data described and audited, the next chapter turns to what was learned from it: the inverse and forward models that form the IF actuator path, the command-conditioned reduced-order model that replaces them in R, and the adaptation module that estimates at runtime what the surrogate above was given.

5. Learned Models and Online Adaptation

Chapter 4 established how the datasets were generated and why both closed-loop operation and open-loop valve excitation were required. This chapter develops the learned representations that consume them: the inverse-dynamics (ID) and forward-dynamics (FD) networks of the IF actuator path, the command-conditioned reduced-order model (ROM) that replaces them in R, and an offline Arm Adaptation Module (AAM) study that tests whether plant context can be recovered from causal history. This offline AAM and its ID adapter are diagnostic prototypes. The deployed AF encoder and residual are trained jointly under PPO in Chapter 6.

The chapter continues one evidence chain. Section 4.11 established the first link on the data itself: a compact plant context exists and improves prediction, and the improvement sits in pressure. What follows is whether that context is recoverable causally from observable history, whether the recovered estimate is as informative as a privileged one, and whether it reaches the command. The chain fails at that last link, and the diagnosis specifies the redesign carried out in Chapter 6.

One commitment organises the chapter: no model is accepted on a supervised score, because such scores repeatedly proved uninformative here, a forward model reaching a one-step R^2 near 0.999 while having learned almost nothing about valve authority (Section 5.3). Every model is therefore presented with its contract, training objective and the offline evidence that accepted or refused it. All models operate on cylinder-space signals.

5.1 Modelling Overview and Common Notation

All learned models act on the four hydraulic cylinders of the MULAG arm, indexed $i \in \{1, \dots, 4\}$ in the fixed order slew cylinder (*Drehzapfen*), boom cylinder I (*Ausleger I*), boom cylinder II (*Ausleger II*) and head pivot cylinder (*Messerkopf-Schwenk*). The passive fifth joint is not actuated and enters no model input. At the learned-model sample time $T = 0.05\,\mathrm{s}$ the discrete index k refers to the cylinder extension $\boldsymbol{x}_k$, the extension velocity $\dot{\boldsymbol{x}}_k$, the chamber pressure difference $\Delta\boldsymbol{p}_k$ and the logged rod force $\boldsymbol{F}_k$, each in $\mathbb{R}^4$. The

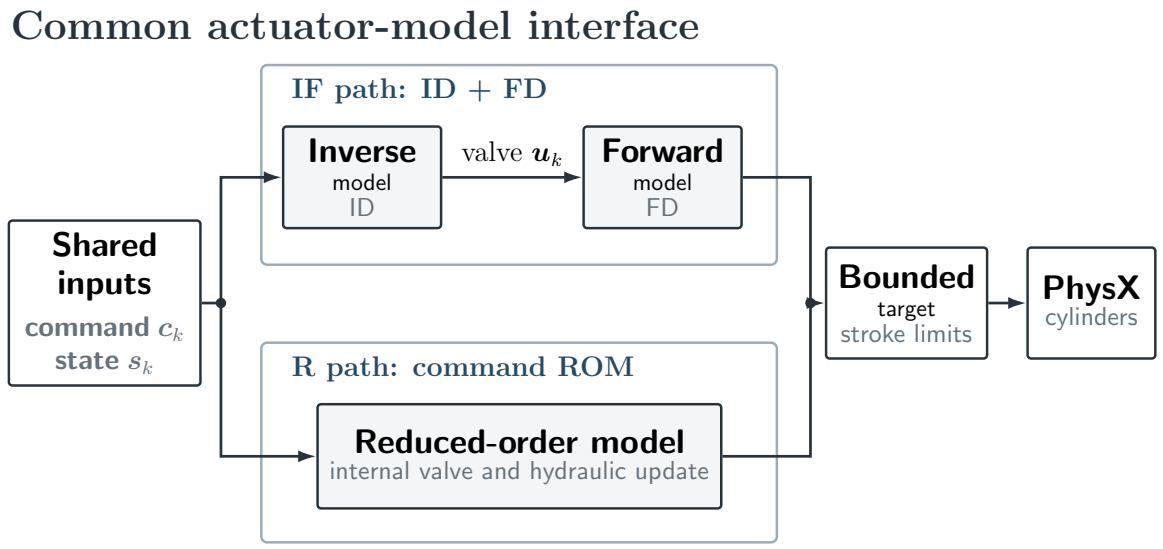

Figure 5.1: Common interface of the two alternative actuator paths. Both receive the desired cylinder command and measured hydraulic state, and both produce the same bounded PhysX cylinder target. The IF path composes the inverse and forward models; the R path uses one command-conditioned reduced-order model.

valve command $\boldsymbol{u}_k \in [-1, 1]^4$ is the normalised spool command of the four proportional valves. The desired cylinder command produced by the task controller is the pair

$$\boldsymbol{c}_k = \left[\Delta \boldsymbol{x}_k^*,\ \Delta \dot{\boldsymbol{x}}_k^*\right] \in \mathbb{R}^8, \tag{5.1}$$

the extension and extension-velocity increments to be realised over the next sample. Predictions carry a hat, and $\Delta(\cdot)_k = (\cdot)_{k+1} - (\cdot)_k$ denotes a within-run one-step increment throughout.

Three conventions apply throughout. Dynamic targets are within-run increments, with $\Delta\hat{\boldsymbol{x}} = \mathbf{0}$ as the persistence baseline. Each checkpoint uses its own training-set normalisation. Inputs at step k are causal, and targets are measured transitions to step $k + 1$.

Simscape supplies the identification data of Chapter 4. The exported arrays are audited, frozen and used to train the models in PyTorch [Paszke 19]; Simscape itself is part of neither training nor deployment.

The closed-loop Simscape export [The MathWorks, Inc. 24a] contains 452 400 transitions at $T = 0.05\,\mathrm{s}$ from 1 040 runs, split by run into 936/52/52 training, validation and test runs. It supplies the supervised ID, FD and command-ROM data. The FD repair uses four open-loop 64-s runs at 0.01 s with rank-4 valve excitation on four plant draws.

5.2 Inverse Dynamics Model

The ID network maps the current hydraulic state and desired cylinder increment to a valve-command change:

$$\Delta \boldsymbol{u}_k = f_{\mathrm{ID}}\left(\boldsymbol{x}_k,\ \dot{\boldsymbol{x}}_k,\ \Delta \boldsymbol{p}_k,\ \boldsymbol{F}_k,\ \Delta \boldsymbol{x}_k^*,\ \Delta \dot{\boldsymbol{x}}_k^*,\ \boldsymbol{u}_{k-1}\right) \in \mathbb{R}^4, \tag{5.2}$$

a 28-dimensional input. The deployed command is $\boldsymbol{u}_k = \mathrm{clip}(\boldsymbol{u}_{k-1} + \Delta\boldsymbol{u}_k, -\boldsymbol{u}_{\max}, \boldsymbol{u}_{\max})$ with per-axis limits $\boldsymbol{u}_{\max} = [0.95, 0.95, 0.8, 0.8]$ stored in the checkpoint. The change formulation retains the previous valve operating point near the direction-dependent dead zone (Section 2.1.2). Logged achieved transitions serve as desired increments because no independent cylinder-space reference was recorded.

The 512-wide residual MLP [He 16] uses SiLU activations, layer normalisation and dropout [Elfwing 18, Ba 16, Srivastava 14]. Adam minimises a smooth-L_1 loss on normalised targets, with saturated samples up-weighted [Kingma 15, Huber 64].

Figure 5.2 separates the three input groups and shows the four residual blocks explicitly. The network predicts a four-axis valve-command increment, which is accumulated and clipped before use.

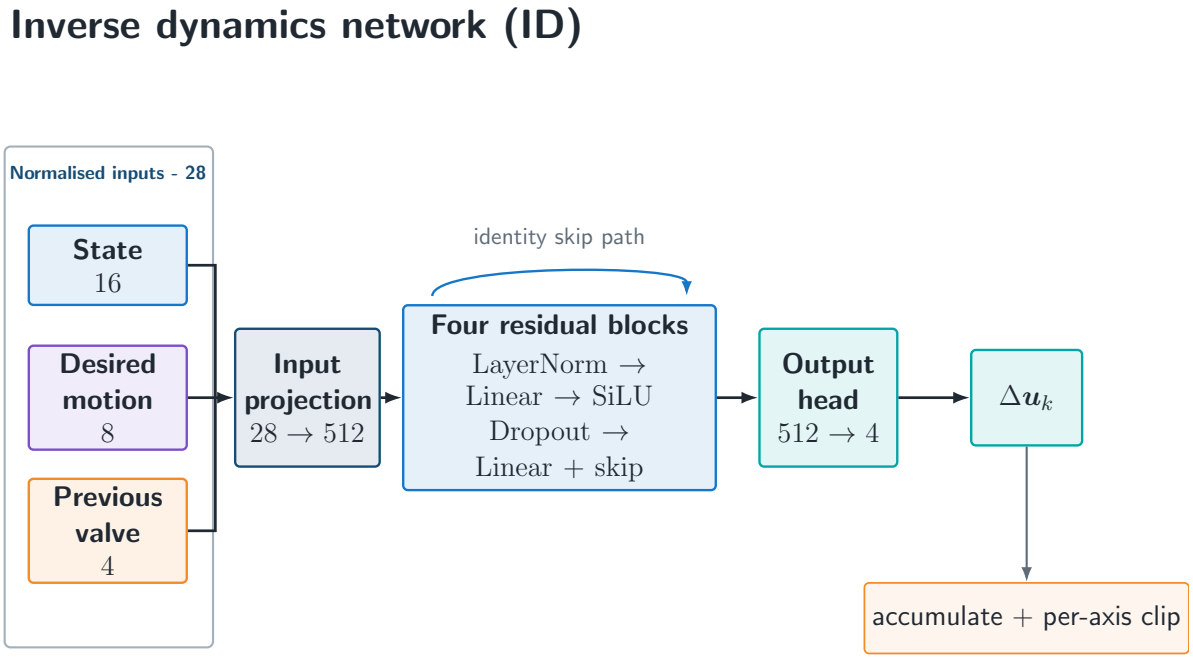

Figure 5.2: Inverse-model architecture. Current hydraulic state (16), desired motion (8) and the previous valve command (4) form the 28 inputs. A 512-wide, four-block residual MLP predicts $\Delta\boldsymbol{u}_k$, which is accumulated and clipped per axis to form the bounded valve command.

Per-valve validation R^2 is 0.9973–0.9996. This shows reproduction of logged commands, not plant authority: in closed-loop data, achieved transitions and logged commands are correlated (Section 2.4.2).

5.2.1 Paired Control Fine-Tuning

The composite is assessed by the *paired differential response*: the FD prediction under the ID command for the requested target minus the prediction under its command for a zero

target,

$$\begin{aligned} \boldsymbol{u}_k^{\mathrm{tgt}} &= f_{\mathrm{ID}}(\cdot,\, \boldsymbol{c}_k,\, \boldsymbol{u}_{k-1}), \qquad \boldsymbol{u}_k^{0} = f_{\mathrm{ID}}(\cdot,\, \mathbf{0},\, \boldsymbol{u}_{k-1}), \\ \Delta\hat{\boldsymbol{x}}_k^{\mathrm{pair}} &= f_{\mathrm{FD}}\Big(\boldsymbol{u}_k^{\mathrm{tgt}}, \cdot\Big) - f_{\mathrm{FD}}\Big(\boldsymbol{u}_k^{0}, \cdot\Big). \end{aligned} \tag{5.3}$$

Before fine-tuning, the composite normalised control error was 1.00, equal to the zero-response baseline, and all regression slopes were below 0.004 (Table 5.1). The networks were fine-tuned jointly on $\mathcal{L}_{\mathrm{ctl}} = \|\Delta\hat{\boldsymbol{x}}^{\mathrm{pair}} - \Delta\boldsymbol{x}^{*}\|^2_{\mathrm{norm}} + \lambda_{\mathrm{ID}}\mathcal{L}_{\mathrm{ID}} + \lambda_{\mathrm{FD}}\mathcal{L}_{\mathrm{FD}}$, with the supervised terms retained. Target blocks were shuffled across valid states to break the logged state–target pairing. After 30 epochs, control error fell to 0.109; all slopes exceeded 0.8 and all correlations exceeded 0.92. This frozen checkpoint is the inverse model of the IF actuator path, and of the latent-conditioned adapter route of Section 5.6. It is not part of AF, whose command path contains no inverse model (Section 6.7.1).

Table 5.1: Effect of paired control fine-tuning on the composite differential response (5.3), per cylinder axis on validation data. Before fine-tuning the composite had essentially no control authority despite near-perfect supervised scores of both networks; the composite normalised control error fell from 1.0011 to 0.1086.

	Regression slope		Correlation	
Cylinder axis	before	after	before	after
1 (slew)	0.003	0.818	0.135	0.920
2 (boom I)	0.002	0.928	0.090	0.967
3 (boom II)	0.001	0.849	0.037	0.933
4 (head pivot)	0.004	0.912	0.249	0.958

5.2.2 Transfer of the Frozen Inverse Model

Because the adapter corrects a frozen ID, its transfer was measured on all 118 frozen runs of Section 4.7, comprising 2 nominal, 40 hydraulic-only, 12 mechanical-only and 64 combined runs, of which 16 are held-out combined-test. These runs lie outside the ID's Simscape-only training distribution and are recorded in the plant the adaptive controller is closed around. Two regimes were scored: a teacher-forced one-step transfer supplying the measured cylinder state at every tick, and a closed-loop transfer feeding back the pair's own prediction while supplying the recorded reference. Pressure and force are held at their training means, as the deployed feature contract requires.

Teacher-forced, the pair transfers well and evenly (Table 5.2). One-step extension error is millimetre-scale in every domain and nearly flat across them, from 2.36 mm on the combined runs to 3.36 mm on nominal, and the held-out combined-test group is no worse than the domains used in training, at 2.34 mm. This is the regime in which IF uses the pair, because the policy closes the loop at every tick and never asks it to predict its own future.

Closed-loop, it does not transfer at all, and the failure is uniform rather than occasional. Every one of the 118 runs leaves the recorded reference by more than one metre, after about 32 s on the combined and hydraulic domains and sooner on the harder ones. The

Table 5.2: Transfer of the frozen inverse and forward pair across deployment domains, all 118 runs of the deployment collection. The one-step columns are teacher-forced; the saturation column is the fraction of pre-clip valve elements at the limit during closed-loop transfer. The ROM valve recorded on these same runs never saturates.

		one-step Δx error		closed-loop
Domain	runs	RMSE	worst run	valve saturation
combined	64	2.36 mm	5.72 mm	0.888
hydraulic	40	2.43 mm	4.96 mm	0.885
mechanical	12	2.81 mm	4.14 mm	0.919
nominal	2	3.36 mm	4.02 mm	0.929
all	118	—	—	0.891

mechanism is saturation: the pair's pre-clip valve command sits at its limit on 89.1 % of elements over the collection, between 88.5 and 92.9 % by domain, while the ROM valve recorded on those same runs never reaches its bound at all. For scale, the measured plant tracks the recorded cylinder reference at about 0.807 mm on the held-out group, so the closed-loop divergence is a failure of model transfer and not a property of the collection.

Two consequences follow. The pair cannot be promoted as an unadapted autoregressive deployment path, because its own predicted state leaves the support its training data covered, though nothing in that result disturbs the one-step use IF makes of it. What it does disturb is adaptation: a bounded correction added to a command already resting on its stops has nothing left to move (Section 6.7.1).

Stated on its own, this is one of the clearer transfer results in the thesis and it is specific to the structure of the map rather than to its accuracy. A model with millimetre one-step error across every domain, including held-out plants, diverges by more than a metre in all 118 of the same runs when asked to close the loop on its own predictions. Accuracy is not the failing quantity. The failing quantity is the accumulate-and-clip contract: the model predicts a command *increment*, the increments are integrated, and sustained demand walks the integral onto the clip and holds it there, at which point the map's one-step fidelity is irrelevant because its output is no longer what reaches the plant. Any inverse model with that contract inherits the result, and a one-step validation score cannot detect it. This belongs with the command ablation of Section 5.3.1 as the second case in this thesis where a supervised score stays high while the property a controller needs is already gone.

5.3 Forward Dynamics Model

The FD network predicts the one-step cylinder response to a valve command,

$$\left[\Delta\hat{\boldsymbol{x}}_k,\ \Delta\hat{\dot{\boldsymbol{x}}}_k,\ \Delta\widehat{\Delta\boldsymbol{p}}_k\right] = f_{\mathrm{FD}}\Big(\boldsymbol{u}_k,\ \boldsymbol{x}_k,\ \dot{\boldsymbol{x}}_k,\ \Delta\boldsymbol{p}_k,\ \boldsymbol{F}_k\Big) \in \mathbb{R}^{12}, \tag{5.4}$$

using the same residual trunk as the ID network. Supervised pretraining on the closed-loop export, with an auxiliary recursive rollout loss [Nagabandi 18], reached grouped one-step R^2 near 0.999 for the extension increments.

Figure 5.3 shows the forward model separately because its input and output contracts differ from the inverse model despite the shared residual trunk.

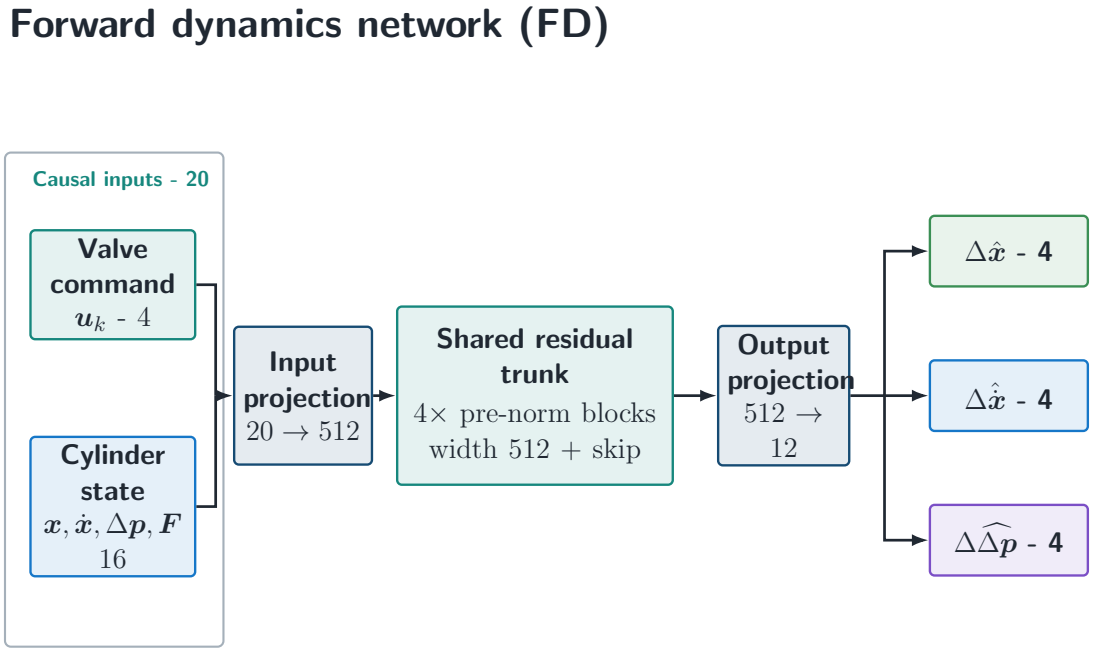

Figure 5.3: Forward-model architecture. The applied valve command and cylinder state condition a one-step hydraulic transition through a 512-wide, four-block residual MLP. Its 12 outputs are extension, velocity and pressure increments. Extension is the integrated controller output; velocity and pressure are diagnostic outputs.

5.3.1 Assessment of the Supervised Score

The supervised score did not establish valve sensitivity. At a 0.5 s horizon, zeroing the valve input changed the rollout-error ratio to only 0.94. The FD had mainly extrapolated the smooth closed-loop kinematics because command and state history were nearly redundant. Closed-loop ablation confirmed that the valve was not identifiable, so repair required new data.

The repair dataset applies rank-4 open-loop valve plateaus to four plant draws. Its trajectories are distinct, valve history predicts pressure with $R^2 \approx 0.80$–0.85, valve-regressor variance-inflation factors [Kutner 05] are 7–10, and relief saturation occupies 9–22 % of samples.

Starting from the paired-control checkpoint, strict retraining minimised

$$\mathcal{L}_{\mathrm{FD}} = \mathcal{L}^{\mathrm{strict}}_{\mathrm{Huber}} + 0.25\,\mathcal{L}^{\mathrm{MAT}}_{\mathrm{Huber}} + 2.0\,\mathcal{L}^{(10)}_{\mathrm{roll}} + 10.0\,\mathcal{L}_{\mathrm{kin}}, \tag{5.5}$$

combining the one-step loss on the strict data, a replay term on the original closed-loop data, a 0.5 s recursive rollout loss, and a kinematic consistency term enforcing the trapezoidal identity [Siciliano 09]

$$\Delta \boldsymbol{x}_k = T\,\dot{\boldsymbol{x}}_k + \tfrac{1}{2}\,T\,\Delta\dot{\boldsymbol{x}}_k, \tag{5.6}$$

which holds in the strict data with $R^2 = 0.994$–0.9997. Valve authority varies by a factor of 15–75 across pressure quartiles and can change sign, so the ROM represents this dependence by operating point [Merritt 67].

Strict retraining reduces extension rollout RMSE from 75.2 mm to 12.4 mm at 0.5 s and from 157.9 mm to 27.3 mm at 1.0 s. The valve-zero error ratio rises from 0.94 to 1.81, and per-axis one-step extension R^2 improves from $[-5.7, 0.60]$ to $[0.76, 0.98]$. Zero-order hold remains competitive below 0.25 s, and velocity increments remain weak on two axes. Deployment therefore integrates only $\Delta\hat{\boldsymbol{x}}$ and retains velocity as a diagnostic output.

5.4 Reduced-Order Model

The R path uses one command-conditioned model instead of the ID–FD composition. Its structure is grey-box: physical equations propagate pressure, force, velocity and extension. Neural heads estimate valve realisation, load and bounded residuals [Karniadakis 21, Lutter 19]. Valve authority is pressure dependent by construction through the orifice-flow and continuity equations [Merritt 67]. A GRU [Cho 14] reads six samples (0.25 s) of cylinder state, previous predicted valve command and desired command. The measured valve is only an internal-head training target [Hwangbo 19].

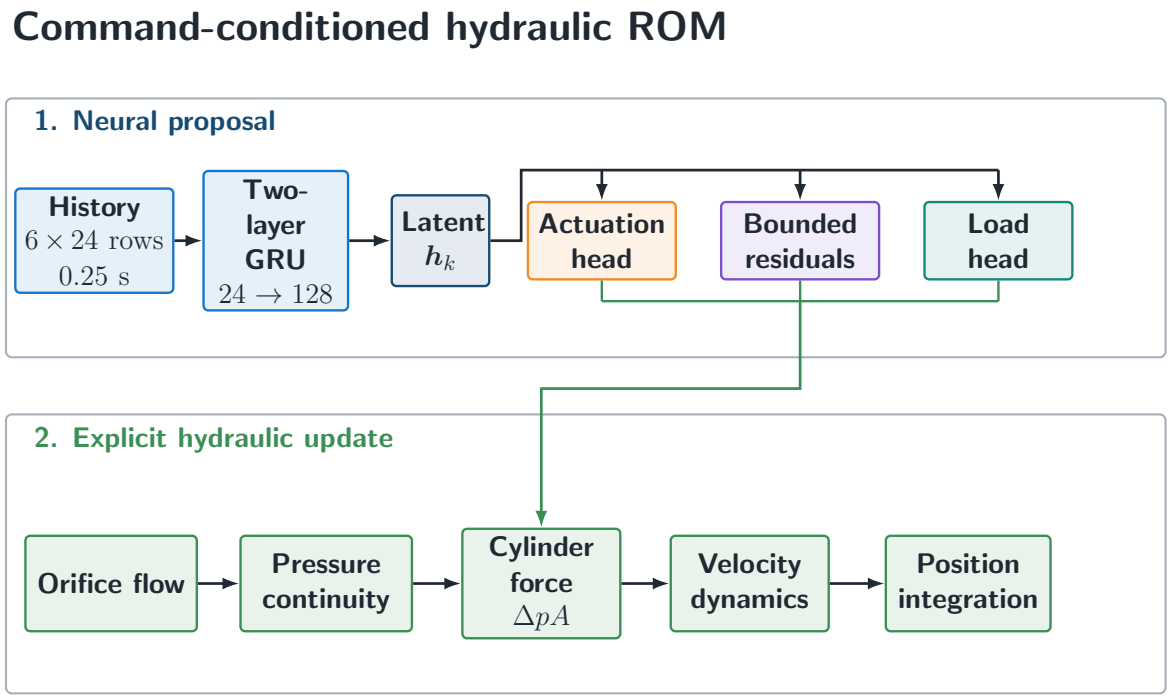

Figure 5.4: Command-conditioned ROM. A recurrent neural proposal encodes a six-step history. Its actuation, bounded-residual and load heads condition the explicit hydraulic update: orifice flow, pressure, force, velocity and position. The explicit chain constrains the predicted transition.

The actuation head uses a learnable smooth dead zone with per-cylinder width and asymmetric gains [Yang 26]. Its output $\hat{a}_i \in [-1, 1]$ drives the orifice-flow and continuity update

$$q_i = \hat{a}_i \, k_{q,i} \sqrt{\tfrac{1}{2}\big(p_{s,i} - \operatorname{sgn}(\hat{a}_i)\, \Delta p_i\big)_+},$$
$$\Delta\hat{p}_{i,k+1} - \Delta p_{i,k} = \frac{4\beta_i}{V_i(x_i)}\big(q_i - A_i\dot{x}_i - k_{\ell,i}\, \Delta p_i\big)\, T \;+\; r_{p,i}, \tag{5.7}$$

where β_i, $p_{s,i}$, $k_{q,i}$, $k_{\ell,i}$, A_i and $V_i(x_i)$ are the effective bulk modulus, supply pressure, flow gain, leakage, piston area and chamber volume. Pressure then produces force and motion:

$$\hat{F}_i = \Delta\hat{p}_{i,k+1} A_i + r_{F,i}, \qquad \ddot{x}_i = \frac{\hat{F}_i - c_{v,i}\,\dot{x}_i - \ell_i(\boldsymbol{h}_k)}{m_i},$$
$$\dot{\hat{x}}_{i,k+1} = \dot{x}_{i,k} + \operatorname{sat}\big(T\,\ddot{x}_i + r_{v,i}\big), \qquad \hat{x}_{i,k+1} = x_{i,k} + \tfrac{T}{2}\big(\dot{x}_{i,k} + \dot{\hat{x}}_{i,k+1}\big) + r_{x,i}, \tag{5.8}$$

Here $c_{v,i}$ is viscous friction, m_i is effective modal mass and $\ell_i(\boldsymbol{h}_k)$ is the learned configuration-dependent load. Physical parameters are trained in log-space inside fixed clamps. Several settle on a clamp bound and are therefore fitting coefficients rather than measured constants: modal mass and viscous friction both reach their upper clamp of 10^7 (in kilograms and in newton-seconds per metre respectively), the piston area reaches its lower clamp of $1\,\mathrm{cm}^2$ against a $64\,\mathrm{cm}^2$ initialisation, and the supply pressure separates onto both clamps, with two cylinders at 1 MPa and two at 50 MPa. The same values appear in every checkpoint of this family. Pressure and velocity residuals are bounded to 10 % of their step limits, and the position residual to 1 cm, capping the magnitude each head may add per step.

The explicit chain and the neural heads contribute to the transition in very different proportions. With the residual heads inactive the explicit chain alone produces no net travel, its velocity increments falling six orders of magnitude below their nominal values: at the fitted parameters the friction term $c_{v,i}\dot{x}_i$ reaches 1.0 MN at 0.1 m/s, against a pressure force $\Delta\hat{p}_i A_i$ of about 10 kN at 100 MPa. Realised displacement is produced by the residual heads. The pressure equation nonetheless remains active and responsive to its own parameters: with the heads inactive, halving $k_{\ell,i}$ still moves Δp by 63 % of its range. Hydraulic parameters therefore reach the predicted motion indirectly, through the pressure state entering the recurrent encoder, rather than through the force balance. The model has a grey-box structure. Its equations fix the transition form and keep it differentiable in named hydraulic quantities, but they do not identify the machine's physical constants. Put bluntly: at these clamped values the explicit chain is a regulariser on the shape of the residual heads' output, not an identified plant, and the residual heads are what actually moves the cylinder.

Training uses rollout horizons $H \in \{1, 5, 20, 50\}$ and scheduled sampling up to 75 % self-fed [Bengio 15]. Residual penalties retain the physical structure. Logged rod force has zero supervised weight because PhysX generates it at deployment and the export fails the piston-area consistency check.

With the true command, mean one-step R^2 is 0.998 for the internal valve, 0.64 for velocity and 0.97 for extension. Zeroing or shuffling the command reduces velocity R^2 below zero and extension to about 0.5; the extension Jacobian is diagonally dominant with condition number 2.14. At 5 s, terminal extension R^2 is 0.998, 0.998, 0.84 and 0.91 across the

cylinders. Pressure-increment R^2 is only 0.06–0.09, so the internal pressure is a dynamic state, not a calibrated pressure estimate (Figure 5.5), consistent with the source Simscape pressure signal's own wrong-level ceiling (Section 3.4.4) and with the AF residual's own deployed pressure state carrying shape rather than level (Appendix B.5). Any later use of this pressure channel, including the seal-condition decoding of Section 5.5.1, inherits this limit and is read accordingly.

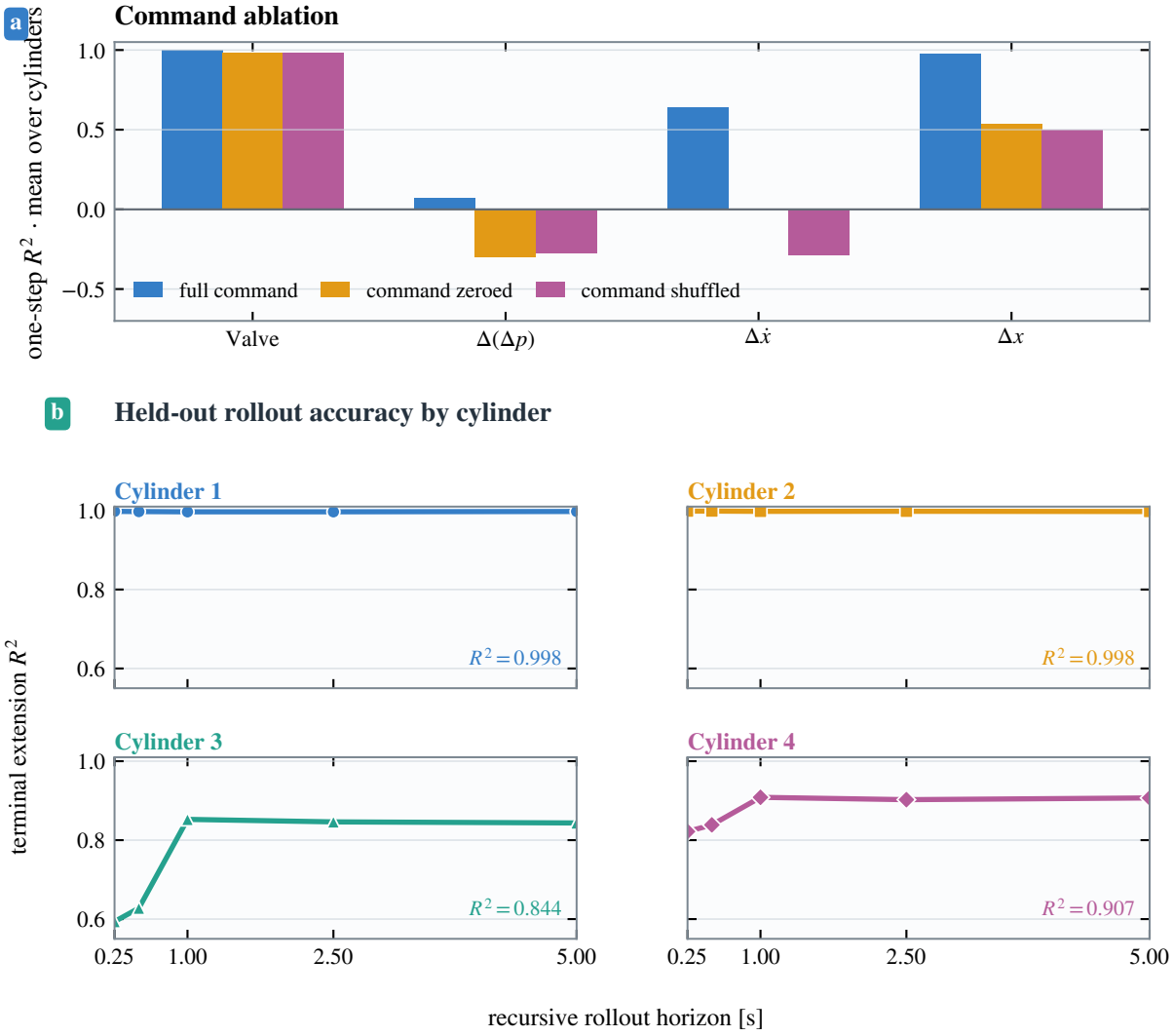

Figure 5.5: Offline validation of the command ROM on held-out runs. (a) One-step R^2 per output group, averaged over the four cylinders, with the full desired command, with the command zeroed, and with the command shuffled across the batch. The collapse of the velocity and extension scores without the true command demonstrates command authority. (b) Terminal extension R^2 of recursive rollouts up to 5 s, shown separately for each cylinder so coincident high-accuracy traces remain visible.

5.4.1 Closed-Loop Calibration Against the Deployment Plant

The Simscape-trained ROM has a systematic extension offset when driving PhysX. The promoted checkpoint adds a bounded, command-gated correction:

$$\hat{x}_{i,k+1} \leftarrow \hat{x}_{i,k+1} + \delta_i(\boldsymbol{h}_k, \boldsymbol{c}_k), \qquad |\delta_i| \le 3\,\mathrm{mm}, \qquad \delta_i\big|_{\boldsymbol{c}_k=\boldsymbol{0}} = 0, \tag{5.9}$$

It is trained on 4 084 IsaacLab transitions from five reachable reference shapes, with Simscape replay. All hydraulic and integration parameters remain frozen. The gate sets the correction to zero for a commanded hold.

The calibrated command ROM serves as the deployment plant in Section 3.4.1. Unlike the ID–FD path, it has seen PhysX data. It is frozen and has no latent input. The earlier prototype assessed in Sections 5.5 and 5.6 modifies the inverse path. Final AF instead omits the inverse model and corrects the frozen actor's valve command, as described in Section 6.7.

5.5 Offline Plant-Context Identifiability Study

This section describes the encoder trained offline against the plant context of Section 4.11, on the frozen collection of Section 4.7. Its purpose is evidentiary and diagnostic: it establishes that plant identity is recoverable from causal observable history at all, and it is the instrument used to judge whether the inverse model can be trusted to carry a correction. It is not a deployed component, and nothing in AF consumes it.

The offline AAM estimates $\mathbf{z}_{\mathrm{AAM}} = \mathrm{AAM}(\boldsymbol{H}_t)$ from deployment history. Each row contains five joint positions, five joint velocities, four cylinder pressure differences and four previous valve commands. A four-layer, width-64 causal convolution processes 20 rows (1.0 s, 360 values).

The pressure values are recursive actuator-model states, not measurements: they come from the command ROM in R and the frozen FD in AF. The factor vector, true latent, reference, future rows and hidden state are excluded. Using the previous valve command keeps each row causal. Validation selected the shortest competitive history from 5, 10, 20 and 40 ticks. Latent RMSE is 0.802 on validation and 0.734 on reserved test.

Table 5.3 collects the trainable size of the four learned models introduced in this chapter, counted directly from each deployed checkpoint's state dictionary rather than estimated from the architecture description. ID and FD share the same 512-wide, four-block residual-MLP family and are consequently within eight parameters of each other despite different input and output widths. The command ROM and the offline AAM are both roughly two orders of magnitude smaller: the ROM's explicit hydraulic chain carries the interpretable structure so its recurrent core need not, and the AAM is deliberately a small probe rather than a capacity-rich estimator, so a positive result could not be attributed to head size alone.

Table 5.3: Trainable parameters of the four learned models, counted from the deployed checkpoint state dictionaries.

Model	Role	Architecture	Parameters
ID	valve-command increment	512-wide, 4-block residual MLP	2 124 292
FD	extension, velocity, pressure	512-wide, 4-block residual MLP	2 124 300
Command ROM	command-conditioned actuator	GRU 128, 2 layers, + hydraulic chain	196 220
Offline AAM	plant-context latent encoder	4-layer causal conv., width 64	42 562

Inference runs at 20 Hz inside the 60 Hz control loop and a causal five-prediction rolling mean adds 250 ms lag. Both belong to this offline encoder, so neither applies to the adaptive result of Chapter 7.

5.5.1 Information Content of the Estimated Latent

Linear decoding from the per-run mean estimate gives $R^2 = 0.956$ for seal condition, 0.448 for compliance, 0.114 for flow, 0.081 for supply pressure, 0.056 for centre-of-mass offset and -0.008 for mass. The latent therefore represents mainly seal condition. It is stable within a run but does not refine over time.

Low-dimensional adapters can lose information when factors act through different channels [Liu 25]. This differs from the rapid-motor- adaptation assumption that only task-relevant factor directions need to be retained [Kumar 21]: here the dominant seal-condition direction acts through pressure, which the controller does not integrate. Estimating named physical quantities would improve interpretation but requires an explicit plant model [Hu 26].

5.6 Why the Correction Does Not Act on the Inverse Command

An earlier prototype conditioned a bounded correction on $\mathbf{z}_{\mathrm{AAM}}$ and added it to the frozen inverse model's valve command. It is reported here in short because its negative result is what fixes the architecture of Chapter 6, not because the mechanism is carried forward.

The frozen ID produces $\boldsymbol{u}_{\mathrm{base}} = \mathrm{ID}(\boldsymbol{s}, \boldsymbol{r})$ and the adapter adds a residual bounded to 0.15:

$$\boldsymbol{u} = \mathrm{clip}\Big(\boldsymbol{u}_{\mathrm{base}} + g_\psi(\boldsymbol{s}, \boldsymbol{r}, \boldsymbol{u}_{\mathrm{base}}, \mathbf{z}_{\mathrm{AAM}})\Big). \tag{5.10}$$

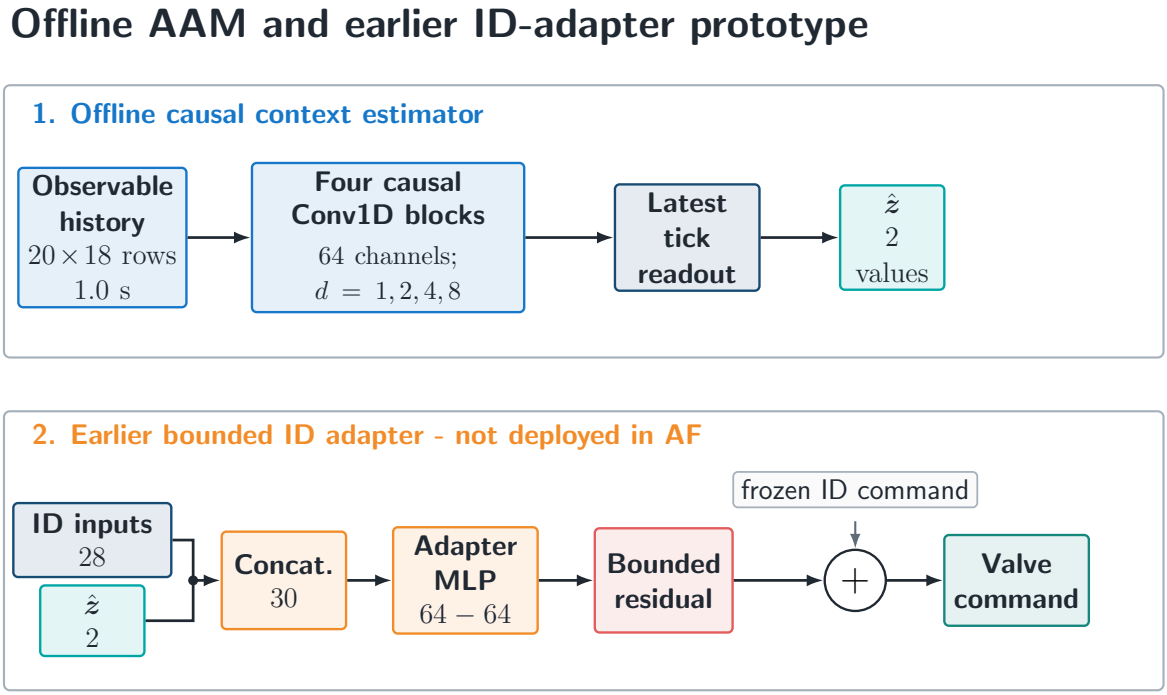

Figure 5.6: The offline AAM and the ID-adapter prototype. Four causal convolutional blocks map a 20-step history of 18 observable features to the two-value estimate $\hat{z}$, and the bounded adapter combines it with the ID inputs to correct the frozen ID valve command. Both are diagnostic. The deployed stack omits the inverse model and conditions its frozen actor on an estimated payload instead (Section 5.7).

5.6.1 The Information Is Present and Usable

On the adapter's own held-out objective, both a privileged and an estimated latent beat a zeroed one in 5/5 seeds, differenced within seed. The effect is consistent but small, and the ordering between the two latent arms is not resolved at five seeds. What the count supports is that a causally estimated latent carries information a model consuming it can use, stated at the level of the adapter objective rather than as a tracking result.

5.6.2 It Does Not Survive the Command Path

Closed-loop ablations against zeroed, shuffled, frozen and mean latents, plus a sham plant change, improve none of lateral error, cut-height error or impacts. Valve saturation explains why: cylinder 1 is saturated for about 92 % of the scored window, cylinder 2 for 67–82 % and cylinder 4 for 50–57 %, and control-effort ratios across the plant change are 1.005–1.009. A bounded additive correction has little authority under those conditions [Sofrony 24]. Reducing the lead cap lowers cylinder-1 saturation from 0.91 to 0.14 but triples cut-height error, so the task sits close to the plant capability limit (Section 3.6.2).

5.6.3 Where the Chain Breaks

Under the frozen protocol with paired bootstrap intervals, neither the true nor the estimated latent improves extension or velocity over a context-free surrogate. Substituting privileged context for the causal estimate recovers nothing, so estimation error is not the bottleneck this test resolves. Every stage up to the command holds and the benefit stops at delivery, against a command already resting on its clip.

Two consequences follow, and both are structural rather than a matter of tuning. A better estimator or a wider latent does not address a limit of this shape, so Chapter 6 removes the inverse model from the adaptive path and leaves the policy commanding the valve directly, where authority remains. And an unconstrained latent is the wrong thing to deliver even once authority exists: what this one carries is mainly seal condition, which acts through pressure, the channel the reduced-order model reproduces worst and the one the controller does not integrate. The deployed mechanism therefore conditions on a single named quantity the controller can act on rather than on a learned code.

5.7 The Deployed Online Payload Estimator

The quantity the deployed controller conditions on is the cutter payload, normalised as $(m - 1)/0.5$ so that a nominal cutter maps to zero. It is estimated at runtime by ridge regression on the 66 task channels of the policy observation, extended with their squares, standardised, and read out as a single scalar. Nothing privileged enters it.

Three choices matter for what the estimate means in the loop. It is fitted *after* policy training, on a mixed-payload rollout of the policy it will serve, because the regression is on the observation distribution that policy induces; each training seed therefore receives its own fit. It is smoothed by an exponential moving average that resets to nominal at every episode boundary, so each episode begins with the machine not knowing what it carries and working it out while driving. And a deadband holds it at exactly nominal until the evidence is strong enough to leave it, which keeps the deployed controller bit-for-bit frozen on a plant that does not need the correction.

The degree-two term is not decoration. A linear fit reads the nominal plant as 1.41×, which tells the policy it is carrying a heavy cutter when it is not; adding the squares brings that to 1.14×. The regularisation strength was not selected on held-out data and is left at its weakest rung, identical across seeds, so it cannot account for differences between them.

What this estimator is worth is not established here. It is established in the closed loop, in Section 7.5, against a frozen arm and against a permuted-payload control.

5.8 Training and Model-Selection Protocol

Adaptation data are split 82/20/16 into training, validation and reserved test by complete run and factor configuration; windows do not cross run boundaries. The 16 test runs are not used for model or hyperparameter selection. SD_z and SD_0 are selected on validation rollout, the offline AAM on downstream rollout, and the earlier ID adapter on validation command error. Table 5.4 records the selected checkpoints.

Table 5.4: Provenance of each model in the modelling chain. Frozen components are loaded unchanged and never receive gradients in this chapter.

Model	Status	Selected checkpoint
Control ID	frozen	paired-control, Section 5.2
Strict FD	frozen	strict-excitation, Section 5.3
Command ROM	frozen	PhysX-calibrated, Section 5.4.1
SD_0 (no latent)	scratch	epoch 58
SD_z + encoder	scratch	epoch 48, frozen after the gate
Offline AAM	scratch	epoch 73, 1.0 s causal CNN
ID adapter	scratch	epoch 116

The ID adapter is trained and selected on recorded valve-command error. Each component is evaluated on its stated training contract. The adaptation residual of AF is deliberately absent from Table 5.4, because it has no supervised checkpoint: its encoder and head are optimised jointly under the reinforcement objective, and their provenance is a training run rather than an epoch number.

With the models identified, audited and frozen, the next chapter builds the controllers that consume them. It assembles the reference and non-adaptive stacks from the models above and introduces the final adaptive stack described in Section 5.6.3.

6. Control Framework and Simulation Integration

Chapter 5 left three actuator representations, each with its own contract and offline evidence. This chapter puts a controller in front of them and turns them into three experiments.

Section 6.1 describes the classical controller, because it is where the competence comes from even though it never runs at deployment. Section 6.2 specifies the policy: what it sees, what it emits, what stops it from emitting something dangerous, and what it is paid for. Section 6.3 explains how the first becomes the second, and reports a result about that transfer which changed the training protocol of this thesis. The individual stacks appear only after those three are fixed.

6.1 The DLS Teacher and Its Two Roles

6.1.1 Scope and Role of the Controller

No damped-least-squares layer runs underneath the policy at runtime and nothing is added to the policy's output; if the policy emits zero, the arm does not move. The DLS controller is nevertheless the reason the policies work at all, and it appears twice, both times at training or collection time and never at deployment (Figure 6.1). It is the *privileged teacher* whose behaviour the policies are distilled from (Section 6.3), and it is the closed-loop controller that drove the identification runs of Section 4.7. Calling it a nominal layer would misdescribe the deployed system and would make every reported result a property of a controller pair rather than of the policy.

6.1.2 Privileged Information

The teacher is *privileged* rather than better tuned: it reads two quantities that no deployable controller on this machine could obtain, namely the exact terrain height directly under the cutter, queried from the simulator rather than inferred from a forward-looking scan,

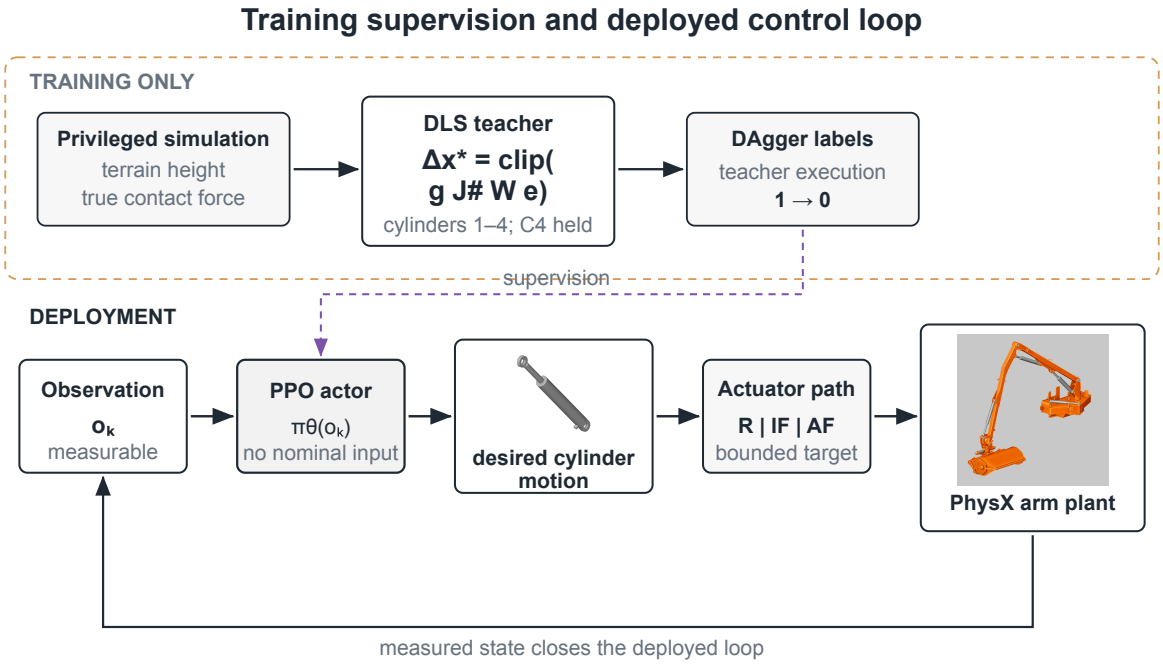

Figure 6.1: Training supervision and deployment. Simulator-only terrain and contact signals drive the DLS teacher, which supplies DAgger labels to the actor. At deployment only the lower feedback loop remains: measured observations pass through the PPO actor, the shared cylinder interface and the selected R, IF or AF actuator path before PhysX updates the arm. No nominal command is added to the policy output.

and the true cutter–ground contact force. Together these turn the hardest part of the task, knowing where the ground is underneath a 1.2 m flail head that is occluding it, into a lookup.

The teacher uses that privilege in a specific way. Rather than commanding a height, it runs a contact regulator: it maintains a vertical offset relative to the ground and adjusts that offset to hold the measured contact force near 250 N within a 100 N deadband, pressing down below the band, backing off proportionally above it, searching downward when contact is lost and retreating immediately on a force spike. The accumulated offset is clamped to $[-0.30, -0.05]$ m so the regulator can neither drive the head into the ground nor lift it clear.

The teacher's advantage is therefore not kinematic. Its inverse is the same DLS solution available to anything; what it has is contact sensing, and that is the quantity the student must infer from a noisy forward scan and its own contact history.

6.1.3 From Task Error to Cylinder Command

The rest of the teacher is conventional and is stated for completeness. A goal point is maintained in the truck frame: laterally it tracks the verge centre line, vertically it follows the measured ground height plus the contact regulator's offset. The task error is the difference between that goal and the measured cutter position, with the lateral component

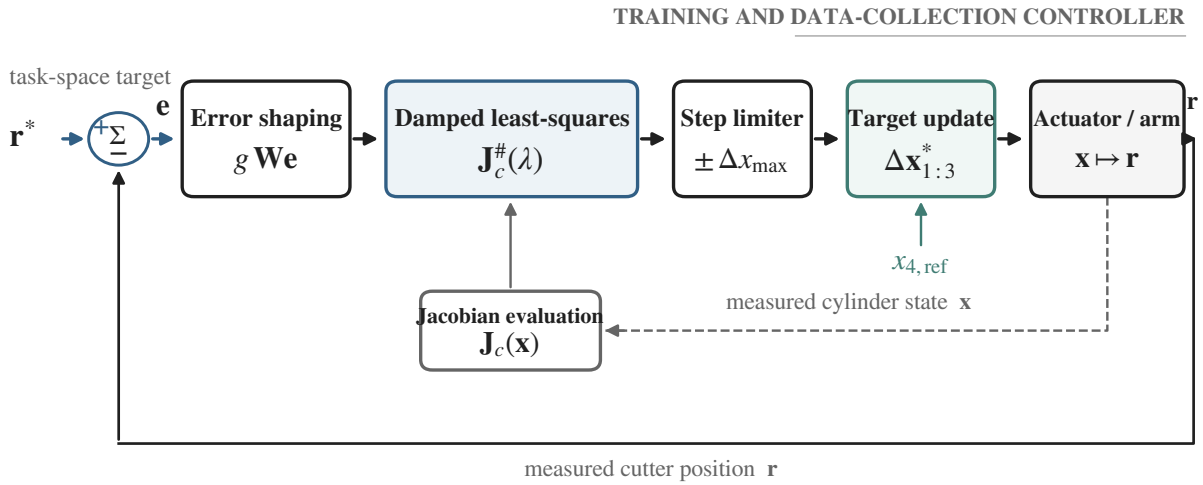

Figure 6.2: Feedback realisation of the damped least-squares teacher of (6.1). The weighted task error is scaled by g, mapped to the first three cylinder coordinates by the measured damped inverse $\boldsymbol{J}_c^{\#}(\lambda)$, and clipped to $\pm\Delta x_{\max}$ per tick. Cylinder 4 is held at its head-down reference. Cutter position closes the task-space loop and the measured cylinder state updates $\boldsymbol{J}_c$ at every step. The loop is used only for teaching and data collection, never at deployment.

weighted by a factor of three, because on this machine lateral error is what the acceptance gate measures and vertical error is largely handled by contact.

That error is mapped to cylinder space through the damped least-squares inverse [Wampler 86, Nakamura 86] of the measured cutter Jacobian $\boldsymbol{J}_c$ of Section 3.2.2,

$$\begin{aligned} \boldsymbol{J}_c^{\#}(\lambda) &= \boldsymbol{J}_c^{\mathsf{T}}\left(\boldsymbol{J}_c\boldsymbol{J}_c^{\mathsf{T}} + \lambda^2\boldsymbol{I}\right)^{-1}, \\ \Delta\boldsymbol{x}^*_{1:3} &= \operatorname{clip}\Big(g\,\boldsymbol{J}_c^{\#}(\lambda)\,\boldsymbol{W}\boldsymbol{e},\ \pm\Delta x_{\max}\Big), \quad g = 0.30, \quad \Delta x_{\max} = 3\,\mathrm{mm}. \end{aligned} \tag{6.1}$$

with $\boldsymbol{W}$ the lateral weighting and the per-step clamp bounding how far any cylinder may be asked to move in one tick. Only cylinders 1–3 are solved. Cylinder 4 is held at the reference extension that keeps the head down, for the reason given in Section 3.1. The gain g is small on purpose: this is a stiff, heavy mechanism in contact, and a resolved-rate controller that closes quickly on the task error simply excites it.

6.1.4 Execution Through the Student Interface

One implementation detail determines whether distillation converges to something usable. The teacher's output $\Delta\boldsymbol{x}^*$ is not written to PhysX directly: it is converted into the same normalised four-value action the student emits and pushed through the student's entire action chain into the same plant, so every label the student imitates is by construction an action it can produce. A teacher writing targets directly to the plant would generate labels outside the student's reachable set, giving the imitation error a floor no training could remove.

6.1.5 Teacher Performance

Quoting the teacher's performance is necessary because distillation cannot exceed it and the gates of Section 3.9 were anchored to it. Table 6.1 reports the privileged teacher

Table 6.1: The privileged DLS teacher, measured on the stage-6 verge task through the command-ROM plant (128 environments, 480 steps, all eleven path families, seed 17). It is competent, not perfect, and several evaluation thresholds sit deliberately just outside what it achieves.

Quantity	Teacher	Gate
Mean $\lvert e_{\text{cut}} \rvert$	0.0234 m	≤ 0.040 m
p95 $\lvert e_{\text{cut}} \rvert$	0.1524 m	—
Mean $\lvert e_{\text{lat}} \rvert$	0.1221 m	≤ 0.300 m
Ground-contact fraction	0.7807	≥ 0.700
p01 head alignment	0.9638	≥ 0.900
Minimum head alignment	0.8317	—
p95 tool acceleration	64.9 m/s^2	≤ 120 m/s^2

measured on the stage-6 verge task through the ROM plant, over 128 environments and 480 control steps.

The teacher is competent but visibly imperfect: it loses ground contact about a fifth of the time and its cut-height tail is six times its mean. This is why Section 3.9 re-derived several thresholds, since a gate set above these numbers could not be passed by the ground-truth controller itself and would carry no information about the policy.

6.2 Policy Formulation

6.2.1 Policy Definition

The deployed controller is a memoryless Gaussian multilayer perceptron mapping a 66-value observation to four normalised actions. Actor and critic are separate networks with hidden widths 256–128–64, ELU activations and empirically normalised observations. PPO [Schulman 17] is supplied by RSL-RL [Rudin 22]. This base policy is shared by all three stacks; AF freezes it and widens the actor input by four plant-context columns, of which only the payload channel carries information here (Section 6.7.3).

The four actions denote desired cylinder increments in R and IF and valve commands in AF (Section 6.7.2). The base policy has no nominal controller beneath it, and the truck's forward motion remains scripted.

6.2.2 The Observation Vector

Figure 6.3 lays out the 66 channels by group. The grouping tells the story of what this controller has to reason about: nine terrain-height samples ahead of the cutter, three task-error channels, sixteen channels of cylinder state and remaining travel, sixteen describing the actuator interface, ten of pose and context, and twelve of model error.

The last twelve channels carry cylinder tracking residual, actuator prediction residual and normalised effort. A memoryless network cannot form these temporal errors from one observation, and without them the error to be corrected is absent from its input. Across ten training blocks, p01 head alignment remained at 0.79–0.80 when only the feedforward source or levelling gain changed.

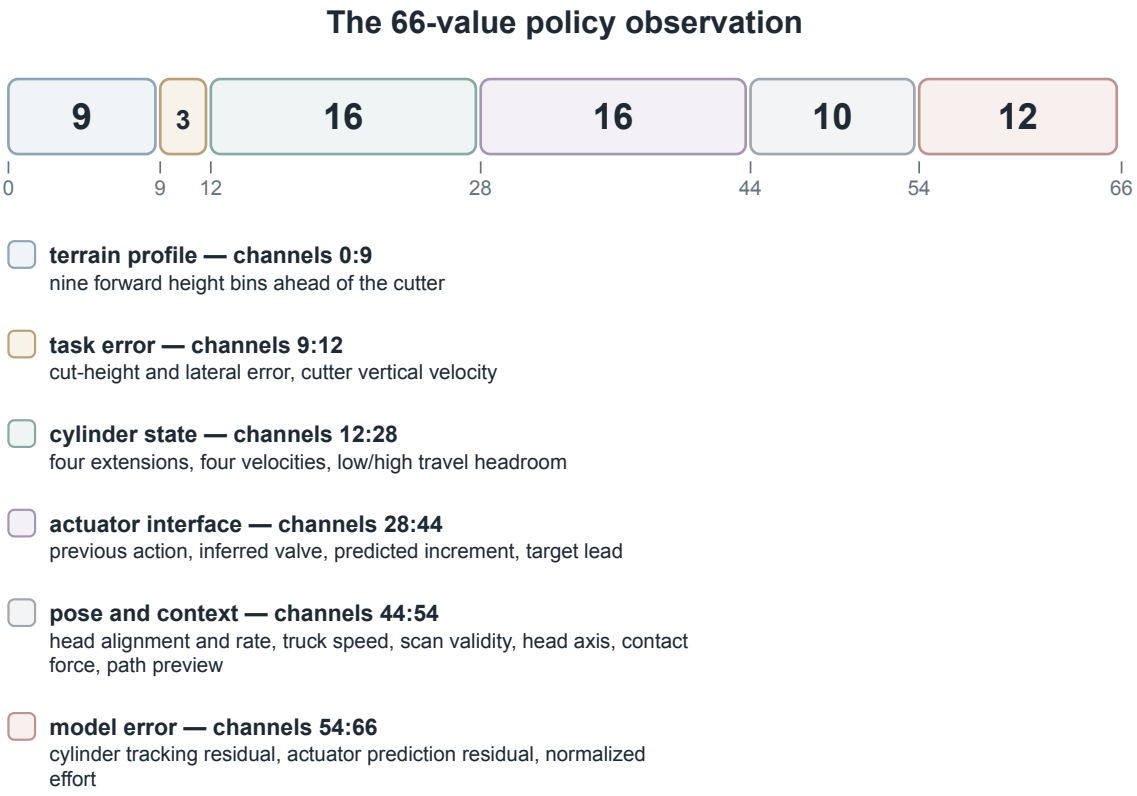

Figure 6.3: The 66-value policy observation, drawn to scale by group. The channels combine measurements, derived quantities and actuator-model estimates. No trajectory-family label or physical factor vector enters the actor. AF appends four context columns to this input, carrying the payload estimated online from these same channels.

This shared contract also limits attribution. The tracking residual is near zero under R but carries the ID–FD path's standing error under IF, giving IF a useful closed-loop correction signal. The R–IF comparison therefore cannot separate the actuator representation from the value of that signal.

6.2.3 The Action and Its Bounds

The policy emits $\boldsymbol{a} \in \mathbb{R}^4$, sampled from its Gaussian at training time and taken at the mean for evaluation. Five bounding stages sit between that sample and the PhysX cylinder target (Figure 6.4). Each can only reduce authority, never introduce motion the policy did not request.

The rate limit bounds $|\Delta \boldsymbol{a}|$ at 0.25 per control step and is held *constant* across every curriculum stage rather than used as a curriculum knob, by the reachability argument above: the teacher's own command stream has a structural per-step bound near 0.20, so a student limit below that makes part of the label set unreachable. The exponential filter ($\beta = 0.70$ at the final stage) smooths the command. The limit shield scales the command down, and only in the direction that consumes headroom, over the last 20 mm of cylinder travel, replacing the teacher's hard clamp into the learned range. The command scale

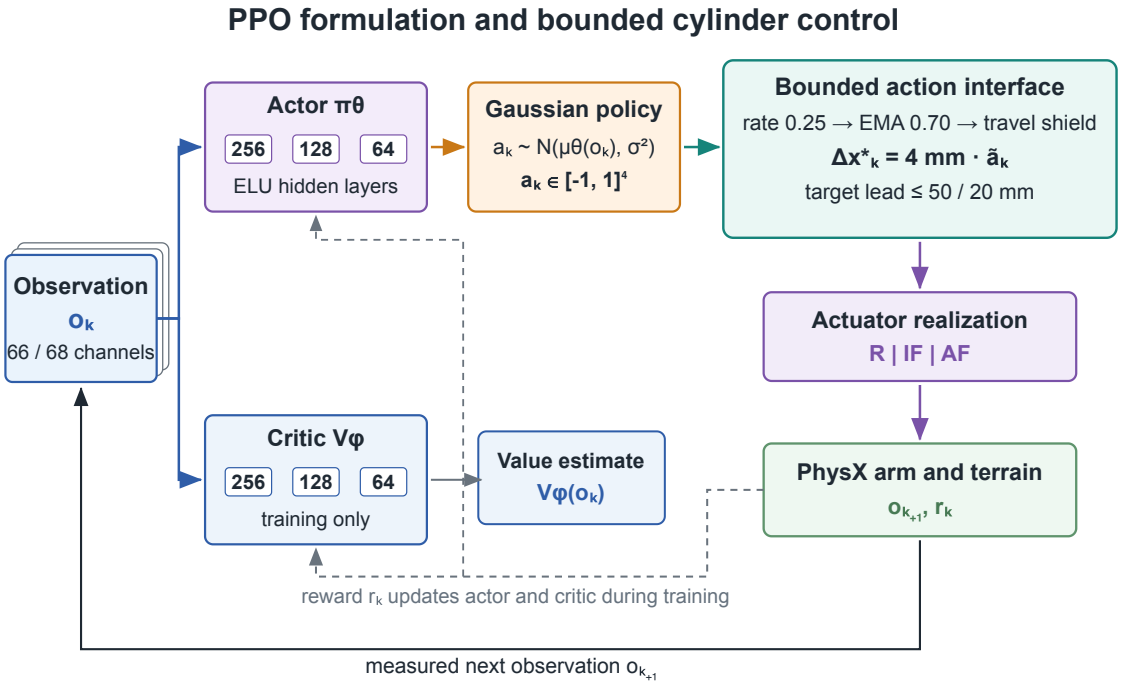

Figure 6.4: PPO actor–critic formulation and deployed control path. The actor maps the observation to a four-dimensional Gaussian action. Rate limiting, exponential filtering, the travel shield, command scaling and the target-lead cap bound that action before the selected R, IF or AF actuator path drives the PhysX arm. The critic and reward update are training-only; measured state closes the deployed loop.

converts the filtered action into 4 mm of cylinder travel per unit action, and the target integrator accumulates the increments in reference space, permitted to lead the measured extension by at most 50 mm on cylinders 1 to 3 and 20 mm on cylinder 4.

This is the cylinder-increment reading of the action, used by R and IF. Under AF the same bounded action is a valve command scaled to span the admissible valve range, so the cylinder-space stages do not carry the same meaning there.

The lead cap is load-bearing. Raising it to 50 mm halved lateral error on `lane_change_hold` (0.4833 → 0.2310) and `boom_sweep` (0.3745 → 0.1602), but applying the same value to cylinder 4 reduced head alignment from 0.9507 to 0.4636. This measured split sets the per-axis limits above.

The chain also fixes the capability ceiling: four millimetres of cylinder travel per step is the entire authority budget, which produces the 0.240 m/s lateral rate limit of Section 3.6.2 from which the reference paths are derived.

6.2.4 The Reward Function

The final R reward has ten bounded terms (Table 6.2): three task rewards and seven behaviour penalties. All retain a gradient beyond the acceptance boundary. Three rules govern them, each adopted after a measured failure.

No Gaussian Shaping

The earlier reward priced tracking as $w \exp[-(e/s)^2]$, which is effectively flat beyond about $2s$ and therefore supplies almost no gradient where the failing path families live. Every shaping term is instead linear or Huber in the error. Writing $x = |e|/s$, the two shapes used are

$$B(e; s, c) = \min(x,\, c), \qquad H(e; s, c) = \min\Big(c,\, \begin{cases} \frac{1}{2}x^2, & x \le 1 \\ x - \frac{1}{2}, & x > 1 \end{cases}\Big). \tag{6.2}$$

B has the same gradient at 0.5 m of error as at 0.05 m, so a policy far from the reference is still told which way to move; H buys quadratic precision near zero where cut height needs it, without a vanishing far field.

Bounded Terms

An unbounded term can dominate the sum and destabilise the critic. Each contribution is clipped, giving the reward a published budget: $6 + 12 + 4 = 22$ for the task terms against -24 for the safety terms. A non-finite value is replaced by -25, just below that worst case, so a numerical fault can never outscore legitimate behaviour.

Fixed Sign Convention

The three task terms are non-negative and the seven safety terms non-positive. This makes worst-case (CVaR) weighting [Rockafellar 00] safe: scaling a non-positive term by a factor ≥ 1 can only lower return. Earlier positive worst-case weights instead rewarded poor environments, while an action-magnitude penalty discouraged the sustained offsets required by three path families.

One apparent inconsistency is deliberate. The tool-acceleration term saturates at 300 m/s^2 while its gate sits at 120 m/s^2, because the reward scale is a saturation scale for a bounded penalty rather than a target: the gate stays the binding criterion and the reward only keeps a gradient against gross excursions. The action-rate term is mild for a related reason, since cylinder 4 owns head pitch and a rate tax falls hardest on the fast corrections that keep the head down.

Every term is logged by name during training, because total reward can rise while the system gets worse. The reward is not the result criterion; promotion uses the eight gates of Section 3.9.

6.2.5 Optimisation

Policies are optimised with clipped-surrogate PPO over generalised advantage estimation, in the form of Section 2.7. The maintained configuration uses clip parameter $\epsilon = 0.2$, discount $\gamma = 0.99$, GAE parameter $\lambda = 0.95$, five learning epochs over four minibatches per update, an adaptive learning-rate schedule targeting a KL divergence of 0.02 from a base rate of 3×10^{-4}, gradient-norm clipping at 1.0, and a clipped value loss with coefficient 1.0. The terrain task collects 32 steps per environment per update. At 512 environments that is 16 384 transitions per update, chosen because the inherited 128-step horizon exceeded two minutes per update on the available GPU without adding useful batch diversity.

Table 6.2: The ten reward terms of the final R task, with weights and shaping scales. B and H are the bounded shapes of Equation (6.2). The lateral and head terms each combine a coarse and a fine scale, so far-field correction and near-field precision are priced separately.

Term	Weight	Shape and purpose
Cut-height tracking	+6	H, scale 0.04 m, cap 2.5; the primary mowing-quality term
Lateral tracking	+12	B, coarse 1.50 m and fine 0.60 m mixed 0.6/0.4; the primary line-following term
Ground contact	+4	pays for the cutter being in contact
Head alignment	−12	B on the deficit below 0.98, fine scale 0.02 and coarse scale 1.00 mixed evenly
Contact loss	−3	charges directly for loss of contact
Action rate	−1	B on the mean absolute command change, scale 0.25; penalizes chatter
Tool acceleration	−1	B, scale 300 m/s^2; mechanical-wear proxy on the cutter
Contact force	−1	B on force above 30 kN, scale 150 kN
Cylinder headroom	−2	B on encroachment inside a 0.02 m margin, scale 0.10 m
Hard impact	−4	indicator on the impact event

The final fine-tuning blocks depart from that configuration because they run under the reference anchor of Section 6.4 rather than an adaptive learning-rate schedule: clip $\epsilon = 0.1$, discount $\gamma = 0.97$, three learning epochs, fixed learning rates of 10^{-5} for the actor and 10^{-4} for the critic, entropy coefficient 0.001, an exploration standard deviation fixed at 0.15 and excluded from the gradient, and frozen observation normalisers.

These departures reflect the locally controlled task and the anchor. The shorter discount credits actions for the terrain state they face, the actor changes more slowly than the critic, and the fixed learning rate avoids competing with the explicit divergence bound. Because zero action leaves the arm stationary, early training used a standard deviation of 0.30 to discover command directions. Distilled terrain policies instead use the smaller fixed value above. Exact run values remain recorded in each frozen manifest.

6.3 Training Protocol and the Distillation Finding

6.3.1 Distillation by DAgger

Policies are initialised by imitation, not by reward. The privileged teacher is rolled out in the student's own environment and the student is fitted to the teacher's raw action by regression, in the DAgger scheme of Ross et al. [Ross 11]: the teacher's probability of *executing* the action, rather than merely labelling it, is annealed from one to zero over the first 70 % of iterations, so later training collects states the student itself visits. Labels are accumulated in a replay buffer of 262 144 transitions and refitted over three epochs per iteration in minibatches of 8 192.

Figure 6.7 shows a complete distillation run for the ROM stack, taken from the stored record. The behaviour-cloning loss falls by more than an order of magnitude while the teacher is still driving, reaches its minimum around the handover, and then *rises* once the student is alone. That rise is the distribution shift DAgger exists to expose rather than a failure: the student is now visiting states the teacher never chose, and the teacher's advice on those states is new information. The closed-loop task metrics remain near the teacher's preflight values without matching them.

6.3.2 Objective Mismatch in Unconstrained Fine-Tuning

Unconstrained PPO fine-tuning from distilled checkpoints increased total reward while degrading the primary tracking objective, in the worst case by a factor of 12.9. The critic remained healthy, with explained variance in the 0.85–0.95 range. This motivates the anchored formulation of Section 6.4, which is used for the final policies.

The reward includes ground contact and cut height as well as lateral tracking, so it can increase while a gate-relevant tracking measure degrades. This is a claim about two functionals over the same policy space rather than about the optimiser. Training maximizes the expected return $J(\vartheta)$, while acceptance is decided by the vector of physical gates $\boldsymbol{G}(\vartheta) \preceq \boldsymbol{g}^\star$ of Section 3.9. Writing $\mathcal{F} = \{\vartheta : \boldsymbol{G}(\vartheta) \preceq \boldsymbol{g}^\star\}$ for the feasible set, the measured content of the finding is

$$\arg\max_{\vartheta} J(\vartheta) \notin \mathcal{F}, \qquad \text{while} \qquad \vartheta_{\text{distilled}} \in \mathcal{F}. \tag{6.3}$$

The teacher places the distilled policy inside $\mathcal{F}$, whereas unconstrained ascent optimises a different functional and may leave it. The protocol therefore reports both checkpoints, logs gate metrics during training and constrains policy drift from the point where reward and evaluation were known to agree.

6.4 Reference-Anchored PPO

Equation (6.3) says that reward ascent leaves the feasible set. It does not say that reward ascent is useless, only that it is unbounded. The construction of this section keeps PPO and constrains how far it may travel from a policy already known to satisfy the gates. Writing π_{ref} for a frozen reference policy and J for the expected return, each training block solves

$$\max_{\vartheta} J(\vartheta) \quad \text{s.t.} \quad \mathbb{E}_{\boldsymbol{o}\sim d_{\pi_\vartheta}}\Big[\mathrm{KL}\Big(\pi_{\text{ref}}(\cdot \mid \boldsymbol{o}) \,\|\, \pi_\vartheta(\cdot \mid \boldsymbol{o})\Big)\Big] \le \delta_{\text{ref}}, \tag{6.4}$$

with the expectation taken over the states the policy itself visits.

6.4.1 Critic Warm-Up

After distillation the actor already behaves sensibly, but the critic has never been fitted to the long terrain return, so the first PPO updates would move a useful actor using a value estimate that is not yet meaningful. The actor is frozen for 150 iterations while only the value loss is optimised, under the asserted invariant $\vartheta_{\text{after}} = \vartheta_{\text{before}}$. The observation normalisers are frozen at the same point, since otherwise the same raw observation is normalised differently from one iteration to the next and the actor's behaviour changes while its weights do not.

6.4.2 The Reference Anchor

Let π_{ref} be a frozen copy of the accepted warm actor. It is queried only on states the live policy visits and never supplies an executed action, so PPO remains the sole controller of the plant. The penalty is the *forward* divergence, which for diagonal Gaussians is available in closed form [Cover 06],

$$D_{\text{ref}} = \mathrm{KL}\big(\pi_{\text{ref}} \,\|\, \pi_\vartheta\big) = \sum_i \left[\log \frac{\sigma_{\vartheta,i}}{\sigma_{\text{ref},i}} + \frac{\sigma_{\text{ref},i}^2 + (\mu_{\text{ref},i} - \mu_{\vartheta,i})^2}{2\sigma_{\vartheta,i}^2} - \frac{1}{2}\right], \tag{6.5}$$

at the cost of one additional forward pass and no sampling. Because both scales are fixed at $\sigma = 0.15$, this collapses to a pure quadratic on the drift of the commanded action,

$$D_{\text{ref}} = \sum_i \frac{(\mu_{\text{ref},i} - \mu_{\vartheta,i})^2}{2(0.15)^2}, \tag{6.6}$$

which is the second reason for fixing σ: with a free standard deviation, entropy collapse shrinks D_{ref} while the commanded action moves, so the constraint would report compliance during the drift it exists to prevent.

Forward KL penalises abandoning any reference behaviour; reverse KL could collapse onto a subset, reproducing the observed trade between competencies. The term is therefore a behaviour-cloning anchor on the live policy's states, following demonstration-regularised policy gradient [Rajeswaran 18].

The actor objective is

$$\mathcal{L}(\vartheta) = \mathcal{L}_{\text{PPO}} + \mathcal{L}_V - \beta\,\mathcal{H}(\pi_\vartheta) + \lambda_{\text{ref}}\,\mathbb{E}_k\Big[\mathrm{KL}(\pi_{\text{ref}}\|\pi_\vartheta)\Big], \qquad \lambda_{\text{ref}} = 0.10. \tag{6.7}$$

6.4.3 Trust Regions and Rollback

A penalty is not a constraint: at a fixed coefficient, a sufficiently large advantage always buys its way past. Feasibility is therefore also verified on the update taken. Two bounds are imposed,

$$D_{\text{update}} = \mathrm{KL}(\pi_{\text{old}}\|\pi_\vartheta) \le 0.01, \qquad D_{\text{ref}} \le 0.05, \tag{6.8}$$

and they do different work. D_{update} is the classical step-size control that keeps the surrogate a valid local model of J. D_{ref} is the new object: because the anchor is *fixed* within a block rather than moving with the policy, it bounds cumulative drift. A sequence of updates each individually satisfying D_{update} can still walk arbitrarily far from feasibility, and that is the trajectory the earlier campaigns took, monotonically, with the reward rising the whole way. Only a constraint referred to a fixed feasible point can observe it.

Enforcement is in two stages. During minibatch optimisation the epoch is early-stopped as soon as D_{update} exceeds its bound, and after the update both divergences are recomputed over the full rollout. If either bound fails, or any quantity is non-finite, the actor and its optimiser state are restored *atomically*. On a divergence rejection the critic fit is retained when finite, since the bounds constrain the policy rather than the value function; on a non-finite update both networks are restored. In the final block, five of one hundred logged updates were rejected, all on the reference bound and none on the update bound, so the drift caught was of the kind ordinary PPO clipping does not see.

6.4.4 The Block Ratchet

The reference is not fixed forever. Within a block it is frozen. When a checkpoint is scored and passes the gates, it is promoted to reference,

$$\vartheta_{\text{ref}}^{(m+1)} = \vartheta^\star, \qquad \vartheta^\star \in \mathcal{F}, \quad \vartheta^\star \in B_{\delta_{\text{ref}}}\left(\vartheta_{\text{ref}}^{(m)}\right), \tag{6.9}$$

and the actor's optimiser moments are reset, because momentum accumulated relative to the old anchor is not meaningful relative to the new one. The critic's are retained: it is fitting the same return, and its calibration is expensive to rebuild.

The anchor advances only through checkpoints verified against $\boldsymbol{G}$, bringing the evaluation criterion into training. Distillation supplies the first anchor; the critic is warmed with actor and normalisers frozen; each later block trains within the fixed bound, rolls back rejected updates, scores checkpoints on at least three held-out meshes and promotes only a passing checkpoint. The last promotion is returned. Algorithm B.1 in Appendix B.6 gives the full procedure.

6.4.5 Theoretical Status

The components are standard and citable. The bound on D_{update} is conservative policy iteration and its trust-region descendants [Kakade 02, Schulman 15]; the atomic rollback is that literature's backtracking line search reduced to a single trial point; the penalty in Equation (6.7) is a Lagrangian relaxation of the constrained problem (6.4) at a fixed multiplier, with the rollback supplying the feasibility enforcement a fixed multiplier cannot.

What does not follow is a certificate. Bounding the divergence from a verified policy does bound gate degradation in form, since the chain rule of KL over trajectories, Pinsker's inequality and the boundedness of every gate metric give $|\mathbb{E}_{\pi_\vartheta}[g] - \mathbb{E}_{\pi_{\text{ref}}}[g]| \le 2G_{\max}\sqrt{H\delta_{\text{ref}}/2}$ for a metric g with $\|g\|_\infty \le G_{\max}$ over a horizon H (Appendix B.7 derives this bound step by step). At the production horizon of 2808 steps and $\delta_{\text{ref}} = 0.05$ the right-hand side exceeds sixteen, which for a metric bounded in $[0, 1]$ is vacuous, and there is a further measure mismatch because the constraint is taken under the live policy's state distribution while the chain rule requires the reference's. The construction is therefore a principled restriction of reward ascent to a neighbourhood of a verified policy, not a guarantee: the gates certify each promotion.

6.5 R: Reference Actuator Baseline

R is the reference condition. The policy emits desired cylinder motion, the frozen command ROM converts it to a cylinder target, and PhysX closes the loop. The scripted truck crosses a 26 m strip at a constant speed drawn per episode from 2 to 7 km/h; path families and terrain meshes are distributed across environments.

It is a reference condition rather than a competitor: ROM-plus-PhysX *is* the deployment plant, so R's actuator coincides with the plant. What that does to its tracking advantage is stated in Section 6.9.2.

The policy follows the chapter's three-stage construction: distillation, frozen actor critic warm-up and reference-anchored PPO. Checkpoints 280–329 were scored on held-out meshes against the eight gates, with worst-mesh and, where applicable, worst-family aggregation. Chapter 7 reports the common-protocol evaluation.

6.6 IF: Non-Adaptive Learned Baseline

IF subclasses the R task and changes exactly one thing. The policy still emits desired cylinder motion, but that command is now realised by the frozen inverse model of Section 5.2, which converts it to a valve command, followed by the frozen strict forward model of Section 5.3, which converts the valve command to a predicted extension increment. PhysX receives the same bounded cylinder target as before. Task, terrain families, meshes, observation contract, action chain, reward, gates and the PPO update are inherited unchanged, so the actuator representation is the only free variable between the two stacks.

Robust here means what Section 2.6 means by it: trained across the randomised plant family, applying one fixed response to whichever member it meets, with no online estimation. The policy is not blind to plant variation, since the twelve model-error channels carry a standing residual that is itself plant-dependent. What it lacks is any representation of *which* plant produced them.

Two properties of the composed path shape its behaviour, both established offline. The valve command is accumulated rather than issued absolutely and is clipped to the per-axis limits $[0.95, 0.95, 0.80, 0.80]$, so the actuator path carries its own saturation boundary independently of the action chain. Only the predicted extension increment is integrated; the velocity output is a diagnostic (Section 5.3).

The stored run manifest matches R field for field on actor learning rate, entropy coefficient, seed, terrain seed, environment count, curriculum stage, frozen normaliser and warm-up, which is what makes the pair a controlled comparison.

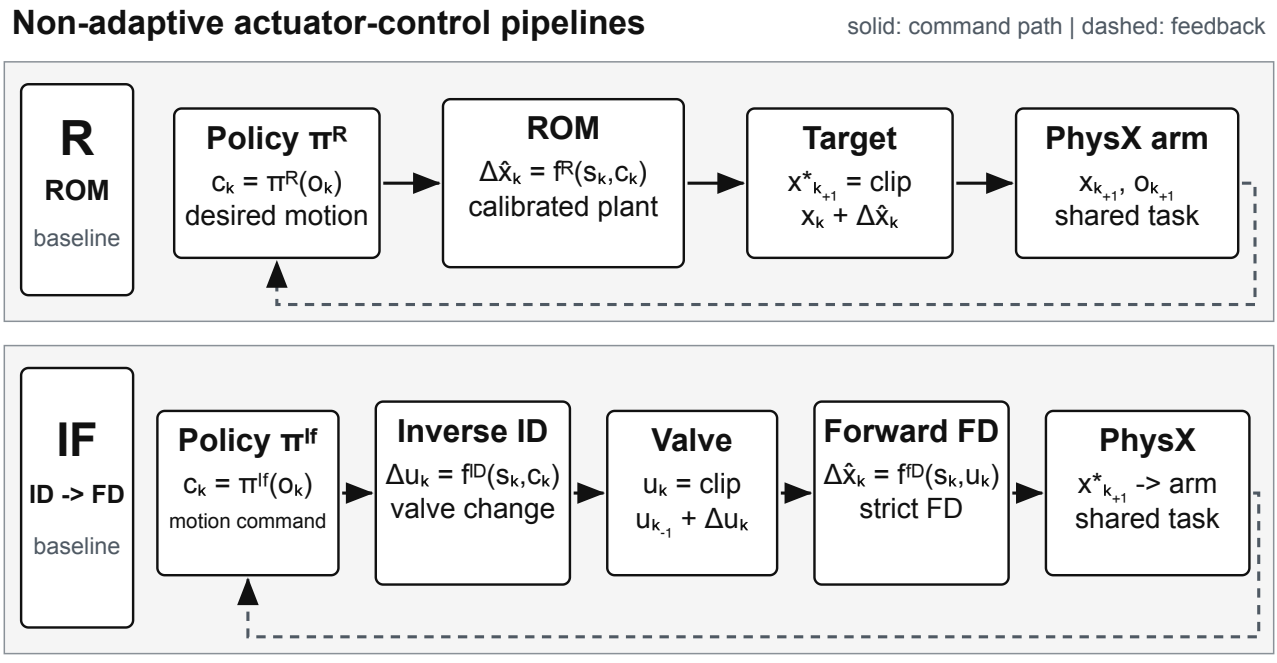

Figure 6.5: The non-adaptive R and IF control pipelines. R sends the policy's desired cylinder motion through the calibrated command ROM. IF composes the independently identified inverse and strict forward models before applying the same bounded cylinder target in PhysX. Both paths use the same observation, action limits and task.

6.7 AF: The Adaptive Stack

Table 6.3 states what AF shares with R and IF and what it does not. AF is not a matched replacement for IF: it answers whether conditioning on an online payload estimate can improve a frozen direct-valve controller once the saturated inverse-model command path has been removed. The matched actuator-representation comparison remains R against IF, and adaptation is evaluated inside AF against its own frozen and wrong-information controls. This is stated once here and not repeated.

Table 6.3: What is matched between AF and the R/IF pair, and what is not. Section 6.9.2 gives the full matching rules.

Dimension	Matched?	Detail
Task, terrain, reward, gates	yes	identical across all three stacks
Base 66-channel observation	yes	AF appends four context columns derived from these channels
Action contract	no	AF emits the valve command directly, not a bounded cylinder increment
Training seed	no	AF trains at a different seed
Actor learning rate	no	lower than the R/IF value
Frozen-actor warm-up	no	AF's base is the already-distilled policy, frozen; there is no separate warm-up stage

AF places online adaptation on the forward model alone. The policy emits the valve command directly, the frozen strict forward model of Section 5.3 converts that command to a predicted extension increment, and PhysX receives the same bounded cylinder target as the other two stacks:

$$\pi_{\text{frozen}}(\boldsymbol{o}, \hat{\boldsymbol{c}}) \longrightarrow \boldsymbol{u} \longrightarrow f_{\text{FD}} \longrightarrow \text{PhysX}. \tag{6.10}$$

The inverse model is absent from the command path. The final AF stack is the frozen direct-valve base actor, four trained context columns carrying the estimate $\hat{\boldsymbol{c}}$, and the frozen strict forward model; the offline AAM of Chapter 5 is not part of it. Zeroing the estimate exactly recovers the frozen base, and permuting it across environments preserves its distribution while destroying its pairing, which are the controlled comparisons used in Chapter 7.

Instrumentation confirms that the direct-valve path applies neither inverse-model valve elements nor adapters at that site. The checkpoint key `id_aam_fd` is retained only for compatibility with earlier runs.

6.7.1 Omission of the Inverse Model

The inverse model belongs in IF and not in the adaptive stack, and the reason is the same measurement in both cases. IF prices the cost of substituting an independently identified pair for the deployment plant, and that question needs the inverse model: a policy commanding desired cylinder motion needs a map from intent to valve command, and supplying that map from identification data alone is the substitution being priced.

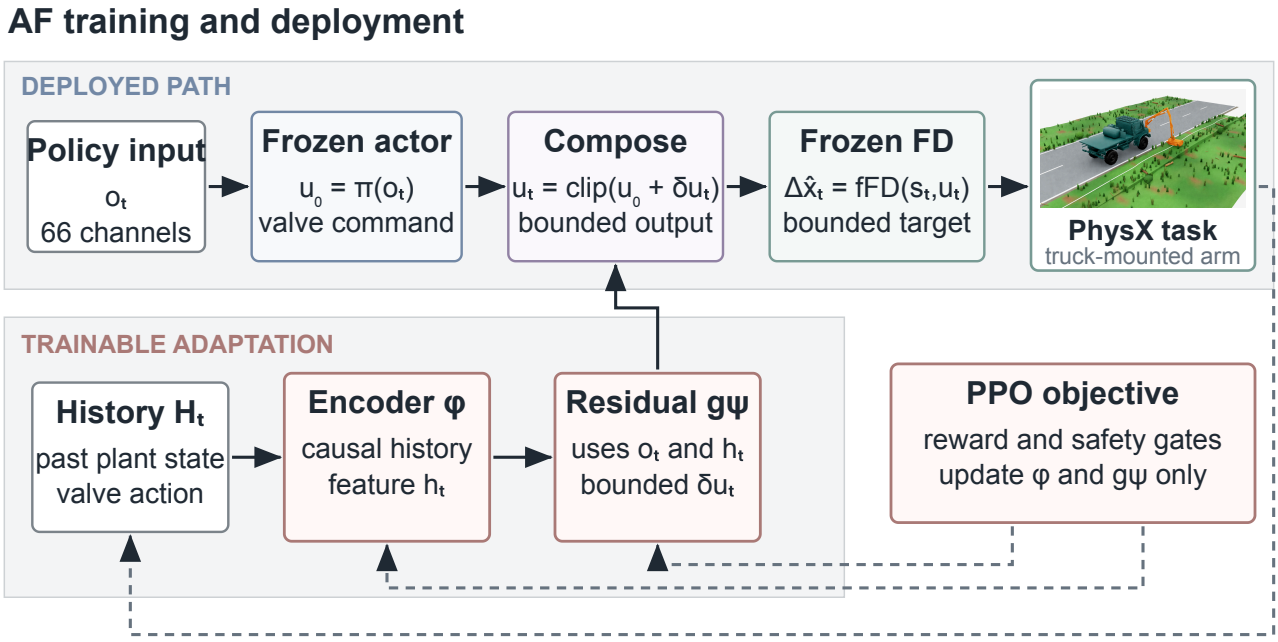

Figure 6.6: Training and deployment path of AF. The frozen base actor emits the valve command $\boldsymbol{u}_0$. A temporal encoder and residual head use causal state–action history to add the bounded correction of Equation (6.11). Only these two modules receive the PPO task gradient; the base actor and strict forward model remain frozen. The composed valve command drives the forward model and the shared PhysX verge-following task. The actor input remains 66 channels; the history window is carried separately and is not part of that count. Dashed paths denote PPO updates and rollout feedback. No privileged factor vector or inverse model enters the AF command path.

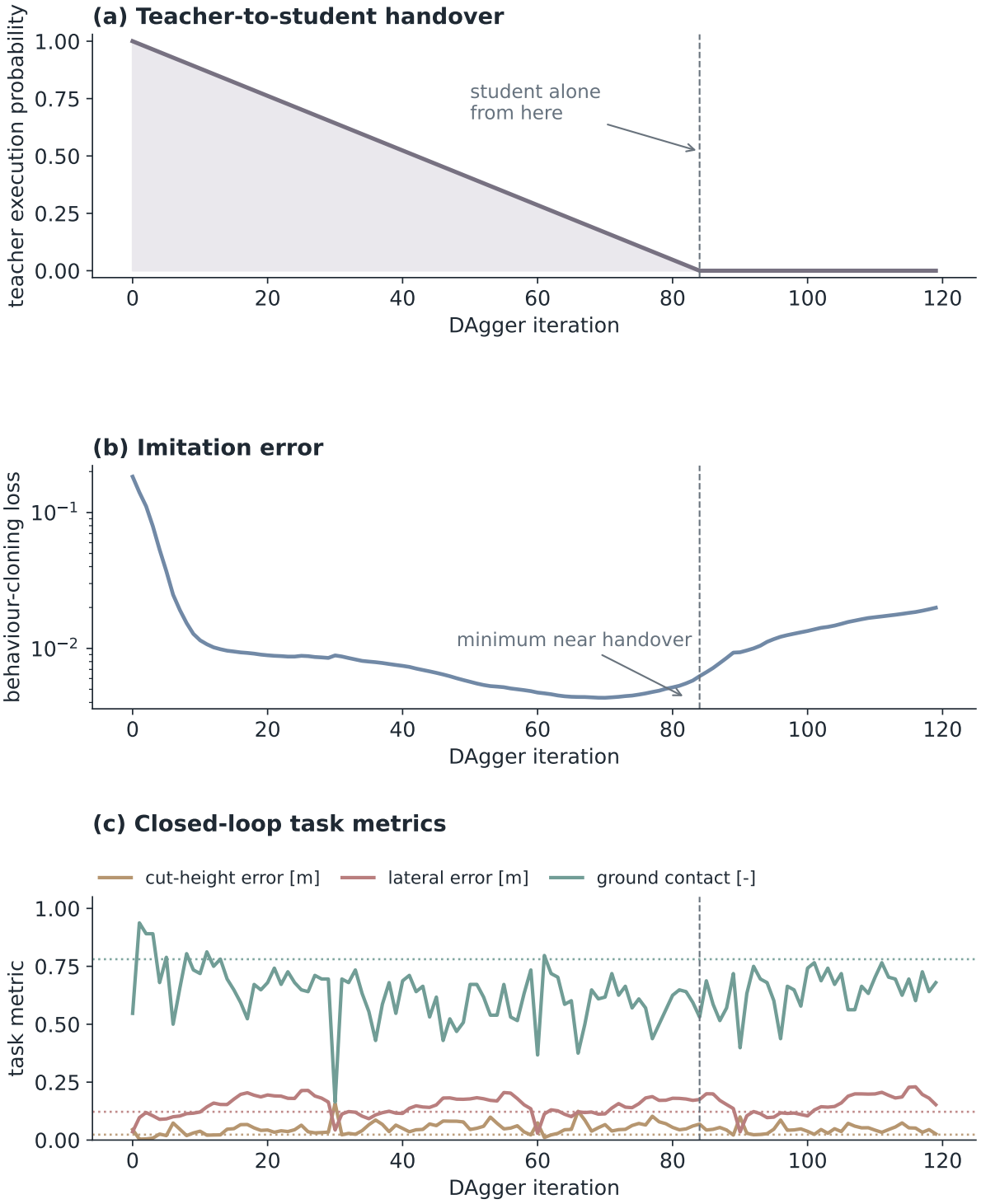

Figure 6.7: A measured distillation run for the R stack. (a) The teacher's execution probability falls to zero over the first 70 % of iterations. (b) Behaviour-cloning loss falls by over an order of magnitude under teacher execution, then rises as the student visits new states. (c) Cut-height error, lateral error and ground-contact fraction; dotted lines mark the teacher's preflight values from Table 6.1.

Adaptation needs something the same map cannot give. Because the model predicts a valve-command *increment* that is accumulated and clipped (Section 5.2), sustained demand walks the command onto its limit and holds it there. Fed its own predictions the composite left the reference in every one of the 118 runs, with the pre-clip command at its limit on 89.1 % of elements while the ROM valve on the same runs never reached its bound. Closed-loop instrumentation over the 33-cell panel says the same thing in adaptation's terms: the command sat at its pre-clip limit on 79.9 % of valve elements, 34.5 % of computed corrections were clipped away, and on 37.9 % of elements the correction could not move the command at all.

The quantity being adapted and the quantity being saturated are therefore the same, which makes a null through that path uninterpretable: it cannot separate a task that is not adaptation-limited from credit clipped before it reached the metric. Nothing here retracts what is claimed for IF, where the command is consumed as issued. The final stack emits the valve command directly, so the conditioning acts on the same bounded command as the policy.

6.7.2 Sufficiency of the Forward Model

Removing one of the two learned models raises the question of whether the remaining one carries the path. The claim is specific: the forward model must convert a commanded valve position into a one-step extension increment with genuine valve sensitivity, because that increment is the only quantity the controller integrates. The evidence is the strict retraining of Section 5.3.1, which raises the valve-zero error ratio from 0.94 to 1.81.

Sufficiency is established on the channel that is used and is not claimed beyond it. Three qualifications carry forward. Only the extension increment is integrated, the velocity output remaining weak on two axes and retained as a diagnostic. Zero-order hold remains competitive below 0.25 s, so the forward model earns its place at the horizons the controller actually spans rather than uniformly. And the valve authority the model represents is strongly operating-point dependent, varying by a factor of 15–75 across pressure quartiles and capable of changing sign, which is why a constant valve Jacobian was rejected.

6.7.3 Conditioning on the Payload Estimate

Adaptation enters through the actor input rather than through its output. The 66-value observation is extended by four normalised plant-context channels, and the actor's first layer is widened to receive them:

$$\boldsymbol{u} = \pi_{\text{frozen}}\big(\boldsymbol{o},\, \hat{\boldsymbol{c}}\big) = \text{base}(\boldsymbol{o}) + \boldsymbol{W}_{\text{ctx}}\hat{\boldsymbol{c}}, \qquad \hat{\boldsymbol{c}} = \mathbf{0} \;\Rightarrow\; \boldsymbol{u} = \text{base}(\boldsymbol{o}). \tag{6.11}$$

Only $\boldsymbol{W}_{\text{ctx}}$ is trained: 1 024 parameters against a 59 588-parameter frozen base, with the input-layer gradient masked to the four context columns and every other parameter left untouched.

The identity on the right is what makes the comparison exact rather than approximate. Nominal is the zero vector, so the frozen arm is not a separately trained control but the same network evaluated at $\hat{\boldsymbol{c}} = \mathbf{0}$, and it is recovered bit for bit. The base task columns are verified identical across training seeds after adaptation training, which confirms the mask held.

The payload channel is normalised as $(m - 1)/0.5$, so a nominal cutter maps to zero and a fivefold payload to eight. The estimate itself is not learned end to end. It is a ridge regression on the 66 task channels, fitted offline on rollouts of the policy it will serve, then applied at runtime and smoothed by an exponential moving average that is reset to nominal at every episode boundary. A deadband holds the estimate at exactly nominal until the evidence is strong enough to leave it, so on a plant that does not need the correction the deployed controller is the frozen one.

6.7.4 Freezing the Base Policy

The base is frozen because anchored fine-tuning suppresses new context dependence, while removing the anchor lets PPO trade line tracking for ground contact. Even a 1000-fold gradient multiplier moved action sensitivity only from 0.001 % to 0.004 % of scale over 400 iterations; masking the anchor reached 0.005 %. Training the context columns against a frozen base avoids that suppression entirely, since the anchor never applies to them.

6.7.5 Training of the Context Columns

The context columns are trained under the reinforcement objective on the true payload, and the estimator that replaces it at deployment is fitted afterwards on rollouts of the trained policy. Separating the two matters: the regression is on the observation distribution that policy induces, so an estimator fitted from one policy does not transfer to another, and each training seed receives its own fit. Nothing constrains which of the four context channels the policy uses. Chapter 7 reports which it selects.

6.7.6 Consistency With the Plant Dynamics

Two consistency requirements follow from the system model of Chapter 3, and both are met by construction.

The first is causality. The estimate is formed from the 66-value policy observation alone, every channel of which the deployed stack already produces at runtime. The privileged factor vector, the true payload and any hidden simulator state are excluded, so nothing enters the deployed path that would be unavailable on the physical machine. The estimate also starts each episode at nominal, which is the sense in which the machine works out what it is carrying while driving rather than being told.

The task reference is not excluded, so the regression could in principle read terrain and task state rather than load. Section 7.5.6 tests that directly by permuting the payload across environments, which leaves the distribution of the context untouched and destroys only its correspondence to the plant it describes.

The pressure differences among those channels are recursive states of the frozen forward model rather than plant measurements. The estimate is therefore faithful only to the extent bounded in Section 6.7.2.

6.7.7 Scope of the Claims

The smoothing rate of the estimate is a tuned hyperparameter, and its effect is not neutral: the gradual arrival of the estimate is part of the mechanism rather than an implementation detail, to the point that supplying the true payload immediately is not an upper bound on what the estimate achieves (Section 7.5). The adaptation claim is made inside AF, on the identified forward model: frozen base, online estimate, and true payload. Chapter 7 separates architecture-level evidence that the conditioning can reach the plant from method-level evidence about whether the improvement replicates across training seeds, and reports the payload regime in which it holds.

6.8 IsaacLab Integration and Timing

Table B.1 lists the concurrent rates: PhysX integrates at 120 Hz, the policy and its payload estimate run at 60 Hz, and the learned actuator models advance at 20 Hz. No single value describes the whole control loop.

The relationship between the 60 Hz and 20 Hz layers is where integration errors hide, so it is fixed by construction rather than by interpolation. A learned actuator model is evaluated on every third control step and its predicted increment is held across the two intervening steps. The action presented to the model is the action of the tick on which it fires, not an average, and the target integrator accumulates in reference space, because integrating against the measured extension turns a static hold into a slow ratchet (Chapter 3).

Within each tick, the encoder consumes history ending at the previous tick before the actuator path runs. Per-environment buffers are cleared on asynchronous resets. A trace replay checks bit-exact agreement with an offline causal recomputation, and checkpoint loading rejects mismatched normalisation, features or history length.

6.9 Constraints and Fair-Comparison Rules

6.9.1 Limits and Safeguards

Authority is bounded by the action-chain rate, travel shield and lead cap, then by per-axis valve limits. The environment terminates on the safety conditions of Section 3.9, and every learned component has a defined fallback for non-finite output.

Preflight also asserts that each reference remains within the measured 0.240 m/s lateral authority ceiling of Section 3.6.2; a faster path would be infeasible regardless of the reward or optimiser.

6.9.2 Matching Rules and Comparison Controls

Two comparisons carry distinct claims; the thesis does not rank all three stacks.

The Matched Pair: R Against IF

The representation claim rests on R and IF, which differ only in the actuator path. Their task, paths, terrain and held-out meshes, reward, action bounds, gates, episode length, networks and training construction are identical. Only this pair supports an across-stack comparison.

One departure is declared. Because the command ROM plus PhysX *is* the deployment plant (Section 5.4.1), R's tracking advantage over IF is not evidence that a reduced-order model approximates the actuator better than an identified pair: one of the two defines the target.

The Paired Comparison: AF Against Its Own Frozen Base

The adaptation claim is made within AF. Its arms share the frozen base, training budget, reward and evaluation conditions, and are the same network evaluated at different context values (Section 6.7.3). Estimated, zeroed and permuted context are compared within seed and paired environment by environment. The primary checkpoint rule uses training reward without consulting held-out performance; the fixed final checkpoint provides a robustness analysis (Sections 7.5.6 and 7.5).

Fine-Tuning Budgets Must Be Matched or Reported

Because reward ascent can move a policy away from the evaluation criterion (Section 6.3.2), fine-tuning budgets must be matched or reported as an additional treatment. The paired AF construction satisfies this rule.

7. Experimental Evaluation: Representation, Robustness, and the Value of Adaptation

This chapter evaluates the closed-loop stacks defined in Chapter 6 as an evidence chain. Sections 7.2 to 7.4 first ask whether the actuator representation matters in closed loop and how much plant variation the trained controllers already absorb. Section 7.5 then asks what remains for a residual to recover inside AF, whether live history explains that recovery, and whether any component varies with cutter mass. Section 7.6 states the conditions under which the evidence supports online adaptation.

7.1 Experimental Protocol

7.1.1 Test Conditions and Comparisons

All closed-loop results use held-out terrain meshes, eleven trajectory families, a truck speed of 4 km/h, a 26 m scoring window, and sixteen reserved plant configurations. The matched comparison is between R and IF. Adaptation is assessed only inside AF, by comparing a frozen base, a capacity-matched residual without history, and a history-conditioned residual.

The results are paired by environment or by mesh and trajectory family. Error metrics are reported separately for lateral tracking and cut height because the two errors describe different aspects of the task. Confidence intervals are percentile bootstrap intervals over the paired units [Efron 93], at $B = 20\,000$ resamples. Where one channel is required only not to get worse, the bound is a non-inferiority margin declared before the comparison [Piaggio 12] rather than a failure to reject. Success rules and checkpoint rules are fixed in writing before the arms are trained, in the sense of a pre-registered analysis [Nosek 18]; the registrations are listed in Appendix B.8. Checkpoints are selected using the validation or training criterion specified for each experiment and are then evaluated once on the held-out set.

Table 7.1: Closed-loop performance over 33 paired held-out conditions. The matched stacks differ only in the actuator representation.

	R	IF
lateral error, mean (m)	0.1026	**0.0703**
lateral error, median (m)	0.0392	**0.0281**
lateral error, p95 (m)	0.4664	**0.2665**
cut-height error, mean (m)	0.0084	**0.0063**
cut-height error, median (m)	**0.0013**	0.0031
ground contact	**0.9623**	0.9529
hard impacts	0.0947	**0.0905**
tool acceleration p95 (m/s^2)	39.90	**39.02**

7.1.2 Comparative Scope and Scoring

R and IF form the cross-stack comparison. AF has a different action contract and is therefore not ranked against that pair. Its results are interpreted only through within-stack comparisons.

One training run exists per stack, so the cross-stack numbers describe two deployed controllers and carry no estimate of policy-training variance. The matched pair differs only in the actuator path, which is what supports the representation claim; the seed-level statements in Section 7.5 come from five runs of a single stack and are paired within seed.

7.1.3 Scoring Horizon

All reported closed-loop results use the full 26 m trajectory rather than a shorter selection window. The earlier 8.9 m selection window covered only about one third of an episode and is therefore treated as an evaluation-protocol limitation, not as evidence about adaptation.

7.2 Closed-Loop Task Performance

The first comparison reports closed-loop tracking of R and IF on the held-out meshes and trajectory families.

The task is scored on two independent error channels, drawn side by side in Figure 7.1: lateral error is deviation from the planned cutting line in the horizontal plane, and cut-height error is deviation from the intended vertical cut. They are kept separate throughout because a controller can improve one while worsening the other. Ground contact, impacts and tool acceleration are reported alongside them as checks on motion quality.

Table 7.1 compares the matched pair over 33 held-out (mesh, family) conditions. Median errors are low for both stacks, and the p95 values move in the same direction as the means, so the improvement also reduces the larger tracking deviations rather than only the average.

Figure 7.2 resolves the aggregate by trajectory family. IF reduces mean lateral error from 0.103 m to 0.070 m and mean cut-height error from 8.4 mm to 6.3 mm over the 33 paired conditions. The advantage is not uniform: IF is better on six of the eleven families and R

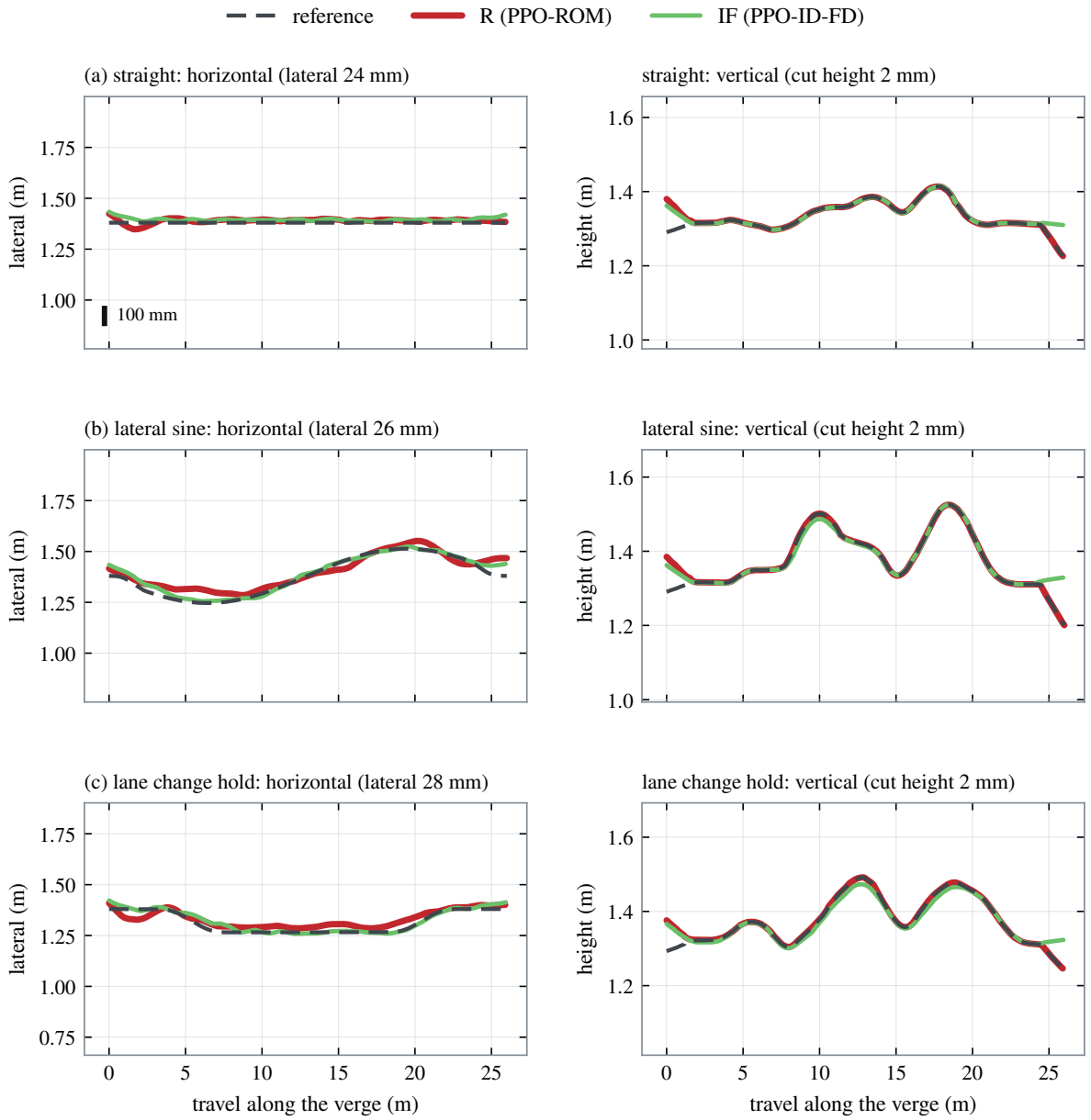

Figure 7.1: Cutter path against the reference on held-out mesh 1337, for three structurally different trajectory families: a flat line, a smooth periodic sweep and a step-and-hold. The left column is the horizontal plane and the right column the vertical plane over the same drive. The vertical reference is not recorded directly and is recovered as $z_{\text{ref}} = z_{\text{cutter}} - e_{\text{cut}}$, so both columns compare reference against actual. In the vertical panels the reference is largely hidden beneath both traces, which is what a millimetre-scale cut-height error looks like drawn at path scale.

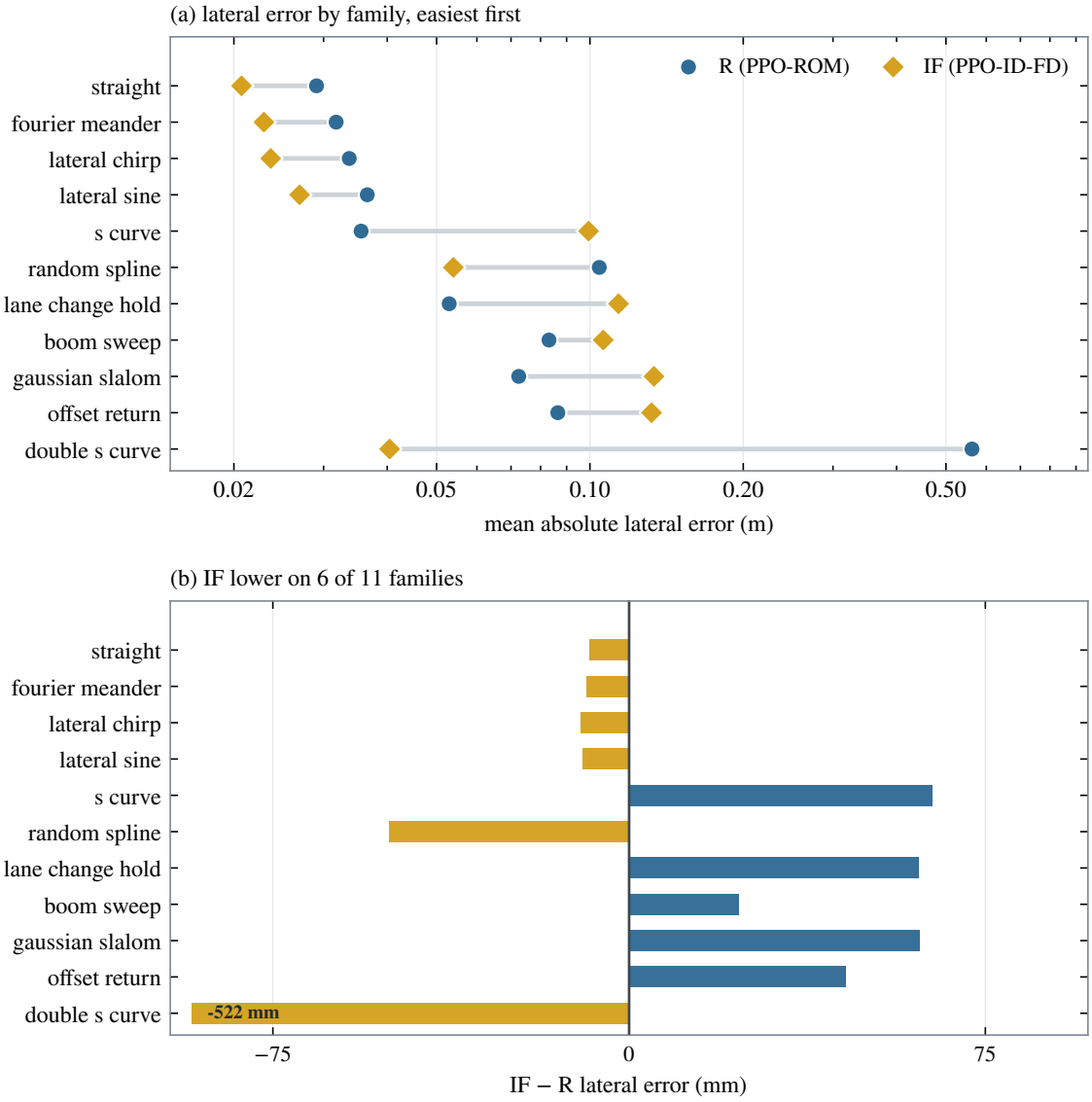

Figure 7.2: Lateral error by trajectory family for the matched pair. The aggregate difference is concentrated in the more difficult families.

Table 7.2: Paired difference between the matched stacks over the 33 held-out cells, in millimetres. A positive value means IF has the lower error. Intervals are percentile bootstrap over the paired cells, $B = 20\,000$. The leave-one-family-out row gives the aggregate that is furthest from the full-set mean.

	lateral	cut height
mean paired difference	+32.3 [−25.8, +98.8]	+2.1 [−3.8, +9.2]
median paired difference	**+9.3** [+7.0, +12.1]	**−1.7** [−2.3, −1.3]
cells with IF lower	26/33	6/33
families with IF lower	6/11	2/11
leave-one-family-out mean	−16.7 [−52.8, +16.2]	−3.2 [−6.4, −0.2]

on five, with the largest difference on the double S-curve. Both stacks follow the simpler paths closely.

That distribution is uneven enough that the mean is the wrong summary, so Table 7.2 reports the paired difference three ways. Three readings follow, and they do not agree. The mean favours IF on both channels, but neither interval excludes zero, so the headline mean is not a resolved difference. The median paired cell favours IF on lateral error by 9.3 mm with an interval excluding zero, and favours R on cut height by 1.7 mm, also excluding zero. The aggregate is carried by one family: the double S-curve contributes 522 mm of lateral difference against ±75 mm everywhere else, and removing it turns the mean lateral difference negative, at −16.7 with an interval crossing zero. A symmetric trim of the two most extreme families leaves −11.5 on lateral and −2.4 on cut height, both crossing zero.

The claim the data supports is therefore narrower than the means suggest. IF has the lower lateral error in the median cell and in 26 of 33 cells, and the advantage is concentrated on the demanding families rather than spread across them. R has the lower cut-height error in the median cell. No aggregate statement about the mean advantage survives removal of the double S-curve, and none is made here. Section 7.1.2 states the further limit that both stacks are one training run each, which no amount of paired cells addresses.

7.3 Actuator Representation and Dynamic Response

The same matched pair is now examined in the actuation channel: transport delay, closed-loop step response, and command effort.

7.3.1 Plant Delay and Step Response

Open-loop excitation measured a command-to-pressure delay of approximately 117 ms. Cylinder extension and joint motion began at the same time, which confirms that the dominant delay is hydraulic rather than kinematic. The similar values across excitation amplitudes indicate that this delay is a stable property of the simulated actuation chain.

The commanded-step experiment measures the closed-loop response to a 100 mm change in the reference line. Rise time is measured from 10 to 90 % of the step. Settling time is the first point at which the error remains within 20 mm for 0.5 s.

IF settles about 0.45 s earlier, whereas R has lower overshoot. The command rate limit alone requires at least 417 ms to complete this step, and both stacks require more than 2 s

Table 7.3: Transport latency along the actuation chain, in milliseconds. Values are medians over open-loop excitation runs at 60 Hz.

	excitation amplitude			
Link	0.05	0.15	0.35	0.70
command → differential pressure	117	117	117	150
command → cylinder velocity	183	167	133	117
command → cylinder extension	117	117	100	100
command → joint position	117	117	100	100

Table 7.4: Response to a 100 mm commanded step. Values are median times over sixteen environments.

Stack	detected response	rise	settled	overshoot
R	333	617	2633	57 %
IF	350	717	2183	73 %

Times are in milliseconds.

to settle. Reducing hydraulic delay alone would therefore not remove the observed settling time; the command rate, the policy response and the reference handling would have to change with it.

Closed-loop settling is much slower than the hydraulic delay. Command-rate limits and policy dynamics dominate the response time. The actuator path changes the shape of the step, not the main settling time.

7.3.2 Control Effort

The reversal count identifies rapid changes in command direction that the root-mean-square magnitude does not reveal, and Figure 7.3 pairs each stack's effort with the tracking error it buys. A lower RMS command is not preferable when it comes with frequent reversals and larger path error.

R uses a smaller command magnitude but reverses almost four times as often. IF uses more actuation, reverses less, and tracks more closely. Effort is therefore reported together with the tracking it buys.

Table 7.5: Control effort on held-out mesh 1337, over 16 environments and 1404 steps. The command and its rate are in normalised command units, so only their ratio between stacks is meaningful. Bold marks the better value of the pair. The lateral errors in the last column are this mesh's, not the 33-cell aggregate of Table 7.1.

Stack	RMS command (norm.)	command-rate RMS (norm. s^{-1})	reversals (s^{-1})	lateral error (mm)
R	0.586	7.55	40.0	120.1
IF	0.827	10.68	**10.6**	**26.8**

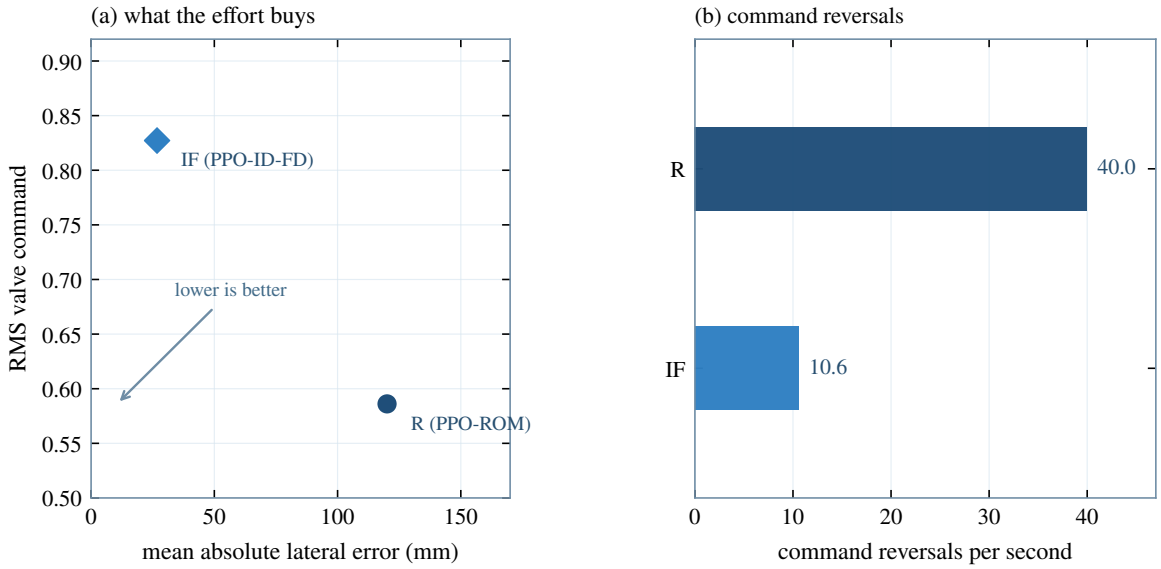

Figure 7.3: (a) Control effort against the tracking error it buys for the matched pair; the lower left is better. (b) Command reversals per second, which the root-mean-square command does not reveal.

This is one mesh, so it is an observation about the recorded traces and not a 33-cell result. The lateral errors it reports, 120.1 mm and 26.8 mm, are further apart than the paired aggregate of Table 7.2, which is what a single condition can do. The reversal ratio is the part worth carrying forward, because it is scale-free and an order-of-magnitude effect rather than a few per cent.

7.4 Generalisation and Robustness

The preceding sections use the held-out evaluation set. This section establishes the robustness already present before adaptation is considered, and then applies a common authority reduction to all stacks.

What is tested here is narrower than the section title, and the limit is stated first because it bounds every number below. Four of the six randomised factors are hydraulic (Table A.3). On the inverse and forward path the hydraulic factors are not applied at all, so the variation that reaches IF is mechanical: cutter mass and centre-of-mass height. On R the hydraulic factors are applied but constrained to the model's identification range. The authority sweep that follows is a generic reduction of realised actuator motion, not plant variation. Nothing in this section therefore supports a claim of robustness to hydraulic variation, and the summary in Chapter 8 is scoped to match.

Within that scope the matched stacks generalise to the held-out meshes, trajectory families and reserved plant configurations used in this chapter, and the randomisation sweep shows little change in tracking over the tested mechanical variation.

The held-out evaluation therefore establishes transfer within the tested simulation population. It does not establish performance on arbitrary hydraulic conditions or physical hardware.

7.4.1 Delivered Stress

The sweep therefore supports robustness to the tested changes in tool mass and centre-of-mass height. It does not establish robustness to unbounded hydraulic variation. A common authority perturbation was used to compare all stacks more directly. This separates a reduction in realised actuator motion from the specific randomisation factors used during deployment.

7.4.2 Actuator Authority

Actuator authority was reduced by scaling the realised actuator displacement:

$$\Delta x_k^{\text{applied}} = \kappa \, \Delta x_k^{\text{actuator}}, \qquad \kappa \in (0, 1]. \tag{7.1}$$

Here, $\kappa = 1$ represents nominal authority and smaller values reduce the motion delivered to the plant. The same perturbation is applied after each stack's actuator representation, which makes the comparison independent of the individual factor parameterisations.

Table 7.6: Authority at which median lateral error first reaches 1.5 times its full-authority value.

Stack	nominal error (mm)	threshold (mm)	onset κ
R	40.69	61.04	0.10
IF	23.66	35.49	0.20
AF frozen	45.18	67.78	0.10

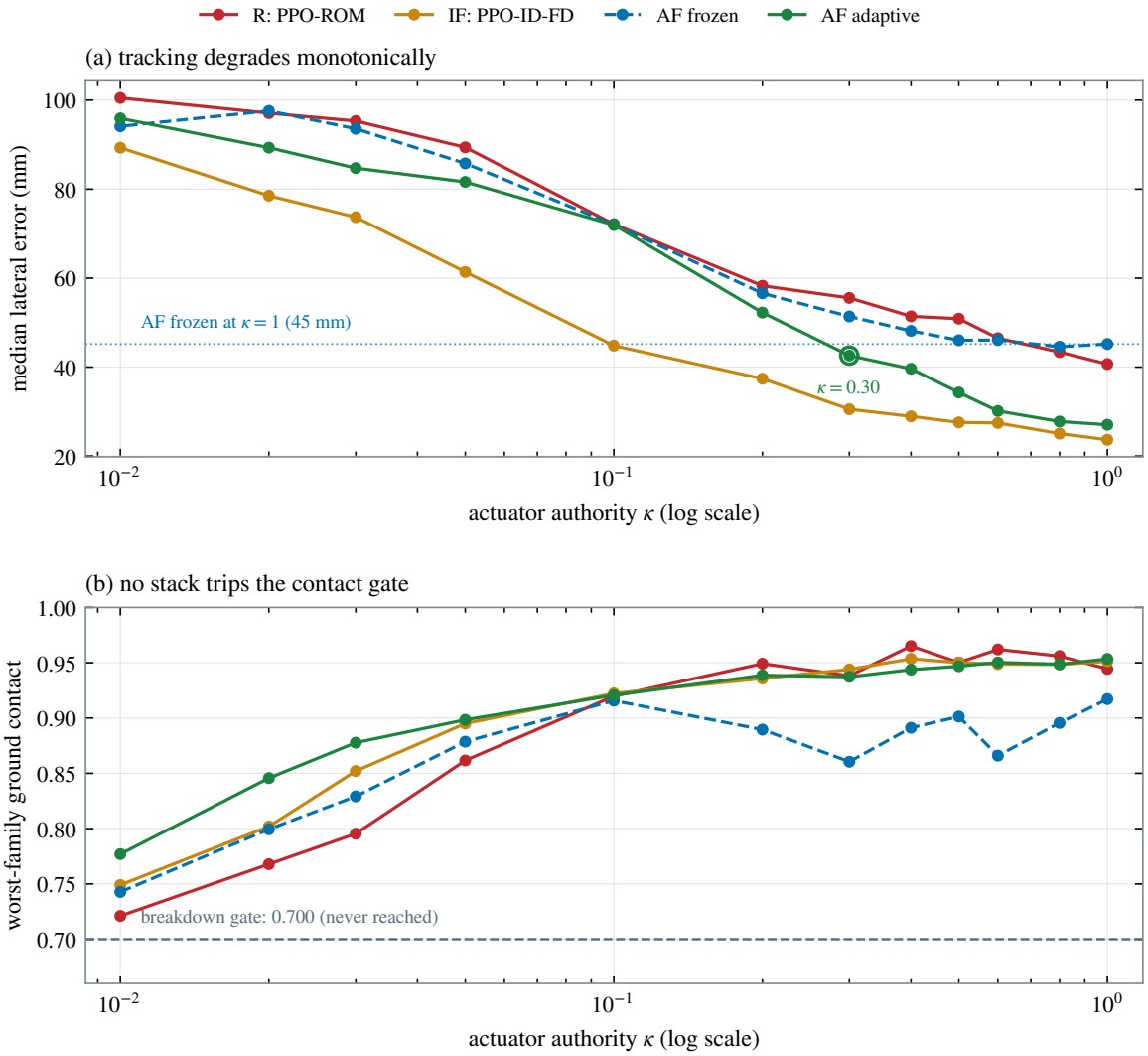

Figure 7.4: Lateral tracking and ground contact as actuator authority is reduced, across the three stacks at nominal payload. The sweep characterises the authority margin of the frozen controllers; the payload-conditioned mechanism is evaluated separately in Section 7.5.

The learned controllers retain a broad margin under authority loss. A hard failure boundary was not reached within the tested range, so the experiment establishes an authority margin rather than a breakdown threshold. IF has the lowest lateral error at every rung of the cross-stack comparison.

No stack reached the contact or safety gate. The sweep therefore bounds the authority margin the frozen controllers carry, and it is that margin the payload-conditioned mechanism is measured against in Section 7.5.

7.4.3 What Robustness Leaves for Adaptation

The robustness result sets the scale for the following experiment. At nominal payload the frozen controllers absorb substantial variation and the authority sweep reaches no contact or safety failure, so there is little there for adaptation to recover. That is precisely why the adaptation study sweeps payload instead: it locates the regime where the frozen controller does fail, and measures the mechanism where a deficit actually exists.

7.5 Evaluation of Online Adaptation

The deployed mechanism conditions the frozen policy on a payload it estimates online from the task observation it already reads (Section 5.7). This section measures what that conditioning is worth, in which payload regime it holds, and how much of it depends on the estimate being correct rather than merely present.

Three arms are compared, all of them the same network at a different context value. *Frozen* zeroes the estimate, which by Equation (6.11) recovers the base policy exactly rather than approximating it. *Adaptive* supplies the online estimate and is the deployable arm. *Oracle* supplies the true payload and is included as a reference point, not as a bound.

7.5.1 Experiment and Measurement

Scoring uses held-out mesh 3141, 176 environments and 900 steps per arm, paired on the environment index so that every comparison holds terrain, trajectory and payload fixed. Each arm runs in its own process, because an earlier layout that swept arms inside one process failed its own A/A check. The statistic is the paired per-environment median difference with a percentile bootstrap at $B = 20000$, reported with the fraction of environments improved.

The median log-ratio is not used as the primary statistic here. It disagrees in sign with the marginal medians in several cells, because with an even sample and a two-lobed paired distribution $\mathrm{median}(x - y)$ and $\mathrm{median}(x/y)$ need not fall on the same side of their null values. The difference and the win fraction are the honest summaries of an effect this heterogeneous.

Three payloads are scored, at 3.0×, 4.0× and 5.0× nominal cutter mass, and three independently trained context-column seeds. The seeds share one frozen base, so they test the adaptation mechanism's sensitivity to initialisation rather than the whole controller's, and are described that way throughout.

7.5.2 The Frozen Controller Has a Payload-Dependent Deficit

The comparison is only meaningful where the frozen policy is actually failing. Table 7.7 establishes that it is, and where.

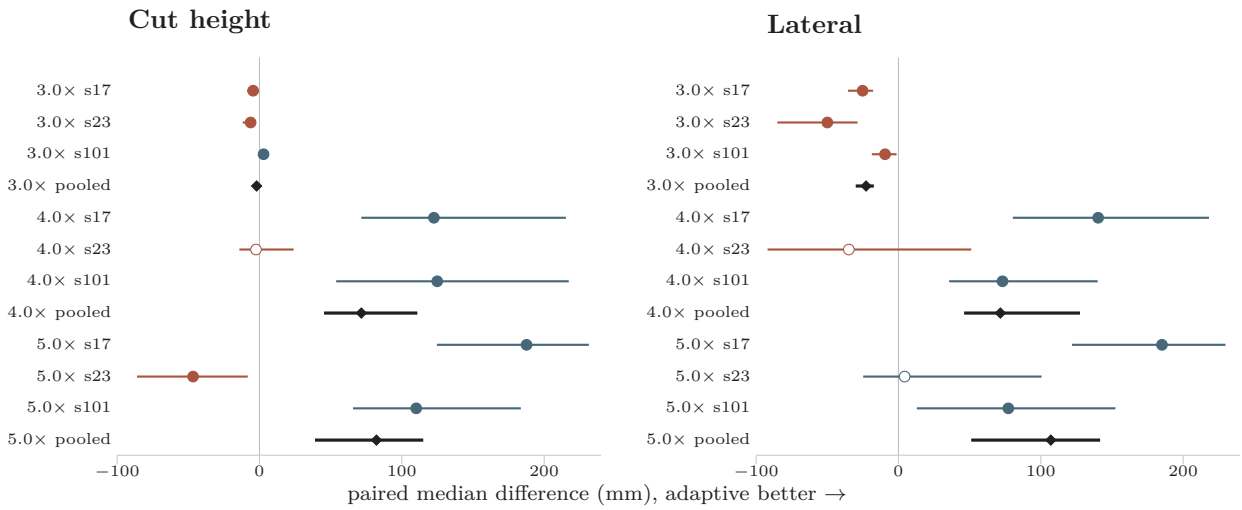

Figure 7.5: Adaptation against the frozen policy, paired per environment on held-out mesh 3141. Points are median differences and bars are 95 % percentile-bootstrap intervals over $B = 20000$; positive is better. Filled markers have intervals excluding zero, hollow markers do not, and diamonds are the across-seed pooled estimate. The sign reverses with payload: at 3.0× the frozen policy still holds ground contact and the conditioning only costs, while at 4.0× and 5.0× both channels improve.

Table 7.7: The frozen policy across payload, held-out mesh 3141. The collapse between 3.0× and 4.0× is the robustness deficit the adaptive arm is asked to repair.

Payload	Cut (mm)	Lateral (mm)	Ground contact	Safety violations
3.0×	10.37	84.37	0.883	0.096
4.0×	167.52	318.63	0.636	0.281
5.0×	267.44	400.89	0.601	0.303

At 3.0× the controller still holds the ground for 88 % of the window and has little to give back. Beyond it, cut-height error rises by more than an order of magnitude and ground contact falls away.

7.5.3 Adaptation Against the Frozen Policy

Figure 7.5 shows every cell and Table 7.8 gives the pooled estimates.

Table 7.8: Pooled paired median difference, adaptive against frozen, across the three seeds. Positive is better. Intervals are 95 % percentile bootstrap.

Payload	Channel	Frozen	Median difference	95 % CI
3.0×	cut (mm)	10.37	−2.03	[−3.51, +0.04]
	lateral (mm)	84.37	−22.75	[−30.04, −17.27]
	contact	0.883	−0.00	[−0.01, +0.01]
4.0×	cut (mm)	167.52	+71.52	[+45.34, +110.88]
	lateral (mm)	318.63	+71.58	[+46.12, +127.55]
	contact	0.636	+0.12	[+0.09, +0.17]
5.0×	cut (mm)	267.44	+82.18	[+39.00, +115.03]
	lateral (mm)	400.89	+107.05	[+51.11, +141.68]
	contact	0.601	+0.09	[+0.05, +0.13]

Online payload conditioning improves cut height, lateral tracking and ground contact together at 4.0× and 5.0×, every channel with its interval excluding zero. Pooled cut-height error falls by 71.5 mm against a frozen baseline of 167.5 mm and by 82.2 mm against 267.4 mm, a reduction of 43 % and 31 % of the frozen error, while ground contact rises by 0.12 and 0.09. Both task channels improve at once, which a single trade-off along the cut–lateral curve cannot produce.

The mechanism is regime-selective, and the sweep resolves the regime exactly. At 3.0× the frozen controller still holds the ground and conditioning costs lateral accuracy; from 4.0× it gains on every channel. The discriminating variable is ground contact, and the deadband of Section 5.7 is what confines the mechanism to the loaded side of it.

7.5.4 One Seed of Three Does Not Benefit

Seeds 17 and 101 carry the pooled effect, improving cut height by 187.6 mm and 110.1 mm at 5.0×. Seed 23 does not: neutral on cut at 4.0× and −46.68 mm at 5.0×, neutral on lateral and contact at both.

The cause is locatable rather than mysterious. All three oracles sit near 750 mm of lateral error, so every seed has the same trade-off curve available, and what differs is where the online estimate places the policy on it: 113 mm for seed 17, 150 mm for seed 101, 368 mm for seed 23. The variation is in the fitted estimator, not in the policy it serves, which makes it a tractable target for future work rather than a property of the mechanism.

7.5.5 The True Payload Is Not an Upper Bound

The deployable arm outperforms the privileged one. Handing the policy the true payload is worse than letting it estimate it, and at the largest load substantially so: seed 101's oracle at 5.0× reads 652.29 mm of cut error at a ground contact of 0.220, while its adaptive arm on the same checkpoint reads 40.46 mm at 0.803. Estimation is not a concession made for deployability here; it is the better controller.

The context enters as $\boldsymbol{W}_{\text{ctx}}\hat{\boldsymbol{c}}$ with the payload channel reaching 8 at 5.0×, and seed 101 carries the largest context weight of the three. The largest weight meeting the largest

input over-drives the policy off the ground. The norm does not order the trade-off at 3.0×, where the smallest-weight seed sits in the middle, so this is a statement about the extreme rather than a monotone law.

The deployable arm escapes it because the estimate is damped: it starts each episode at nominal and arrives through an exponential moving average, so it never reaches the region the true value walks straight into. The gradual arrival is part of the mechanism, not an implementation detail.

7.5.6 Correct Payload Against Wrong Payload

The remaining question is whether the benefit comes from the payload *information* or merely from the presence of a plausible context. The control permutes the true payload across environments on a mixed-payload plant, which preserves its distribution and magnitude exactly and destroys only its pairing with the environment it describes. The permutation is redrawn every step and is a derangement, so no environment keeps its own value. At a fixed payload this control would be vacuous, because every environment shares one mass and a permutation is the identity; the mixed plant is what gives it content.

Table 7.9: Wrong-information control, mixed payload 1.0–5.0× resampled per episode, paired per environment. Positive is better.

Comparison	Cut (mm)	Median difference	Environments better
frozen → true	17.75 → 1.63	+12.66	94.9 %
frozen → permuted	17.75 → 6.43	+11.10	84.7 %
permuted → true	6.43 → 1.63	+3.55	83.0 %

The decisive row is the third: the correct payload beats a wrong one drawn from the same distribution by 3.55 mm, interval $[+3.01, +4.14]$, improving 83.0 % of environments, and by 0.03 on ground contact with $[+0.02, +0.03]$. The mechanism uses payload information, and the falsification test confirms it.

The same table sizes the two components. A permuted payload recovers 11.10 of the 12.66 mm median gain, so conditioning on a plausible load supplies most of the effect and correct identification supplies the remainder. Both are real and the experiment separates them, which is more than a single aggregate number would have shown.

7.6 What Determines Effective Online Adaptation

Three conditions govern whether online adaptation adds value beyond a robust learned controller, and the experiments above separate them.

A control-relevant deficit must remain. This is the condition the payload sweep isolates most cleanly. The same mechanism, on the same checkpoints, improves every channel at 4.0× and costs lateral accuracy at 3.0×, and the discriminating variable is whether the frozen controller is losing ground contact. Adaptation is not uniformly good; it is good where robustness has run out.

The correction path must retain authority. The earlier explicit mechanism produced no closed-loop effect because clipping removed the correction before it reached the plant, although its estimate was informative (Section 5.6.2). Conditioning the policy rather than correcting its command avoids that interface entirely: there is no bounded term to clip, because the estimate changes what the policy computes rather than what is added to its output.

The estimate must be delivered at a rate the plant tolerates. The oracle result shows that correct information supplied instantaneously can be worse than an approximation supplied gradually. Damping is not a safety margin bolted on afterwards; it is the reason the deployable arm outperforms the privileged one.

The result is that a controller estimating its own payload from signals it already carries improves cut height, lateral tracking and ground contact together, in the loaded regime where a robust frozen controller is failing, and outperforms the same controller handed the true value. The conditions above say where it works and the controls say why, which is what makes it deployable rather than merely favourable.

8. Conclusion and Future Work

This thesis asked how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, and under what conditions online adaptation adds a further increment to the robustness those learned models already provide. The six research questions of Section 1.4 are answered in the order they were asked, which is also the order in which the answers depend on one another.

8.1 Summary of Findings

The findings are stated in the order of the research questions. The first four concern how hydraulic actuation is learned, transferred and controlled, and the last two concern robustness and online adaptation. Figure 8.1 makes the dependency between those experiments explicit.

The inverse and forward representation identified in Simscape transfers to the deployment simulation without retraining (RQ1, RQ2). The inverse model was trained on Simscape data alone and evaluated on all 118 IsaacLab deployment runs, drawn entirely outside that training distribution, where its teacher-forced one-step extension error is millimetre-scale and nearly flat across domains: 2.36 mm on the combined runs against 3.36 mm on nominal. The claim holds only in that one-step regime; closing the loop on its own predictions diverges in all 118 runs (Section 5.2.2). The reduced-order model received a small bounded calibration against the deployment plant.

Predictive accuracy must be accompanied by a control-authority test (RQ1). A learned actuator must be tested for dependence on the command, not only for prediction error. With the accepted excitation design, removing the valve input increases rollout error by a factor of 1.81; before it, by 0.94. This is a general validation requirement for learned models used in control.

Two actuator representations both support closed-loop control, and they do not produce the same controller (RQ3, RQ5). The command-conditioned reduced-order model reaches $R^2 = 0.998$ on the internal valve state and 0.97 on extension, and runs inside GPU-parallel training. The inverse–forward chain composes two separately

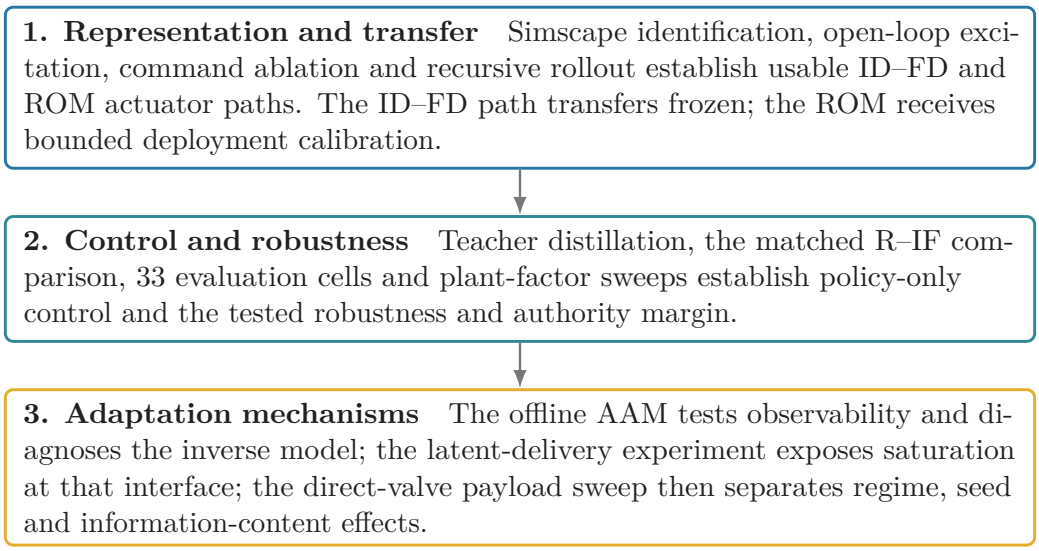

Figure 8.1: Experimental evidence chain used to answer the research questions. Each stage constrains the claim made at the next stage; the final adaptation claim is interpreted only after transfer and robustness have been established.

learned models in series, which could have compounded their errors and did not. The two induce different control strategies from the same policy interface. On the recorded traces of one held-out mesh, R spends the least command and reverses about 40 times per second against 10.6 for IF, which converts greater actuator effort into lower lateral error (Table 7.5). Over the 33 paired cells the median cell favours IF on lateral error by 9.3 mm with the interval excluding zero and favours R on cut height by 1.7 mm, while neither mean difference is resolved and the mean changes sign if the double S-curve family is removed (Table 7.2). The choice of representation therefore affects the resulting controller and is not an implementation detail, but the size of that effect is not established.

R and IF match on recorded training fields and differ only in the actuator path. AF uses a different seed and training configuration and is therefore evaluated through the paired within-stack comparison of Section 7.5.6, not ranked against them. With one R and one IF training run, their seed variance remains unknown.

A learned policy can replace the classical controller at runtime (RQ4). The damped-least-squares controller supplies privileged demonstrations during training and is then removed; the distilled policy emits desired cylinder motion directly into the learned actuator and is the entire deployed controller. Over 33 evaluation cells IF holds a mean lateral error of 70.3 mm, a median of 28.1 mm and 95.29 % ground contact (Table 7.1). The deployed policies remain anchored to the demonstrated behaviour because the reward is an imperfect proxy for the evaluation gates.

The robust stacks retain a broad margin against mechanical variation and authority loss without online adaptation (RQ2, RQ5). On held-out meshes, eleven trajectory families and sixteen reserved factor configurations, IF meets the lateral criterion in 32 of 33 cells at a median of 0.0281 m. Its frozen inverse model transfers to all 118 deployment runs with millimetre-scale one-step error.

The factor sweep supports the tested mechanical variation, but not unbounded hydraulic variation. On the inverse and forward path, hydraulic factors are not applied; on R they

remain inside the identification range (Section 7.4.1). Reducing realised actuator motion to one per cent of nominal did not trigger the safety or ground-contact gates. The result therefore establishes an authority margin, not a hard robustness boundary (Section 7.4.2).

A controller that estimates its own payload online improves task tracking where robustness runs out (RQ6). The deployed mechanism conditions the frozen policy on a payload regressed at runtime from the task observation it already carries. Against that frozen policy, paired per environment on a held-out mesh, it improves cut height, lateral tracking and ground contact together at 4.0× and 5.0× nominal cutter mass, every channel with its bootstrap interval excluding zero: pooled cut-height error falls by 71.5 mm and 82.2 mm against frozen baselines of 167.5 mm and 267.4 mm, and ground contact rises by 0.12 and 0.09.

The effect is regime-selective, and the sweep locates the regime. At 3.0×, where the frozen controller still holds the ground, the same conditioning costs lateral accuracy; from 4.0×, where it is losing contact, every channel gains. Two of the three training seeds carry the pooled effect. Permuting the payload across environments, which preserves its distribution and destroys only its pairing, is beaten by the correct estimate on cut height and ground contact, so the mechanism uses the information rather than merely the presence of a context.

One result was not anticipated. Supplying the true payload is not an upper bound: at the largest load it can drive the tool off the ground, while the damped online estimate on the same checkpoint tracks well. The gradual arrival of the estimate is part of the mechanism.

Table 8.1 separates the experiments behind that conclusion.

8.2 Design Implications

The design implications follow the same sequence: actuator identification, policy training and plant adaptation.

Identification and Transfer

Design the excitation for identifiability, and test the model by input ablation. Task-control data can make state and command nearly redundant. Dedicated excitation makes command effects identifiable. A learned actuator should therefore be accepted on both prediction accuracy and loss of performance when the command input is removed.

Embed known physics and bound learned corrections. The reduced-order model propagates pressure, force, velocity, and extension through explicit updates. Its learned terms are bounded, preventing unphysical state changes inside the training loop. This structure also represents the pressure dependence of valve authority directly.

Prefer a frozen model with a bounded, gated correction to retraining. The inverse model transferred at millimetre scale and remained frozen. A 3 mm correction absorbed the deployment offset, vanished at zero command and preserved the original data through replay, confining deployment-specific behaviour to a bounded component.

Table 8.1: Experimental decomposition of the adaptation claim. The rows answer different questions at different levels of the control chain and are not interchangeable evidence.

Experiment	Controlled comparison	Supported conclusion
Latent adapter objective	estimated, privileged and zero latent on the adapter's own objective	Online plant context is usable at the model level: both latent arms beat the zero-context control on five of five seeds.
Latent-delivery test	estimated, privileged and zero latent through the inverse-model correction path	Better plant context does not recover extension or velocity; saturation leaves insufficient correction authority at that interface.
Payload sweep	estimated payload against the zeroed-context frozen policy, paired per environment at 3, 4 and 5×	Cut height, lateral error and ground contact all improve at 4× and 5× with intervals excluding zero; at 3× the conditioning costs lateral accuracy.
Seed replication	three independently trained context-column seeds, pooled and individually	The pooled effect is large and significant, and is carried by two seeds of three.
Wrong-information control	true payload against a payload permuted across environments on a mixed plant	Correct identification beats a wrong value of the same distribution on cut height and contact; conditioning on a plausible load supplies the remainder.
Privileged reference	true payload against the online estimate on the same checkpoint	The true value is not an upper bound and over-drives the policy at the largest load; the damped estimate does not.

Control Methodology

Constrain policy updates when the reward is not the evaluation criterion. The physical evaluation gates and the reward are not identical. A forward-KL anchor to a verified reference policy bounds policy drift, and the anchor advances only through checkpoints that satisfy the gates. Reward terms should represent quantities that are also treated as evaluation constraints.

Where no baseline is inherited, the action contract is the safety argument. Zero action leaves the arm stationary. A rate limit, filter, travel shield, command scale and per-axis anti-windup cap bound its authority. The cap must be resolved per axis because the value that recovers boom offsets damages head alignment on cylinder 4 (Section 6.2).

Plant Adaptation

Measure identifiability and correction authority before training an estimator. The identifiability envelope determines whether plant identity is present in the data. The delivered-stress check determines whether a perturbation reaches the plant and whether a correction can influence the command. Both tests should precede estimator training.

Filtering may be needed when a changing estimate enters a memoryless policy, but its time constant remains to be determined on the deployed stack.

8.3 Contributions

Section 1.4 stated six contributions. Their standing at the end of the work is as follows, with the supporting evidence as summarised above rather than repeated.

Transferable hydraulic actuator learning is delivered through two Simscape-identified representations realised in IsaacLab; the inverse model transfers without retraining. The **reduced-order actuator model** is a bounded, differentiable surrogate suitable for GPU-parallel training and serves as the calibrated deployment plant. Its clamped constants are not identified physical parameters (Section 5.4).

The **control-relevant identification and validation** procedure combines input ablation, which exposed a forward model without command authority, with an identifiability audit of the excitation envelope.

Policy-only control through learned hydraulics is demonstrated. After the classical teacher is removed, IF holds a mean lateral error of 70 mm and a median of 28 mm over 33 held-out cells at 95 % ground contact (Section 7.2). Worst-cell aggregation still misses two gates; that bound is stated in Section 8.4.

The **decision procedure for online adaptation** establishes plant sensitivity, observability and correction authority before attributing any gain to adaptation. It also exposes when the optimiser trades away a gated quantity that the reward does not price.

Where online adaptation is effective is answered by the payload sweep. Conditioning the frozen policy on an online payload estimate improves cut height, lateral tracking and ground contact together once the load is heavy enough to cost the frozen controller its ground contact, and costs lateral accuracy below that point. The mechanism therefore belongs in the loaded regime, and the deadband keeps it there.

Two enablers are recorded: the causal estimator and bounded adaptation path use no privileged or future information and revert safely on invalid input. Five of the eight promotion gates separate the matched pair (Section 3.9.1).

8.4 Limitations

Scope of plant validation. The evidence is simulation-to-simulation, and the deployment plant is itself a learned model. No claim here is a claim about hardware.

Scope of hydraulic observability. The differential pressure the estimator consumes is produced by the actuator model rather than measured, so transferring the model-level result to hardware requires instrumentation the FME 500 does not carry. An observation contract remains unresolved alongside this: every result uses the pressure-inclusive row, while the stricter deployable alternative excludes measured differential pressure. Since the estimator's advantage is concentrated in that channel, the deployable variant should be expected to perform worse, and by how much is not established.

Scope of controller comparison. On worst-mesh, worst-family statistics both stacks fail the lateral gate and IF does not satisfy the head-alignment gate under worst-case aggregation (Section 7.2), so neither stack satisfies the whole eight-gate evaluation contract. The closed-loop comparisons resolve differences to a few per cent over 33 paired conditions and do not establish equivalence. The available plant variation bounds the disturbance rather than the architecture, and more tightly than the sweep's labels suggest. A stack has been shown to tolerate what actually reached it, which on the inverse and forward plant is a mechanical perturbation only, plus a generic reduction of realised actuator motion. Median lateral error first reaches 1.5 times its full-authority value at $\kappa = 0.20$ for IF and $\kappa = 0.10$ for R, and no stack reaches the pre-declared breakdown gate down to $\kappa = 0.01$ (Section 7.4.2). That is an authority margin, not a demonstration of hydraulic robustness. The three stacks are matched on evaluation but not on training budget, so the comparison describes three deployed controllers rather than three equally tuned training procedures.

Scope of training variation. Policy-training variation is not quantified for the matched R and IF pair, and it bounds two claims differently. R and IF are each represented by one training run, so the 33 paired cells measure terrain and trajectory variation and say nothing about how far a second run of either would move; the actuator-representation result rests on that assumption. The adaptation result uses three context-training seeds on one shared frozen base, so it bounds the mechanism's sensitivity to initialisation and not the whole controller's, and the pooled effect is carried by two of the three.

Scope of the adaptation evidence. The commanded step cannot finish faster than 417 ms at the command-rate limit while the plant answers in about 117 ms, so this loop favours continuous conditioning on the operating point over reaction to a single event, and the mechanism is built and measured that way. The result is scored at three payload levels on one held-out mesh; the regularisation of the payload regression was left at its weakest setting, identical across seeds.

8.5 Future Work

The evidence in this thesis is simulation-to-simulation, and the three experiments that would most change what it can claim are also in simulation. They are listed first for that reason.

Close the gap between the estimate and the true payload. All three seeds share one trade-off curve, and what separates them is where the fitted estimator places the policy on it. The regression is deliberately simple, fitted once per seed on a mixed-payload rollout with its regularisation unselected. A held-out selection of that strength, or a short temporal window in place of the per-frame fit, addresses the one component the results identify as the source of seed-to-seed variation.

Train more than one seed per representation. R and IF are one training run each, so the cross-stack comparison of Section 7.2 describes two deployed controllers and cannot separate the representation from the seed. Three or more runs per stack, scored on the same 33 cells, would convert that observation into a result at moderate cost.

Ablate reference-anchored PPO. Section 6.3.2 establishes that unconstrained fine-tuning fails; it does not establish that this particular construction is what fixed it. A three-arm comparison against plain PPO with early stopping and against the un-fine-tuned distilled policy would separate the block ratchet from the constraint it sits inside.

Then the machine. Hardware evaluation begins by adding chamber-pressure and extension sensing and a rod load pin to the FME 500. The open-loop excitation schedule and identifiability audit of Chapter 4 can then be applied to machine data. Any correction should remain bounded, retain the identified hydraulic parameters and original data for replay, and pass input ablation. The reference-anchored policy provides the verified starting checkpoint.

The central design lesson is conditional. Robust training absorbs a large part of the plant variation before online adaptation is introduced. A remaining correction is useful only when the variation is observable, the fixed controller leaves a deficit to recover, and the command path retains authority. Apparent adaptation gains must also be separated from gains due to residual capacity. Evaluating these conditions together provides a defensible basis for deciding when online adaptation is justified on a hydraulic manipulator.

A. Dataset Ranges and Audit

Chapter 4 describes how the Simscape and IsaacLab collections were generated. This appendix retains only the numerical plant ranges, the open-loop audit and the frozen deployment split needed to interpret or reproduce those collections.

A.1 Randomised Plant Ranges

Table A.1: Randomised hydraulic parameters. The seal-wear coordinate is drawn independently for each cylinder and drives three physically linked quantities.

Parameter	Nominal	Randomised range	Acts on
Entrained air fraction ε_{air}	0.005	0.001 – 0.080	fluid compliance
Dead-volume fraction ε_{vol}	0.020	0.005 – 0.080	fluid compliance
Discharge coefficient c_d	0.70	0.56 – 0.84	valve flow gain
Seal wear $w_{\text{seal},i}$	0	0 – 0.25 per cylinder	clearance, leakage, spool lag
Relief setting p_{set}	27.6 MPa	24.8 MPa – 30.4 MPa	pressure ceiling

Table A.2: Randomised mechanical parameters. Rigid-body perturbations are applied independently to each of the fourteen bodies.

Parameter	Nominal	Randomised range
Body mass	CAD value	±20 %
Centre-of-mass offset	CAD value	±5 %
Inertia tensor and products	CAD value	consistent with mass draw
Joint stiffness, J1 and J2	60 kN/m	40 – 120 kN/m
Joint stiffness, J3 and J4	60 kN/m	40 – 120 kN/m
Joint stiffness, J5	25 kN/m	15 – 60 kN/m
Joint damping	250	100 – 600

Algorithm A.1: Closed-loop curriculum generation.

input : run budget $N = 1000$; joint limits $\boldsymbol{q}_{\min}, \boldsymbol{q}_{\max}$; nominal parameters $\boldsymbol{\theta}_0$
output : dataset $\mathcal{D} = \{(\text{trajectory}_i, \boldsymbol{\xi}_i)\}_{i=1}^{N}$

$\mathbf{L} \leftarrow \textsc{LatinHypercube}(N, 12)$ `// one design for the whole batch`
for $i \leftarrow 1$ **to** N **do**
  assign i to a curriculum block (nominal / hydraulic / mechanical / full)
  $\boldsymbol{\xi}_i \leftarrow \textsc{DrawParameters}(\mathbf{L}_i, \text{block})$ `//` $\varepsilon_{\text{air}}, \varepsilon_{\text{vol}}, c_d, w_{\text{seal}}, p_{\text{set}}$ `; masses, CoM, inertia, stiffness`
end
for $i \leftarrow 1$ **to** N **do**
  inject $\boldsymbol{\xi}_i$ into the Simscape model
  $n_w \leftarrow \textsc{RandInt}(12, 20)$; $\boldsymbol{W} \leftarrow \textsc{LatinHypercube}(n_w, \text{JDOF})$ within $[\boldsymbol{q}_{\min}, \boldsymbol{q}_{\max}]$
  assign each segment a speed class (slow / medium / fast / variable); **no** dwell
  $\boldsymbol{q}_{\text{ref}}(t) \leftarrow \textsc{Interpolate}(\boldsymbol{W})$ over 22 s
  simulate with `ode23t`; log on the 0.05 s grid
  store $(\boldsymbol{u}, \boldsymbol{x}, \dot{\boldsymbol{x}}, \Delta\boldsymbol{p}, \boldsymbol{F}, \boldsymbol{q}, \dot{\boldsymbol{q}}, \boldsymbol{q}_{\text{ref}}, \boldsymbol{\xi}_i)$
  if $i \bmod 100 = 0$ **then** checkpoint $\mathcal{D}$ to disk
end

Table A.3: Six physical factors in the IsaacLab deployment collections. One factor vector is drawn per run and held fixed throughout that run.

Coordinate	Range	Domain	Acts on
e_{flow}	$[-0.25, 0.25]$	hydraulic	valve flow gain
$e_{\text{compliance}}$	$[-0.25, 0.10]$	hydraulic	effective chamber compliance
e_{seal}	$[0, 1]$	hydraulic	seal wear, leakage, spool lag
e_{pressure}	$[-0.15, 0.15]$	hydraulic	supply/relief pressure level
e_{mass}	$[0.8, 1.5]$	mechanical	cutter mass scale
e_{COM_z}	$[-0.10, 0.05]$	mechanical	cutter centre-of-mass height

A.2 Closed-Loop Curriculum Generation

The procedure that produced the 1 000-run Simscape curriculum of Section 4.2. No reported model is trained on it; it is recorded so the collection can be reproduced and audited.

A.3 Open-Loop Export Audit

Table A.4: Audit of the strict open-loop excitation export. The extraction stage computes and stores every check.

Check	Result	Interpretation
Rows, runs, columns	25 600 / 4 / 49	four complete runs
Median sample time	0.01 s	as commanded
Maximum sample-time error	9.1×10^{-15} s	numerically exact grid
Maximum Δextension error	0	targets match state differences
Maximum Δvelocity error	0	targets match state differences
Maximum $\Delta(\Delta p)$ error	0 Pa	targets match state differences
Non-finite values	0	no solver escapes
Valid fraction	0.937	startup, gaps and shutdown excluded
Command rank	4	all four channels independent
Lagged excitation rank	80	history matrix full rank
Valve saturation	8.5 % per channel	limits reached but not dominant
Stroke coverage	12 % – 33 %	above floor, below target

A.4 Frozen Deployment Collections

The task-controller and open-loop context-identification collections share the same factor design, domain composition and run-level split. Each contains 118 runs with one fixed factor vector per run. Model-level identification results use the open-loop collection for the identifiability reason given in Section 4.7.5.

Table A.5: Composition of each frozen deployment collection. The 102 training runs are divided into 82 fitting and 20 validation runs; the remaining 16 runs form the held-out test split.

Domain	Runs	Factors varied
`combined`	64	all six
`hydraulic`	40	flow, compliance, seal, pressure
`mechanical`	12	mass scale, centre-of-mass height
`nominal`	2	none
Total	118	

Splitting is performed at run level. Since the factor vector is constant within a run, a window-level split would place near-duplicates of the same plant in both fitting and evaluation sets. The frozen manifests store the run identifier, factor key and split label, making the separation directly auditable.

B. Implementation and Reproducibility Record

The methods chapters define the architectures and training procedures. This appendix retains only the timing contract, exact input layouts, frozen artifact identifiers and reproduction entry points that are cumbersome to include in the main text.

B.1 Timing Contract

Table B.1: Rates of the simulation, control and learned components. The offline AAM and deployed AF encoder are listed separately because they use different history grids.

Layer	Rate	Period
PhysX physics substep	120 Hz	8.33 ms
Policy and control step	60 Hz	16.67 ms
AF payload estimate and context	60 Hz	16.67 ms
Inverse, forward and ROM models	20 Hz	50 ms
Offline AAM inference	20 Hz	50 ms
Simscape logging grid	20 Hz	50 ms

The offline AAM reads 20 rows on the 20 Hz grid, giving a 1.0 s window, and its five-estimate smoother adds 250 ms of delay. Deployed AF instead reads 20 control-step rows, giving a 333 ms window, and applies no output smoother.

B.2 Frozen Artifacts

Table B.2: Frozen artifacts and their SHA-256 digests, truncated to the first 32 hexadecimal digits. The context-column checkpoints share the frozen AF base and differ only in their 1 024 trained parameters.

Artifact	Role	SHA-256 (truncated)
`aam.pt`	offline AAM	`cd6a7f0b118b96c7ce66b889b9b3e422`
`zhat_adapter.pt`	ID adapter	`30629a4f83db5c2bed3c777bcf82dc57`
`best_id_cylinder_control.pt`	inverse model	`8e26c52e82d89efe193e001d0a8b07a6`
`best_fd_cylinder_strict.pt`	forward model	`1f41e44309a8f320430a3382ca88ce5f`
`rom_s6_aligned/model_598.pt`	R policy	`ce7e3ab9649b1193f123f9f91107bbf8`
`idfd_s6_aligned/model_598.pt`	IF policy	`06533b8e8614b0ad982dd8c238a17772`
`distilled_v1/phase1_init_70.pt`	AF frozen base	`641e476cfcd765a8d68a17980d6cbbf2`
`ctxonly17/model_1596.pt`	AF context columns, seed 17	`805da12505d214ca9f40c405cf02e505`
`ctxonly23/model_1596.pt`	AF context columns, seed 23	`79778e16722bbf2ace1b5c0e680a47dd`
`ctxonly101/model_1596.pt`	AF context columns, seed 101	`80b15d39d0b61ed3ae5ac72b248d9329`
`s17_fit.npz`	payload estimator, seed 17	`1b48070e035f05b07f52449ee2a42b83`
`s23_fit.npz`	payload estimator, seed 23	`422a5f498b03426ab2fe26d52d724174`
`s101_fit.npz`	payload estimator, seed 101	`d7abd08b30d0489f8da593f3c5c0541e`

B.3 Learned-Actuator Interfaces

Both actuator networks use a linear projection to 512 units, four pre-norm residual blocks, final normalisation and a linear output projection. Their exact interfaces are given in Table B.3.

Table B.3: Input and output layouts stored in the frozen inverse and forward model checkpoints.

	Inverse model	Forward model (strict)
Input width	28	20
Input layout	extension(4), velocity(4), $\Delta p(4)$, force(4), target Δextension(4), target Δvelocity(4), $\Delta t(4)$	valve(4), extension(4), velocity(4), $\Delta p(4)$, force(4)
Output width	4	12
Output layout	valve command(4)	Δextension(4), Δvelocity(4), $\Delta(\Delta p)(4)$
Target mode	normalised Δ valve command	normalised Δ cylinder state
Parameters	2 124 292	2 124 300

Only the predicted extension increment of the forward model is integrated in the control loop. Its velocity output is retained as a diagnostic.

B.4 Policy Observation Layout

The deployed actor receives 66 task values in every stack. AF appends four plant-context columns, giving a 70-value input. The final twelve task values make actuator prediction and cylinder tracking errors visible to the otherwise memoryless actor.

The context width is read off the frozen checkpoint rather than asserted. In `model_1596.pt` of each context-column arm the actor's first layer is 256×70, of which the last four columns are the context and carry 1 024 trainable parameters; the preceding 66 columns are bit-identical across the three seeds, which is the check that the gradient mask held. Only the payload channel is non-constant in this campaign, and the exclusion contract of Section 4.8.2 applies to the offline AAM only.

Table B.4: Policy observation layout. Indices are zero-based and half-open.

Index	Width	Contents
0–9	9	verge height profile ahead of the cutter
9	1	cut-height error
10	1	lateral error
11	1	cutter vertical velocity
12–16	4	measured cylinder extensions
16–20	4	measured cylinder velocities
20–24	4	cylinder headroom, low side
24–28	4	cylinder headroom, high side
28–32	4	previous applied action
32–36	4	current valve commands
36–40	4	actuator-predicted Δextension
40–44	4	commanded target lead
44	1	head-down alignment
45	1	head-down alignment rate
46	1	truck speed
47	1	scan valid fraction
48–50	2	cutter local $+Z$ axis in world XY
50	1	normalised cutter-terrain interaction force
51–54	3	path preview at 0.5, 1.0 and 2.0 m
54–58	4	measured minus commanded cylinder state
58–62	4	actuator prediction residual
62–66	4	normalised cylinder effort

B.5 Internal Model Pressure State, Not a Physical Pressure

The pressure-state analysis is mechanistic support only. It verifies that the internal hydraulic state changes across the load event; it is not used to quantify adaptation performance or hydraulic accuracy. Figure B.1 shows the internal state per cylinder for all three stacks on the same plant and aligned on the same event.

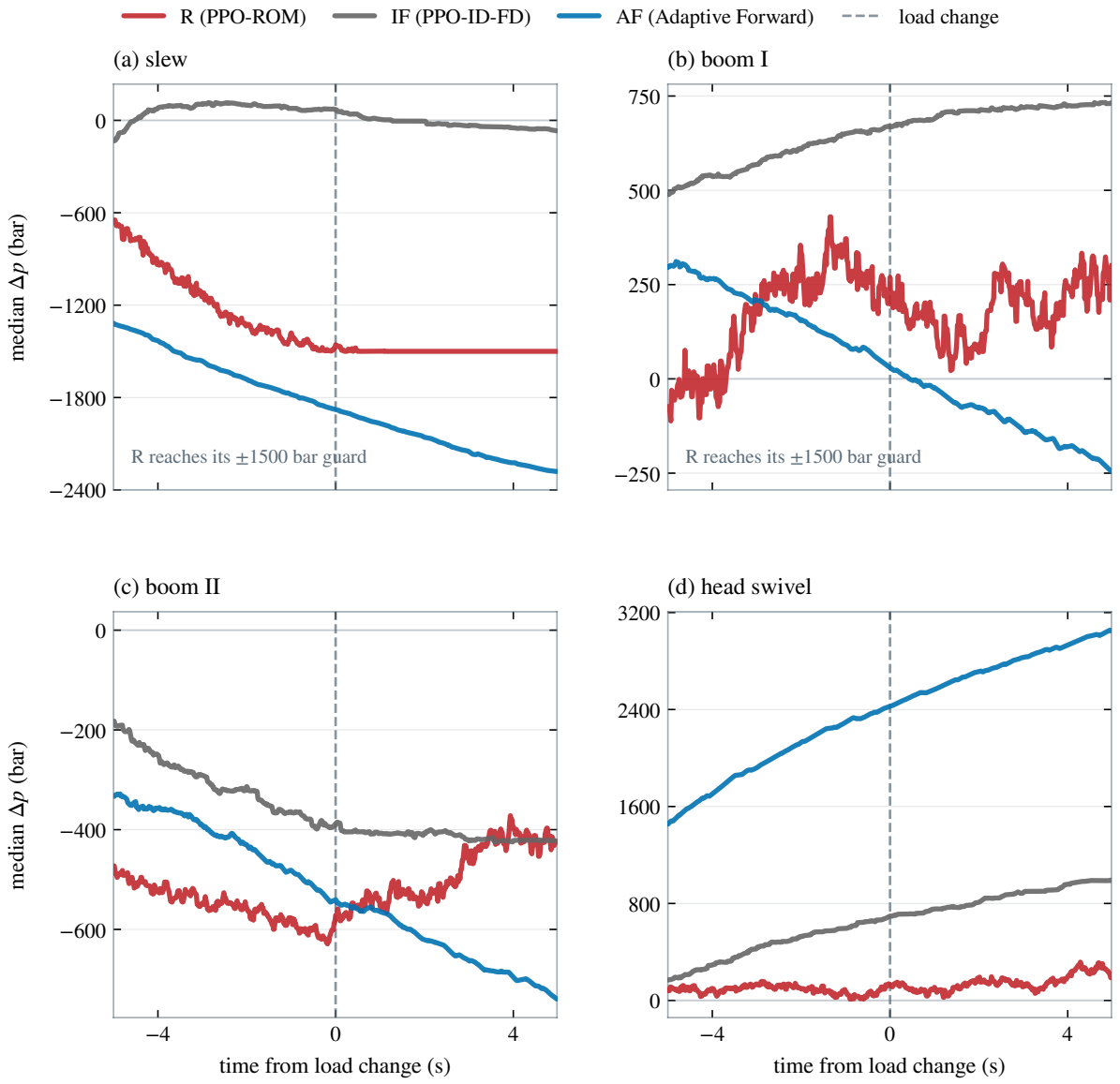

Figure B.1: Median differential pressure per cylinder through the change of load, 151 environments aligned on their own flip instant. The quantity is each stack's own actuator-model state, internal to the command ROM for R and the forward model's recursive state for the other two, so the panels show what each controller believes about the hydraulics rather than a common measurement, and each panel carries its own scale. The slew and boom I panels mark where R sits on its 1500 bar numerical guard.

Two properties of the signal govern how the figure is read. The pressure state is not a calibrated pressure on any stack: in this window the medians run to 2295 bar and the peaks to 3784 bar, which is past what a cylinder of this size could develop. The rod-force channel carries no supervised weight and pressure-increment accuracy is only $R^2 = 0.06$ to 0.09 (Section 5.4). On R the excursions are additionally clipped by the 1500 bar numerical guard, which binds on the slew and the first boom cylinder and not on the other two. There is therefore no common physical level the panels can be read against, and each

cylinder is drawn on its own scale. What the figure supports is the shape of the response either side of the change, not the level.

Read that way, the traces show that the model-derived pressure state responds through the event. The adaptive advantage is already present before it, which is consistent with continuous conditioning on the operating point. The uncalibrated state is not evidence for the magnitude or cause of the tracking improvement.

B.6 Reference-Anchored PPO in Full

The five-stage procedure summarised in Section 6.4. The teacher is used only in Stage 1; every PPO rollout is controlled by π_ϑ alone, and the frozen reference advances only through checkpoints that pass every gate.

B.7 Derivation of the Reference-Divergence Gate Bound

Section 6.4.5 states, without showing the steps, that a bound on the per-step reference divergence D_{ref} implies a bound on gate degradation of the form $|\mathbb{E}_{\pi_\vartheta}[g] - \mathbb{E}_{\pi_{\text{ref}}}[g]| \leq 2G_{\max}\sqrt{H\delta_{\text{ref}}/2}$. This appendix derives that bound and recomputes it at the production horizon, since the number quoted in the main text does not match evaluating the formula it cites.

Setup

Let P_π denote the distribution over length-H trajectories $\tau = (s_0, a_0, \ldots, s_{H-1}, a_{H-1})$ induced by rolling out policy π under the fixed environment transition kernel. Because the transition kernel is shared by π_{ref} and π_ϑ, it cancels from the log-ratio between the two trajectory distributions, and the chain rule of KL divergence reduces to a sum of per-step, per-state action divergences:

$$\mathrm{KL}(P_{\pi_{\text{ref}}} \,\|\, P_{\pi_\vartheta}) = \mathbb{E}_{\tau\sim P_{\pi_{\text{ref}}}}\left[\sum_{t=0}^{H-1} \mathrm{KL}\Big(\pi_{\text{ref}}(\cdot \mid s_t) \,\|\, \pi_\vartheta(\cdot \mid s_t)\Big)\right]. \tag{B.1}$$

This step is exact only if the per-step divergences on the right are evaluated on states drawn from π_{ref}'s own rollout distribution. In the actual training loop D_{ref} is evaluated on states visited by the *live* policy π_ϑ (Section 6.4), not by π_{ref}, so what follows is the idealised argument the main text already flags as carrying a measure mismatch, not a derivation that closes that gap.

Chain rule bound

The chain rule for KL divergence is standard [Cover 06]. Bounding every one of the H per-step terms by the training-time average bound δ_{ref} of Equation (6.8) gives $\mathrm{KL}(P_{\pi_{\text{ref}}} \| P_{\pi_\vartheta}) \leq H\,\delta_{\text{ref}}$.

Pinsker's inequality

[Cover 06] For any two distributions P, Q on a common space, total variation distance $\delta_{\text{TV}}(P,Q) := \sup_A |P(A) - Q(A)|$ satisfies $\delta_{\text{TV}}(P,Q) \leq \sqrt{\frac{1}{2}\mathrm{KL}(P\|Q)}$. Applied to the trajectory distributions,

$$\delta_{\text{TV}}(P_{\pi_{\text{ref}}}, P_{\pi_\vartheta}) \leq \sqrt{\tfrac{1}{2}H\delta_{\text{ref}}}. \tag{B.2}$$

Algorithm B.1: Five-stage reference-anchored PPO procedure. The teacher is used only in Stage 1. All PPO rollouts are controlled by π_ϑ alone, and the frozen reference moves only when a checkpoint passes every gate.

input : teacher π_T; gate test $\boldsymbol{G}(\vartheta) \preceq \boldsymbol{g}^\star$; M PPO blocks; anchored loss and divergence bounds from Equations (6.7) and (6.8)
output : last promoted actor $\vartheta^\star$ satisfying the gate test

`// Stage 1 - Distil a feasible actor`
$\mathcal{D} \leftarrow$ DAgger states labelled by teacher mean action $\boldsymbol{y}^T$
$\vartheta \leftarrow \arg\min_\vartheta \mathbb{E}_{(\boldsymbol{o},\boldsymbol{y}^T)\sim\mathcal{D}}\big[\|\mu_\vartheta(\boldsymbol{o}) - \boldsymbol{y}^T\|_2^2\big]$
remove π_T from the controller `// all later rollouts are teacher-free`
`// Stage 2 - Warm the critic and set the first anchor`
freeze ϑ and the observation normalisers
for $j \leftarrow 1$ **to** 150 **do**
  roll out π_ϑ and set $\psi \leftarrow \psi - \eta_{\text{critic}} \nabla_\psi \mathcal{L}_V(\psi)$
end
verify $\vartheta_{\text{after}} = \vartheta_{\text{before}}$
$\vartheta_{\text{ref}} \leftarrow \vartheta$; $\vartheta^\star \leftarrow \vartheta$ `// initial verified anchor`
for *block* $m \leftarrow 0$ **to** $M - 1$ **do**
  `// Stage 3 - Train inside the frozen reference bound`
  freeze ϑ_{ref} for the whole block
  for *each PPO iteration in block* m **do**
    collect an on-policy rollout with π_ϑ as sole controller
    compute and normalise GAE; save snapshot $\mathcal{S}$; set $\vartheta_{\text{old}} \leftarrow \vartheta$
    for 3 *epochs over the rollout minibatches* **do**
      if $\mathrm{KL}(\pi_{\text{old}} \| \pi_\vartheta) > \delta_{\text{upd}}$ **then** stop the update epochs
      update actor and critic using Equation (6.7)
    end
    compute $(D_{\text{upd}}, D_{\text{ref}})$ on the full rollout `// full rollout`
    if D_{upd} *or* D_{ref} *is non-finite* **then**
      restore the complete snapshot $\mathcal{S}$
    else if $D_{\text{upd}} > \delta_{\text{upd}}$ ***or*** $D_{\text{ref}} > \delta_{\text{ref}}$ **then**
      restore the actor and its optimiser moments from $\mathcal{S}$; retain the new critic fit
  end
  `// Stage 4 - Evaluate the block and promote a passing checkpoint`
  score every saved checkpoint on at least three held-out terrain meshes
  aggregate each gate to the worst mesh and, for tracking gates, the worst path family
  $\mathcal{C} \leftarrow \{\vartheta_c : \boldsymbol{G}(\vartheta_c) \preceq \boldsymbol{g}^\star\}$
  if $\mathcal{C} \neq \varnothing$ **then**
    choose $\vartheta^\star \in \mathcal{C}$ using the fixed checkpoint selection rule
    load $\vartheta \leftarrow \vartheta^\star$ and set $\vartheta_{\text{ref}} \leftarrow \vartheta^\star$
    reset the actor optimiser moments; retain the critic state
  else
    leave ϑ_{ref} unchanged; do not promote the block
  end
end
`// Stage 5 - Return the deployable policy`
return $\vartheta^\star = \vartheta_{\text{ref}}$

Bounded-functional bound

For a trajectory functional $g : \mathcal{T} \to [0, G_{\max}]$, such as any of the eight gate metrics, the standard identity $\sum_\tau |P(\tau) - Q(\tau)| = 2\,\delta_{\mathrm{TV}}(P, Q)$ gives

$$\left|\mathbb{E}_{P_{\pi_{\mathrm{ref}}}}[g] - \mathbb{E}_{P_{\pi_\vartheta}}[g]\right| \le G_{\max} \sum_\tau |P_{\pi_{\mathrm{ref}}}(\tau) - P_{\pi_\vartheta}(\tau)| = 2\,G_{\max}\,\delta_{\mathrm{TV}}(P_{\pi_{\mathrm{ref}}}, P_{\pi_\vartheta}). \tag{B.3}$$

Combining Equations (B.2) and (B.3) gives the quoted bound,

$$\left|\mathbb{E}_{\pi_\vartheta}[g] - \mathbb{E}_{\pi_{\mathrm{ref}}}[g]\right| \le 2\,G_{\max}\sqrt{\tfrac{H\,\delta_{\mathrm{ref}}}{2}}. \tag{B.4}$$

Numeric value at the production horizon

At $H = 2808$ steps and $\delta_{\mathrm{ref}} = 0.05$, $\sqrt{H\delta_{\mathrm{ref}}/2} = \sqrt{70.2} \approx 8.38$. For a gate metric normalised to $[0, 1]$, $G_{\max} = 1$, so Equation (B.4) evaluates to $2 \times 8.38 \approx 16.76$, the value quoted in Section 6.4.5. A bound of 16.76 on a $[0, 1]$-valued metric is vacuous, and it does not close the measure mismatch noted above. The construction remains a principled restriction of reward ascent to a neighbourhood of a verified policy, not a certificate.

B.8 Reproduction Entry Points

The closed-loop results of Chapter 7 are generated by the scripts under `experiments/E1` through `experiments/E6`. Each experiment directory records the evaluated model, mesh set, trajectory families, carrier speed and scored horizon in its manifest. AF arms are trained and scored under `PPO_AAM_FD/experiment_scripts`; the paired seed statistics are recomputed from `outputs/rif_delay` by `experiments/E3/code/seed_table.py`. The paired cross-stack statistics of Table 7.2, including the leave-one-family-out aggregate, are recomputed from the frozen E1 reports by `experiments/E1/code/rif_paired.py`. Model-level results use the frozen collections described in Appendix A.4.

Each pre-registered comparison has a dated design note written before its arms were trained, under `reports/`: the success rule, the checkpoint rule, the statistic and the declared outcomes are fixed there and are not re-decided after the seeds are seen. Together with the artifact digests in Table B.2, these records identify the data, models, rules and evaluation code behind the reported results.

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